Relational Frictions Along the Supply Chain: Evidence from Senegalese Traders*

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Abstract

We study search and trust frictions in international sourcing, and whether the rapid growth of "social commerce" in lower-income countries can alleviate them. Guided by a dynamic model of relational contracting, we run a field experiment with 1,862 garment firms in Senegal, randomly matching them to suppliers in Türkiye (search) and varying information about supplier type (adverse selection) and incentives (moral hazard). New supplier connections expand access to foreign varieties and quality, but the additional information about supplier trustworthiness is necessary for building lasting and profitable relationships. Structural estimates imply that both adverse selection and moral hazard substantially limit trade.

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1 Introduction

Search and trust frictions have historically made it hard for small firms to access foreign inputs. Consider a clothing wholesaler in Senegal who wants to start selling high quality European-made jeans. First, they must find a supplier and some way of seeing what this supplier sells—a classic search problem. Second, even if they find one and place an order, they typically pay before observing quality—exposing them to trust barriers in the form of both adverse selection (some suppliers may be more reliable than others) and moral hazard (suppliers may have incentives to shirk). These frictions may be particularly severe in lower-income settings, whose large informal sectors complicate information aggregation and lower state capacities make contracts hard to enforce. While there is a long theoretical tradition on search (Stigler, 1961) and trust (Shapiro, 1983), separate identification of these frictions and of how they interact remain rare because researchers typically do not observe variation that is both exogenous and maps to the theoretical primitives.

In recent years, rapid growth in smartphone ownership in lower-income countries has changed how small firms buy and sell. Across Africa, online sales have expanded quickly—B2C e-commerce revenue doubled between 2019 and 2024 (Statista, 2025)—and much of this activity runs through social media rather than traditional marketplaces. Indeed, in a survey of MSMEs in six African countries, GSMA (2023) report that social media is the most widely used online channel, regardless of the size of the business. This growth in "social commerce" may reflect the fact that social media can alleviate both search and trust frictions in supply chains: suppliers can broadcast assortments at low cost (search), and buyers can share information and coordinate action to deter cheating (a modern analogue of Greif (1993)). However, despite its prevalence, evidence on the extent to which social media can serve as a trade-enhancing infrastructure remains limited.

In this paper, we provide experimental evidence on the magnitude, mechanisms, and remediability of both search and trust frictions in access to foreign inputs, and evaluate whether widely used social media platforms can relax them. We run a field experiment in a large international import market that exploits common features of social media to lower search costs and manipulate trust primitives. Specifically, we randomly allocate 1,862 small garment firms in Senegal to treatment arms that connect them to new suppliers in Türkiye (search), and cross-randomize information about the types (AS) and incentives (MH) of these suppliers. We measure how these interventions affected their access to foreign goods, supplier relationships, and profits and sales, using data from a mystery shopping exercise, real-time transactions from Senegal's largest mobile payments provider, and a follow-up survey.

We first conducted a large-scale baseline survey of 1,862 garment firms in Dakar, Senegal's capital. We provide novel descriptive evidence that firms use social media extensively in ways consistent with alleviating search and trust frictions. For search, 86% of firms are in WhatsApp groups managed by suppliers (both domestic and international), with the median firm in four. These groups help firms see what suppliers sell, but it is unclear how much of the friction it alleviates as firms must still find the suppliers in the first place. For trust, 23% of firms are in WhatsApp groups with other firms for the purpose of sharing business information, and 67% have recommended or warned against particular suppliers to other firms in the past year. However, these networks are often highly fragmented, so it is unclear how relevant this information is to the average firm. While our focus is on WhatsApp, we also document that other apps, such as TikTok, Instagram, and Facebook, are also popular, each used by around a third of firms. Motivated by the widespread use of "social commerce" among businesses, we design an experiment that leverages social media to create exogenous variation in search and trust frictions in international trade.

To be precise about what we mean by search and trust frictions, we develop a model of relational contracting with sequential search for suppliers and both adverse selection and moral hazard. We use this to discuss how social commerce may affect the frictions, how experimental variation might separate them, and how they interact with each other. A firm can pay a search cost to match with a random supplier abroad, who can take a costly but unobservable action that increases the probability of high quality output. Bad-type suppliers will never do this, while good-type suppliers will only do so if the future value of the relationship exceeds the current cost. We derive a wedge equation showing that the trust frictions distort quantity downwards on the optimal contract. Moreover, we show that interventions that alleviate them are complements: if the firm's beliefs about a supplier's type exogenously improve, then they buy more today, thereby flattening the quantity schedule, weakening the supplier's dynamic incentives, and increasing the return to alleviating moral hazard. We view social commerce as making this type of remote contracting feasible (so search costs are finite), facilitating social learning (alleviating adverse selection), and facilitating joint punishment strategies (alleviating moral hazard).

The model highlights the types of variation that the ideal experiment aiming to isolate the three frictions would generate. To isolate search frictions, treatments should either create matches or lower the cost of finding a random new supplier. To isolate adverse selection, treatments should either directly give information about a particular supplier or improve the ability to learn this information over time, but should not affect suppliers' incentives. To isolate moral hazard, treatments should strengthen the incentives of the supplier or firms' perceptions of these incentives, but should not provide any other information. Understanding the relative importance of each friction is not only of theoretical interest but also has direct policy implications, since they require different solutions.

The experiment comprises three treatments that aim to separate the frictions, as well to be of real-world interest in their own rights by mirroring common ways that firms use social commerce. The treatments focus on WhatsApp as it is particularly convenient for generating separating variation, but the mechanisms generalize to other social media platforms. In the search treatment, we add treated firms to the supplier WhatsApp groups of three different suppliers in Türkiye, the second-largest source of ready-to-wear garments in Dakar. We inform firms that the suppliers were recruited by a local team in Türkiye and export to Senegal, but do not provide any further details. Thus, the treatment creates new matches between firms in Senegal and suppliers abroad.

We then cross-randomise the adverse selection and moral hazard treatments among the firms in the search arm. In the adverse selection treatment, we add firms to a fourth WhatsApp group with other buyers matched to the same suppliers. The purpose of this group is for them to meet other firms with whom they can privately share information about supplier quality. We also seed these groups with initial information: each firm receives a positive recommendation for one supplier, based on real mystery orders we commissioned prior to the study. This treatment thus directly provides information about types and facilitates social learning.

In the moral hazard treatment, we inform firms that we will ask them to rate the study suppliers, and that any supplier that consistently receives negative feedback will be removed from the study and thus lose access to a large number of potential clients. We emphasize that the suppliers know this and therefore have strong incentives to exert effort. Thus, the treatment aims to shift firms' perceptions about the suppliers' incentives by invoking a joint punishment strategy with the experimenter acting as the coordination device.

Altogether, we have five equally sized groups: Pure Control, Search Only, Search + Adverse Selection, Search + Moral Hazard, and Search + Adverse Selection + Moral Hazard.

We measure outcomes using three independent data sources. The first is a revealed preference measure of access to foreign goods. We designed a mystery shopping exercise in which trained surveyors, acting as real customers, buy goods from the study firms. This allows us to measure access to foreign goods using real behavior, and to separately measure gains along a horizontal dimension (access to a wide set of differentiated varieties), a vertical dimension (access to high quality varieties), and a price dimension (cheaper access to otherwise available varieties). On the horizontal dimension, each good that we try to buy is defined by 5 criteria, such as color and sleeve style, and our outcome is an indicator for whether the firm has a good matching at least 3 criteria. On the vertical dimension, the outcome is an index that aggregates three measures, including a detailed quality scorecard that we designed together with hired experts. We pre-specified these outcomes and the regression specification that we use throughout the paper in our Pre-Analysis Plan (PAP).

We find that the treatments generate large and significant effects on access to foreign goods. Pooling the four treated groups (all of whom received the search treatment), treated firms are 7.1 percentage points more likely to have a suitable good: a 23.1% increase ($p < 0.01$). Conditional on having a relevant good, the vertical index increases by 0.412 standard deviations ($p < 0.01$). We find no significant change in prices, suggesting that gains stem from access to more and higher-quality varieties rather than cheaper ones. Across the four groups, we do not find significantly larger effects in the trust-treated arms relative to Search Only. Given that the mystery-shopper orders were small, this pattern is consistent with trust frictions becoming binding only for larger, more consequential transactions.

The mystery shopping results suggest that there are gains from trade available from being connected to suppliers abroad. However, to realise these gains in practice, firms must develop these connections into relationships. We measure this using data from our two other sources: a follow-up survey that we conducted after 3 months, and real-time, transaction-level administrative data from the largest mobile money provider in Senegal, tracking a share of transactions with study suppliers for up to 22 months after the study started.

In the survey data, we find that (pooled) treatment increases the likelihood of having a regular supplier in Türkiye by 3.7 percentage points ($p = 0.088$) relative to a control mean of 16.7%. Disaggregating the treatments matters: the effect comes primarily from the groups with the trust treatments, and is largest in the Search + Adverse Selection + Moral Hazard group at 7.5 percentage points ($p < 0.01$), which is consistent with the implication from the model that the trust treatments should be complements. We find no effect on the total number of suppliers, suggesting that firms have substituted away from a local wholesaler and towards importing directly. The mobile money data exhibits a very similar pattern (although the estimates are often not significant): the trust groups, and in particular the Search + Adverse Selection + Moral Hazard group, order considerably more from study suppliers than Search Only. The fact that we find this pattern in two completely independent datasets leads us to conclude that the trust treatments were important in causing these connections to develop into meaningful relationships.

To understand whether alleviating these frictions ultimately flows through to producer surplus, we collect standard summary measures of monthly profit and sales in our follow-up survey. Pooling the treatments together, we find increases of 82.4 USD ($p = 0.025$) in profits and 245.2 USD ($p = 0.051$) in sales. These are 43.8% and 40.2% increases relative to control. The coefficient in the Search + Adverse Selection + Moral Hazard is, again, much larger than the others. The implied increases are very large, and the magnitudes reduce by around half (but remain significant) when we winsorize at the 1% level. We find distributional effects that are consistent with the model and that help with interpreting the OLS magnitudes: limited effects at the bottom half of the profit distribution, economically and statistically significant gains in the order of 15% to 20% in the upper third, and possibly very large, but imprecise, effects at the very top. Overall, we conclude that firms are able to realise meaningful gains from accessing a new supplier abroad by using social media to overcome search and trust frictions.

Finally, we use the experimental treatment effects as moments to estimate the model. We use the Recursive Lagrangrian formulation from Marcet and Marimon (2019) to solve the model numerically and calculate theoretical treatment effects, which we match to the experimental ones using Simulated Method of Moments. The estimates imply that firms behave as if just over half of suppliers in the marketplace are bad types, and that the moral hazard parameter is close to one, as commonly assumed by models of dynamic moral hazard. To our knowledge, these are the first experimental estimates of these structural parameters, which can be readily used in quantitative models. Estimated search costs are large but imprecise. We then conduct counterfactuals for a formal social commerce platform or online matching service that can (i) pre-screen suppliers to remove some bad types and (ii) implement review-based incentives at scale. Pre-screening has a large effect both alone and joint with incentives, whereas incentives alone have a much smaller effect. This reflects the complementarity in the model, for which the experiment provides suggestive evidence: at the estimated parameters, adverse selection is severe enough that firms rarely want to order large quantities initially anyway, so learning itself creates important dynamic incentives and thus the incentive compatibility constraint rarely binds.

Taken together, the evidence points to a dual constraint: reducing search frictions expands access to varieties and quality, but alleviating trust frictions—especially when done jointly—is necessary for these matches to develop into lasting and profitable relationships. We show how firms use social media infrastructure to find, learn about, and build relationships with suppliers.

This paper contributes to several strands of literature. First, we contribute to the literature on information and contracting frictions in buyer–seller relationships, which has highlighted the importance of search and matching frictions (Bergquist et al., 2024; Cai et al., 2024; Rudder and Dillon, 2024; Startz, 2025) and of trust barriers stemming from asymmetric information or contracting frictions (Antràs and Foley, 2015; Macchiavello and

Morjaria, 2015; Monarch and Schmidt-Eisenlohr, 2023; Bai, 2025; Startz, 2025). We advance this literature in two ways. First, trust frictions can stem from adverse selection or moral hazard, and, to our knowledge, this is the first paper to (i) experimentally vary both and (ii) study how they jointly shape the formation and evolution of new firm-to-firm relationships. Analyzing them together matters because the two components of trust interact: the gains from mitigating one type of friction depend on the extent of the other. We also embed this in an international setting, where long distances and cross-border contracts may make the frictions most severe, and where both exogenous variation in matches and the ability to understand the conditions under which relationships form are rare. We show that creating matches (alleviating search frictions) improves initial access but is not sufficient for sustained trade unless trust frictions are simultaneously relaxed. Second, as emphasized in Szeidl (2025)'s review, we provide both descriptive and causal evidence on how a new and widespread technology—ubiquitous social media infrastructure—shapes these frictions. We show that in low-trust, high-friction environments, such infrastructure can alleviate both search and trust frictions and thereby support trade.

Second, we contribute to the growing literature on digital trade. Prior work has documented how the internet (Freund and Weinhold, 2004; Fernandes et al., 2019; Akerman et al., 2022) and formal platforms like Alibaba (Lendle et al., 2016; Chen and Wu, 2021; Carballo et al., 2022; Bai et al., 2023) reduce the impact of distance and facilitate cross-border trade. However, most of this evidence comes from formal platforms in high- or middle-income settings. In contrast, we show that, in a lower-income setting, the vast majority of firms operate outside these platforms, relying instead on informal, flexible, and highly relational tools like WhatsApp and Facebook—a form of "social commerce" rather than traditional "e-commerce." We experimentally show that, in such environments, social media serves as a way to mitigate similar information frictions that formal platforms aim to alleviate.

Finally, we contribute to a literature in trade and development that emphasises the importance of networks when formal institutions to solve information frictions and enforce contracts are inadequate (McMillan and Woodruff, 1999; Rauch, 2001; Karlan et al., 2009; Fisman et al., 2017; Cai and Szeidl, 2018; Boken et al., 2024).[1] In a classic article, Greif (1993) highlights how 11th-century Maghribi traders sustained a multilateral punishment system for overseas agents through informal information flows over social networks. While our

[1] There is also a wide literature outside of economics that analyzes how shared ties substitute for formal institutions among particular groups or geographies. For example, Cohen (1969) studies the Hausa trade diaspora in Yoruba towns in 1960s Nigeria, Weidenbaum and Hughes (1996) studies the "bamboo network" of Chinese entrepreneurs across Southeast Asia in the latter half of the 20th century, and Chin et al. (1996) studies the role of Korean immigrants in Los Angeles in 1968-1977 in facilitating trade in the wig industry.

study does not focus on the role of networks *per se*, one of the main channels through which social media may alleviate trust frictions is exactly the mechanism in Greif (1993). Our study thus highlights how social media facilitates a modern manifestation of this idea.

The rest of the paper is organised as follows. Section 2 presents the setting and descriptive evidence on how firms use social media in supply chains. Section 3 describes the model. Section 4 describes the experimental design. Section 5 describes the data and methods. Section 6 presents the results. Section 7 presents the model estimation. Section 8 concludes.

2 Setting

2.1 Ready-to-Wear Garments in Dakar

Our study focuses on the ready-to-wear garment industry in Dakar, the capital city and economic hub of Senegal. The ready-to-wear garment industry exhibits substantial horizontal and vertical differentiation, making it ideal for our study: horizontal differentiation (a wide range of varieties) is well-suited for studying search frictions, and vertical differentiation (the presence of high and low qualities) is well-suited for studying trust frictions. It is also a large and important industry in its own right: in a consumer survey that we conducted with 400 households in Dakar, ready-to-wear garments represented an average of 48% of clothing expenditure and 6% of total household expenditure.

Within the ready-to-wear garment industry, our study places particular emphasis on goods made in Türkiye. We chose Turkish-made goods for two reasons. First, Türkiye is the second largest source of ready-to-wear garments in Dakar (after China). Second, Turkish-made goods have a reputation for being higher quality than goods made in China, which is ideal for studying trust frictions. In fact, in this setting, requesting a good "Made in Türkiye" is a common way to signal willingness to pay for quality.[2]

2.2 Sample

Firms in Senegal The main subjects of the study are 1,862 small firms in the ready-to-wear garments industry in Dakar. These firms are typical of small, informal, owner-operated businesses in many large cities in lower- and middle-income countries. Our sample includes both firms that operate a physical store in a market (33%) and online-only firms that sell exclusively through platforms like WhatsApp and Facebook (67%). Firms

[2]To quantify this, in our consumer survey we showed households an image of a product and randomised whether we said the good was made in Türkiye or made in China. We then asked for their willingness to pay. We plot the CDF of willingness to pay in Appendix Figure A3, Panel (a). The Türkiye CDF is shifted uniformly rightward relative to the China CDF, with an average premium of 34% ($p < 0.01$).

with a physical store were recruited through a census in selected markets that sell both high and low quality goods while online-only firms were identified through a combination of advertisements on Facebook and snowball sampling.

Importantly, firms operating exclusively online are typically not captured in existing datasets, despite their growing prevalence in African cities. Our study offers an opportunity to document and understand the behavior of these firms. 93% of online firms and 83% of firms with a physical store agreed to participate when contacted. To ensure the relevance of the study, we conducted a short screening survey with some eligibility criteria (see Appendix F.1), which were met by 85% of online firms and 89% of physical-store firms.

At baseline, 91% of eligible firms sell Turkish-made goods, and Turkish-made goods represent 44% of sales on average. Average monthly profit is 227 USD and 33% of firms sell wholesale. 10% have traveled internationally for business in the past 5 years, so most buy their goods from other firms in Dakar or through e-commerce. 22% of firms have at least one regular supplier based in Türkiye. As we show in Section 2.3, these firms ubiquitously use social media to receive information from suppliers.

Firms typically have negative views about unknown suppliers abroad. 60% know more than one other firm that have had bad experiences ordering from a supplier online. To measure beliefs, we asked them to consider a scenario in which they made orders from 10 unknown suppliers abroad, and to opine as to how many such orders would arrive with the anticipated quality. The median firm's view was that this would happen only 50% of the time.

Suppliers in Türkiye The study involves connecting firms in Senegal with suppliers in Türkiye. Our experiment involved 30 suppliers. We conducted a census of two neighborhoods of Istanbul that are textile wholesale and export hubs to many parts of the world, including to several countries in West Africa. Among the 89 suppliers that met our inclusion criteria—including being based in Istanbul, regularly exporting ready-to-wear garments to West Africa, of Senegalese nationality, and using WhatsApp for commerce (see Appendix F.2 for further detail)—we then conducted a mystery shopping exercise to screen out any suppliers that were unwilling in practice to engage with new customers or were otherwise unresponsive. We conducted this mystery shopping in waves (with order randomized) and invited suppliers to participate in the study on a rolling basis, with recruitment closing upon reaching our target of 30 suppliers. The mystery shopping exercise also provides baseline information, which we use in the Adverse Selection treatment (as described in Section 4.1). Since our selection criteria were primarily about being active, rather than about quality, and, in any case, the mystery shopping exercise provides only a limited signal, we expect the supplier pool to be a reasonable representation of the type distribution.[3]

The suppliers in our study are not manufacturers—rather, they are agents of Senegalese nationality, based in Istanbul, that specialize in connecting the goods sold by Turkish manufacturers or wholesalers with African markets. We chose to focus on such agents for three reasons. First, these types of agents with shared ties play an important role in trade, both present and historical, as has been documented by economists and other social scientists (see Rauch (2001) and references therein for several case studies of trade intermediaries and diaspora-mediated networks). Indeed, among the 31% of firms in our survey that have a supplier abroad, for 72% of them at least one of these suppliers abroad is a Senegalese national. Second, this significantly reduces logistical barriers, such as language differences and incompatible payment technologies, which ensures our study can focus on the search and trust frictions of interest, which as we show, remain important barriers even within the same culture and country of origin. Finally, these agents accept payment through Senegal's largest mobile money platform, which provides a unique measurement advantage as we have access to the platform's administrative data and thus can track transactions directly.

2.3 Social Media in Supply Chains

In this section, we present new descriptive statistics on how small firms use social media to interact with and learn about suppliers. While the growth of firms using social media as a way to buy and sell goods—often called social commerce—has attracted considerable media attention, there are few statistics available to formally document this phenomenon. One recent exception, a survey of six African countries by GSMA (2023), finds that, among firms that have at least some online presence for selling to customers, 60% use social media as their *only* channel of doing so. We are not aware of any comparable statistics for B2B trade, and the evidence we present below aims to help fill this gap.

Suppliers, both domestic and international, use social media in a variety of ways that can complement or substitute for face-to-face trade, ranging from public-facing activities that expand reach, such as managing a Facebook business page or advertising products on TikTok, to more private activities, such as using WhatsApp groups to engage with regular customers. Indeed, social media companies are creating features to facilitate this. For example, WhatsApp Business, launched in 2018, allows a supplier to maintain a profile with product catalogs, run targeted advertisements on Facebook and Instagram that click through to this profile, and manage inquiries at scale (see Panel (a) of Figure 2 for an exam-

[3] While our mystery shopping screens out any suppliers who are unresponsive or would not engage with new customers, as described in Appendix F.2, we do not think that this significantly affects the relevant distribution of trustworthiness as these features would also be visible to new customers.

ple of a profile). Firms can also use social media to exchange information about suppliers, and the fact that firms can broadcast negative experiences means that suppliers' reputations are at least somewhat public (see Appendix Figure A1 for two examples of firms broadcasting warnings about particular suppliers). Our experiment focuses on WhatsApp as suppliers and firms actively use WhatsApp groups in two particular ways that appear closely related to alleviating search and trust frictions. This is useful as we can mirror these groups to provide a natural way to experimentally isolate different frictions (as opposed to, for example, experimentally creating Facebook business pages, which do many things at once). We first describe these two types of groups, and then present statistics on both WhatsApp and broader social media use.

Supplier Groups Many suppliers in our setting—both domestic and international—operate WhatsApp groups with their clients to advertise goods, post prices, and highlight new items in stock. We refer to these as "supplier groups" (see Panel (b) of Figure 2 for an example). A typical group has one supplier and 50-100 clients, most of whom are regular or repeat customers. These are not discussion groups, and usually only the supplier has permission to post. Buyers can send a private message to negotiate prices or make inquiries. We view these groups as digital storefronts, where clients can browse suppliers' offerings much as they would on other social media platforms. This dissemination leverages canonical social media features such as social connectivity (group membership, contact lists, forwards), platform engagement (notifications, read receipts), and identity/reputation (supplier profiles visible to clients).

Supplier groups primarily reduce search frictions by allowing firms to observe varieties and prices from suppliers at considerable distance, with minimal cost required to negotiate and follow up. Moreover, these groups are designed to integrate with the large growth in B2C social commerce: firms can directly forward supplier images to their own customers via social media to gauge interest or make targeted sales. To a lesser extent, supplier groups may also mitigate trust frictions: a large, active group could raise the reputational cost of dishonest behavior, and suppliers can use these groups to build a brand over time.

Firm Discussion Groups Firms also participate in WhatsApp groups aimed at sharing information—for example, about suppliers—with other firms, which we refer to as "discussion groups". These groups primarily address trust frictions by allowing firms to exchange experiences and warnings. In our sample, 23% of firms are members of at least one group whose explicit purpose is information sharing between businesses. More generally, firms exchange information about suppliers through their networks: 67% have recommended or recommended against a particular supplier to another firm in the past year.

Statistics on Social Media Usage In Figure 3, we present statistics from the 1,862 firms in our baseline survey. We present sample-wide averages here, but we also show in Appendix Figure A2 that the results are identical among firms with and without a physical store, reflecting the widespread nature of this phenomenon. In Panel (a), we plot the share of firms that reported using different types of social media to obtain information about suppliers, such as learning about new varieties or price information. WhatsApp is ubiquitous: 92% of firms use WhatsApp Status (a feature, used less often in the United States, that broadcasts content to all contacts for 24 hours), and 86% use supplier groups. TikTok, Instagram, and Facebook are also popular, each used by about a third of firms. Panel (b) shows the distribution of the number of unique supplier WhatsApp groups that firms belong to. Firms are in many supplier groups, with almost half of firms in 5 or more, allowing them to continuously observe information about different varieties and prices. Finally, firms are not simply passive observers: the distribution of the number of groups that they have bought goods from in the past year is almost identical.

To understand why such groups are so widely used, in Panel (c) we present the responses to a survey question asking about the main advantages of supplier groups. Firms highlight that it allows them to keep up with new or recent varieties, to see a wider range of varieties than would otherwise be possible, and to compare prices across suppliers. Finally, in Panel (d), we show the location of the suppliers running these groups. The majority of firms (81%) are in a group with a supplier located in Senegal, showing that social commerce is used actively for domestic trade, and a large minority (27%) are in at least one group with a supplier located abroad. 21% are in a group with a supplier in Türkiye, 12% are in a group with a supplier in China, and 6% are in a group with a supplier in Dubai.

Firms are therefore familiar with the concept of using supplier WhatsApp group to transact with suppliers, but, since only 21% are in a group with a supplier in Türkiye, our experiment is still able to generate meaningful variation.

2.4 e-Commerce in Supply Chains

Formal B2B e-commerce platforms, such as Alibaba, also aim to alleviate search and trust frictions. Yet, in this setting, they are seldom used: 85% of firms have never bought from these platforms, and, of those who have, half have done so less than once a year.[4] This is not because they have not heard of them (88% have), and reflects a broader trend of large e-

[4]Regular engagement with e-commerce is rare: only 6% of firms purchase online at least monthly. This rate is essentially flat across the size distribution—neither baseline employment nor profits predict regular use (coefficients near zero). If anything, online firms are 6pp more likely to use e-commerce regularly than physical stores. Indeed, just 1% of the representative sample of physical stores report regularly using it. If anything, these 1% regular e-commerce users are smaller on average, although imprecisely estimated.

commerce companies having limited success at penetrating African markets.[5] In response to a our survey question asking why they do not use B2B platforms, apart from the 39% who gave no specific reason, the two most common answers were that firms find them too complicated to use (40%)—often due to language barriers and bank access requirements—and that firms do not trust them (33%).

The statistics presented in this section confirm the observations that led us to run this study: small firms use social media extensively in ways that appear related to search and trust frictions, including a sizable share that use it as a means of doing international trade.

3 Theory

In this section, we describe a dynamic model of relational contracting with adverse selection and moral hazard, embedded within a sequential search framework. Defining the frictions precisely allows us to more clearly discuss how social commerce may affect the frictions, analyze how the frictions interact with each other, and highlight what types of variation would separate them. In Section 7, we use the experimental treatment effects as moments to estimate the parameters that govern the frictions. While the forces in the model are all canonical, we are not aware of existing work that combines them in this way.[6]

3.1 Setup

Time is discrete and infinite horizon. All players are risk neutral, with common discount factor δ. Every period, a firm with productivity z buys high quality inputs q_t^j from one or more suppliers $j = 1, 2, ...$ and sells them downstream according to a downward-sloping demand curve that generates revenue $zr(q_t^1, q_t^2, ...)$, where r satisfies gross substitutes and is strictly increasing, concave, and with $\lim_{q\to\infty} \partial r/\partial q = 0$ in each argument. Inputs may also be of low quality, and for simplicity we set the productive value of low quality inputs

[5] B2C marketplaces have expanded but remain small in absolute terms. Jumia—the largest pan-African platform—serves roughly 6 million consumers across nine markets in 2024, still modest relative to national retail sales. Local B2B distribution apps are also emerging—e.g., Twiga and Wasoko–MaxAB in Kenya, or OmniRetail and TradeDepot in Nigeria—but these mainly link domestic FMCG suppliers to micro-retailers rather than enabling direct cross-border e-imports. Because social-commerce transactions are largely informal, official value data are scarce, though surveys document substantial use among firms and consumers (e.g., GSMA (2023); International Trade Centre (2023)). In our representative sample of garment retailers, we observe no adoption of these B2B apps. We view documenting the scale of social commerce and local B2B distribution as an important avenue for future research.

[6] The dynamic moral hazard with endogenous quantity part is closest to Ray (2002), and the adverse selection part is closest to Yang (2013). The sequential search problem accounts for the cutoff entering the value function directly via the outside option (due to relational contracting), but is otherwise standard. We are unaware of literature that explicitly studies how the dynamic frictions interact with each other, nor, with the exception of Brugues (2025), structurally estimates such a model. We also intentionally abstract from behavioral frictions—e.g., overly pessimistic priors—that can depress search; see Szeidl (2025) for a review.

to zero. The firm can always buy high quality inputs without frictions from local suppliers at constant price c_l, which can be thought of as visiting the wholesale market in Dakar and directly inspecting quality.

3.2 Three Frictions

Sequential Search The firm can also engage in international trade. To do so, the firm must pay a search cost s to match with a random supplier abroad. Upon matching, both parties observe a draw of a fixed match-specific productivity shifter, $\psi \sim G$, that transforms inputs q_t into effective inputs ψq_t. The rationale for engaging in international trade may be horizontal (if ψ is interpreted as representing newer varieties or if the revenue function r features love-of-variety), vertical (if ψ is interpreted as a quality shifter), or to obtain lower prices (if ψ is interpreted as a price shifter).[7] We provide a formal proof in Appendix B, but standard arguments imply that, if the firm ever searches, they will do so until they obtain a draw that exceeds an endogenous cutoff, and we write the expected utility of doing this as $\bar{U}(z)$. As the search cost is fixed, there exists a cutoff $\bar{z}$ such that only firms with $z > \bar{z}$ ever search, an importing equivalent of the result in Melitz (2003).

Moral Hazard If the firm matches with a supplier abroad, they play a repeated game. The stage game has three sub-periods. First, the firm orders a quantity q_t at price p_t and pays $p_t q_t$ upfront. Second, the supplier chooses an unobservable action $a_t \in \{0, 1\}$. If they choose $a_t = 1$, they pay cost cq_t and the goods are high quality with probability 1, where $c > 0$ is a constant marginal cost. If they choose $a_t = 0$, they instead pay a lower cost $(1 - \xi)cq_t$ with $\xi \in (0, 1)$ but the goods are high quality only with probability λ. Avoiding the action and cutting a share ξ of the cost may, for example, be interpreted as purchasing the goods from a cheaper but riskier manufacturer that only delivers with probability λ. Third, the good arrives and the firm observes the quality. We focus on grim trigger strategies, in which both parties take their outside option if a low quality realization occurs, because these strategies provide maximal incentives.

While the action a_t is unobservable, the firm can incentivize the supplier to choose $a_t = 1$ by promising future rewards tied to repeated high quality realisations. In particular, any equilibrium in grim trigger strategies that sustains $a_t = 1$ must satisfy the following

[7] We do not model multi-variety firms as this would introduce considerable complexity that our data are not well suited to test. Our mystery shopping exercise helps understand the mechanisms for potential gains from trade, but we would need strong assumptions to extrapolate from this to a welfare-relevant horizontal demand parameter without detailed data on consumer demand for varieties. We thus interpret ψ broadly as a shifter that can encompass horizontal, vertical, or price-based gains as the rationale for forming relationships.

Dynamic Incentive Compatibility Constraint (DICC),

$$(1-\lambda)\delta V_{t+1} \geq \xi c q_t \tag{DICC}$$

where $V_{t+1} \equiv \sum_{n=1}^{\infty} \delta^n \left(p_{t+n} - c\right) q_{t+n}$ is the supplier's (endogenous) discounted sum of future profits from the relationship. Intuitively, the supplier has already received the payment $p_t q_t$, so they will only choose $a_t = 1$ if the increased probability $(1-\lambda)$ of obtaining the future value of the relationship, δV_{t+1}, exceeds the additional cost $\xi c q_t$. We can see that the constraint holds trivially for $\xi = 0$, and that larger ξ implies a more binding moral hazard problem. This constraint highlights two important features of any equilibrium. First, the supplier needs to be paid above marginal cost at some point to provide incentives. Second, the value of the relationship provides a (time-varying) ceiling on contractible quantity, as going beyond this would make the supplier too tempted to cheat.

Adverse Selection There are two types of suppliers abroad: a share $1-\mu$ are good and the remaining μ are bad. A supplier's type is fixed over time, known to the supplier, and unobservable to firms. Good type suppliers are as described above. Bad types, in contrast, are unable to choose $a_t = 1$ (for example, because they are unskilled). We assume that λ is sufficiently low that, if the firm knew that the supplier was a bad type, they would prefer to exit the relationship and take their outside option. We denote the firm's beliefs that a given supplier is a bad type after observing t high quality realisations as μ_t, with $\mu_0 = \mu$. The firm updates this belief each period using Bayes' Rule. To prevent unrealistic equilibria in which the firm can learn the supplier's type without taking any risk by repeatedly ordering $\epsilon > 0$, we also impose that, for order sizes below some small $\underline{q}$, learning is an increasing function of quantity, $\mu_{t+1}(q_t)$, with $\mu_{t+1}(0) = \mu_t$ and $\mu_{t+1}(\underline{q}) = \mu_{t+1}^{\text{Bayes}}$, where μ_{t+1}^{Bayes} is the posterior implied by Bayes' Rule.

3.3 Relationship with Social Commerce

We interpret the growth of social commerce as affecting these three information frictions as follows. For moral hazard, it facilitates a shift towards a joint punishment equilibrium. For simplicity, suppose there are N symmetric firms for whom the technology allows them to coordinate behavior through social networks. Then, the supplier risks losing not just V_{t+1} but NV_{t+1} if they cheat. This strengthens suppliers' incentives, and, from inspecting the DICC, is equivalent to a reduction in the contracting friction parameter from ξ to $\xi' = \xi/N$.

For adverse selection, we interpret social commerce as facilitating social learning. For simplicity, suppose that N firms that match with the same supplier agree to share quality realisations, which they may choose to do as they are jointly better off coordinating if competition is not too strong (e.g., if they operate in geographically dispersed markets), or because they share social ties that yield non-monetary payoffs.[8] Then, the firm learns faster: for example, if the supplier is a good type, the firm's posterior after one period updates to μ_N rather than μ_1.

Finally, for search, social commerce both lowers the search cost s—for example, by allowing suppliers to create business profiles that appear on searches on Facebook or Instagram—and, at more fundamental level, helps make this type of remote contracting feasible by significantly lowering the cost for firms to interact and exchange media bilaterally with suppliers abroad.

3.4 Further Contracting Details

Outside Options: The firm has outside option $\bar{U}$, given by the maximum of the expected value of searching again or of buying locally forever. Suppliers have outside option 0, reflecting the idea that relationships are separable for the supplier.

Contracts and Enforcement Constraints: In period 0 of the relationship, the firm makes a take-it-or-leave-it offer of a long-term contract to the supplier, which is a sequence $\{q_t, p_t\}_0^\infty$. The contract is relational, meaning that it is not enforceable in court: both parties can agree on a long-term plan, but are unable to commit to it. This implies that both parties face a Dynamic Enforcement Constraint (DEC) that requires that in every period t it must be optimal for them to not walk away: $U_t \geq \bar{U}$ and $V_t \geq 0$ for all t. The enforcement constraint prevents the firm from (for example) promising very large bonus payments to the supplier in the distant future, because when the distant future arrives the firm will prefer to renege.

Limited Liability: The contract must satisfy $p_t q_t \geq c q_t$ for all t; that is, the supplier must make weakly positive profits period-by-period. We impose this restriction to prevent large period-0 rent extractions that we rarely see in our empirical setting. Instead, in our discussions with suppliers, we often heard of selling at cost early in the relationship to establish a reputation, which this restriction permits (and which will occur in equilibrium).

Equilibrium: We focus on Perfect Public Equilibria in which the good type supplier chooses $a_t = 1$. We also restrict attention to pooling equilibria, which we show in Proposition 1 in Appendix B is without loss. Intuitively, the firm cannot commit to not immediately termi-

[8] A more formal and general treatment of firms' incentives to share information in a dynamic setting can be found in Golosov et al. (2014). If deviations from truth-telling are at least occasionally discoverable, then the group can exclude any firm that deviates from accessing future information revealed in the group. Hence, for sufficiently patient firms, truthful, state-by-state disclosure is optimal, so full revelation can be sustained in equilibrium with repeated cheap talk, including under competition or when disclosure carries stigma (e.g., revealing having been cheated).

nating the contract if it learns that the supplier is a bad type. Bad types thus earn 0 upon revealing their type, but can earn a positive payoff by mimicking the good type.

3.5 Results

The optimal contract is the solution to the dynamic program formed by the firm maximising its discounted lifetime profits subject to the DICC, the DECs, and limited liability. We state this program explicitly and characterize the solution in Appendix B.[9] Here, we highlight three aspects that are relevant to the interpretation of the experiment.

Result 1 (Intensive Margin)**.** *Trust frictions (adverse selection and moral hazard) distort quantity and value purchased from the supplier abroad downwards.*

This result can be seen most intuitively in the following wedge equation relating marginal revenue and marginal cost, derived from the FOCs of the dynamic program:

$$r'(q_t^*) = \underbrace{\frac{1}{1-\mu_t(1-\lambda)}}_{\text{Adverse Selection}} \underbrace{(1+\xi\rho_t^*)}_{\text{Moral Hazard}} c, \tag{1}$$

where ρ_t^* is a term proportional to the Lagrange multiplier on the DICC. Both wedge terms are weakly greater than 1, and so distort q_t^* downwards. The adverse selection term captures the firm's incentive to lower quantity because with probability $\mu_t(1-\lambda)$ the supplier is a bad type and delivers low quality. The moral hazard term captures the firm's incentive to lower quantity (and increase the price) to ensure the supplier is not tempted to cheat by avoiding a share ξ of the cost, which matters in proportion to how binding the DICC is (captured by ρ_t^*). Treatments that alleviate these frictions should therefore increase quantity on the intensive margin.

Result 2 (Extensive Margin)**.** *The buyer's period-0 relationship value is decreasing in μ and ξ. Moreover, the productivity cutoff for importing, $\bar{z}$, is increasing in μ and ξ.*

This result highlights that adverse selection and moral hazard limit the gains from trade, which prevents some otherwise productive matches from developing relationships and

[9] In brief, the optimal contract is backloaded, with both quantities and markups rising over time. In an initial phase, quantity builds up ($q_{t+1} > q_t$) and the supplier invests in the relationship by selling at cost ($p_t = c$). Second, in a mature phase, quantity converges to a long-run q^{LR} and the supplier charges positive markups to reap the rewards of their reputational investment, i.e., $p^{LR} > c$. The long-run price premium makes the supplier the residual claimant: p^{LR} is as large as possible to maximise dynamic incentives while ensuring that the buyer still prefers to participate. Intuitively, the contract is the dynamic equivalent of selling the firm to the agent, where the buildup phase is a second-best way for the firm to extract surplus when charging the full rent upfront is prohibited by limited liability. This type of contract is a very general prediction of models of dynamic moral hazard without commitment (Ray (2002), Martimort et al. (2017)).

causes some firms to opt out of searching for an international supplier altogether. Treatments that alleviate these frictions should increase the share of firms that import directly.

Result 3 (Relaxing Trust Frictions is Complementary)**.** *The buyer's period-0 relationship value satisfies* $\partial^2 U_0(\mu, \xi)/\partial\mu\partial\xi > 0$.

This result shows that interventions that lower the trust frictions are complements. The intuition is as follows. When adverse selection is severe, the quantity profile is steep: as Equation (1) shows, quantity grows over time as beliefs about the supplier improve. This steep profile itself generates strong dynamic incentives for the supplier, which limits the marginal benefit of alleviating moral hazard (lowering ξ). By contrast, if adverse selection is exogenously alleviated (so that μ falls), quantity rises in earlier periods and the quantity schedule thus becomes flatter. This weakens the learning-induced dynamic incentives, makes the DICC more binding (so ρ_t^* rises), and therefore increases the return to alleviating moral hazard. This complementarity is captured by the positive cross-partial above.

3.6 Implications for the Experiment

The model makes precise the three frictions that we will study in the experiment: search, adverse selection, and moral hazard. Search frictions limit the share of firms that import from suppliers abroad and Results 1 and 2 show that adverse selection and moral hazard not only reduce the share of firms that import but also lower the quantity and value ordered among firms that do import. To relax the search friction, we need a treatment that either lowers the cost of matching with suppliers abroad or improves the likelihood of finding a suitable match. To relax adverse selection, we need a treatment that improves beliefs (or improves the ability to learn) about a supplier or suppliers in general. To relax moral hazard, we need a treatment that either improves suppliers' incentives or changes firms' perceptions of suppliers' incentives. Finally, since our experiment will cross-randomize treatments for both adverse selection and moral hazard, Result 3 highlights that these treatments are complements in the model.

In addition to these predictions, the model also implies a variety of dynamic patterns, such as value traded building up over time. While we do find descriptive and anecdotal support for these patterns, which we discuss and relate to the literature in Appendix G, we do not emphasize formally testing them because statistical power would depend on the number of relationships, not the number of firms, in the study. In contrast, we designed the experiment to generate exogenous variation to test outcomes concerning access to foreign goods and relationship *formation*, which are powered at the firm level.

4 Experimental Design

4.1 Treatment Conditions

The goal of the experiment is to generate variation that identifies the three frictions: search, adverse selection, and moral hazard (we refer to the latter two jointly as trust frictions).

Search 80% of firms receive the Search treatment. The model highlights that search costs can be identified by lowering the cost of matching with a supplier in Türkiye, so this treatment directly creates matches. We add treated firms to the supplier WhatsApp groups (described in Section 2.3) of 3 different suppliers, which we interpret as paying the search cost s three times on behalf of the firm. In particular, we divide the 30 study suppliers into 10 buckets of 3, and allocate each treated firm at random among the buckets that match their sector.[10] As is standard, only the supplier can post in the group, so this is a channel for firms to see what the supplier sells. We intentionally disclose nothing about the suppliers beyond noting that they were recruited in Türkiye in a similar manner to how the firm itself was recruited and that they are seeking customers in Senegal, explicitly to avoid any inference of endorsement by the research team.[11] We communicate to the control group that unfortunately we cannot add them to any supplier groups at this time, but that we might do so at the conclusion of the study.[12]

Adverse Selection (AS) 50% of firms in the Search condition are treated with the Adverse Selection treatment. As the model highlights, identifying adverse selection requires either improving the ability to learn or providing information directly. This treatment does both. We add treated firms to a fourth WhatsApp group. This group does not contain any suppliers, but instead contains 20-30 other firms in the study that were matched with the same suppliers. We explain that all members of this fourth group have been matched with the same three suppliers and that the purpose is to put such firms in touch with each other to share information about the suppliers (either directly in the group or privately). A member of the study team moderates discussion and periodically encourages firms to share information. As discussed in Section 2.3, firm discussion groups of this kind are common.

[10] As a robustness check, in Appendix Table D5 we show that controlling for supplier fixed effects makes no difference to the results.

[11] As one way to evaluate whether firms interpreted these matches as coming with implicit endorsements, we conducted a brief phone survey with all firms one week after recruitment and how much, on a scale of 1 (very high trust) to 5 (no trust at all), they would trust one of the matched suppliers if they were to make an order. Only 32% of firms reported a trust level of 1 or 2. Moreover, such endorsement effects would, if anything, lead us to underestimate the effect of the trust treatments.

[12] In this Search arm, as in any WhatsApp group, the member list is visible. While only the supplier can post in-group, firms could in principle contact one another privately. We did not observe any evidence that this occurred. If anything, such peer contact would attenuate our estimated AS effects below.

Since no firms have experience with the supplier at this point, we seed the groups with initial information. Treated firms receive a phone call 2-3 days after recruitment from a recommender.[13] The recommender is part of a team of firms—who are not subjects in the study—that we hired prior to the study to make mystery orders from all of the suppliers. The recommender describes a positive experience ordering from one of the suppliers that the firm was matched with and sends a photo of the item that they ordered (we designed the buckets of 3 suppliers in a way that ensures a positive recommendation is always possible). They explain that they are also in this fourth WhatsApp group, and post a similar message there. To ensure firms were not surprised to receive this call, we asked all firms at the end of the baseline survey if they would be willing to call a few other firms to discuss their experiences working with the study suppliers. The information sharing in the group could in principle lead to more or less trade, depending on whether positive or negative information is shared. But given the fact that the seed always provided a positive and specific signal, we expect this arm to increase trade. Moreover, we deliberately implemented this treatment by having firms share information among each other—rather than the experimenter providing information directly—to capture the spirit of social learning.

The AS arm is designed to shift buyers' information about supplier types while minimizing any impact on suppliers' incentives. To that end, firms are explicitly told that suppliers are not informed that the buyer-only groups exist. Although firms could, in principle, try to convince suppliers that these groups exist to influence their behavior, this may be hard to do credibly, and we find no evidence of such attempts in practice. These design choices aim to limit—though not necessarily fully eliminate—buyer updating about incentives; to the extent it occurs, it would attenuate the differential effects between the AS and AS+MH arms.

Moral Hazard (MH) 50% of firms in the Search treatment are treated with the Moral Hazard treatment, cross-randomised with the Adverse Selection treatment. As the model highlights, identifying moral hazard requires either shifting the suppliers' incentives or shifting firms' perceptions of suppliers' incentives. Recruitment and survey procedures were identical across arms; the only addition for MH was a statement at the end of the baseline survey that emphasizes that the study has given the study suppliers strong incentives to encourage reliable fulfillment, deliberately avoiding any language suggesting pre-screening. The full statement is printed in Appendix F.3. In brief, we explain that we will ask the firms to rate the study suppliers, and we will remove suppliers that receive bad feedback, meaning they will lose access to the large number of potential customers

[13] All firms not in this treatment condition instead receive a "placebo" phone call from a surveyor, asking them for their opinion about supplier WhatsApp groups in general.

we have matched them with.[14] Importantly, we stress that the suppliers know this and therefore have strong incentives. We deliberately used the loss of future business from study firms as the incentive effect—rather than, say, a financial bonus for good behavior—to capture the spirit of a joint punishment strategy, with the experimenter serving as the coordination device.

After delivering this message, the surveyor provides a business card to the firm. The business card has a phone number to call, and prominently highlights that this number should be used to rate the suppliers and/or to signal any problems. Untreated firms receive a similar card, but without any mention of ratings or suppliers—instead saying that the phone number is for questions about the study. Both cards can be seen in Appendix Figure A4.

The experiment does not randomise the incentives provided to suppliers; instead, all suppliers face the same high-powered incentives and we randomise whether this is made salient to firms.[15] This allows us to keep all interventions at the firm-level, which greatly simplifies interpretation and significantly improves power. This is equivalent in the model because untreated firms would not propose or agree to a contract that they believe violates the Dynamic Incentive Compatibility Constraint, which also means that this perception can persist over time as the firm never observes how the supplier would have behaved under a weaker contract. This also implies that we should not expect this treatment to affect supplier defaults (conditional on a relationship forming). As above, we explain that we are not endorsing the suppliers: we have simply provided them with incentives. To the control firms, we explain that our only involvement is the initial connection. Overall, the primary function of this treatment is to shifts firms' views about the suppliers' incentives while providing minimal information about the suppliers' types.[16]

[14]This punishment is credible as the supplier WhatsApp groups used in the Search treatment were created specifically for the study, meaning that we are administrators. We can therefore remove a problematic supplier and replace them with a supplier from the wait list.

[15]We intentionally did not use a Karlan and Zinman (2009)–style two-stage design. That approach requires the experimenter to set and randomise *post-match* contract terms at the relationship level to separate selection from behaviour. In our B2B context, transaction terms are multidimensional (price, delivery, payment type, and quality), privately negotiated, and often renegotiated, so we could not easily re-set terms for a subset of relationships. We thus instead used buyer-side information (relevant in a world of AS) and incentive salience (relevant in a world of MH) as levers. Cross-randomizing AS and MH aims to identify the two primitives without intervening on contract terms and, by operating within widespread WhatsApp groups, mirrors the buyer-driven approaches that actually occur in this market.

[16]In principle, since bad types will eventually draw a low quality realisation, firms could wait and attempt to infer types by observing whether the supplier is still around after several periods. We do not think this happens for two reasons. First, firms typically decide quickly whether to initiate a relationship with the study suppliers (this is also embedded within the model as the firm makes a take-it-or-leave-it offer upon matching). Second, the script does not imply that the enforcement process is particularly fast.

Sub-Treatments We cross-randomised two additional sub-treatments within the pure control group. The first sub-treatment aims to test whether the binding constraint behind the limited use of formal B2B e-commerce platforms is that firms do not know how to use them. We thus provide a short training on Alibaba that covers how to install the app, how to search for products, how to contact suppliers, and how to make purchases and arrange delivery. The training was short and lasted about 5-10 minutes, covering an installation and an app demo that all firms in this arm received. The second sub-treatment is a placebo check for the fact that, in the Adverse Selection treatment condition, we have connected firms to each other. To ensure that results are not driven by connecting firms *per se* (as in Cai and Szeidl (2018)), we thus also create similar groups here, where none of the firms have been connected with any suppliers.[17]

4.2 Randomisation and Balance Check

Overall, there are five equally likely groups: Pure Control, Search Only, Search + Adverse Selection, Search + Moral Hazard, and Search + Adverse Selection + Moral Hazard. We randomly assigned firms to one of these five groups, stratifying on product (men's clothing, women's clothing, shoes & bags), whether the firm has a physical store, and whether the firm had prior direct importing experience. Any misfits, due to integer indivisibility or other issues, were unconditionally randomised across the five cells.

Since this is an RCT, treatment is independent of pre-randomisation covariates by construction, absent errors in the randomisation protocol. To check that the randomisation operated as expected, we report a balance check in Appendix Table A1. The differences in means across groups are all small and insignificant, and a joint test has p-value 0.958.

5 Outcomes, Data, and Empirical Methodology

5.1 Data and Outcomes

Consumer Survey In March 2024, we conducted a 15-minute survey with 400 households in several areas of Dakar. Enumerators selected respondents by starting from a random point and then visiting every fifth household along the route. We use this to calculate two sets of summary statistics. First, we use it to measure the relationship between consumer willingness to pay and our measures of garment quality (described below). Second, we use it to calculate statistics on household clothing expenditure.

[17] Consistent with our prior, we find no meaningful effects of the placebo groups (see Appendix Table A12). The nature of our intervention is quite different from Cai and Szeidl (2018), where firms met in person every month for a year. In contrast, we connected firms only via a WhatsApp group, which allows them to pool information about a specific topic but is less likely to generate broader networking interactions.

Baseline and One-Week Surveys Upon recruiting a firm (see Section 2.2), we conducted a 30 minute baseline survey, between November 2023 and January 2024. The survey contained questions on their supplier relationships, social media usage, e-commerce usage, and 30-day profits and sales. In a short additional follow-up survey one week later, we asked for their views about their access to Turkish-made goods, and, if applicable, their beliefs about the trustworthiness of one of the suppliers we matched them with.[18]

Access to Foreign Goods To evaluate whether social media reduces search and trust frictions in a way that expands firms' access to foreign goods, we designed a novel mystery shopping exercise. This allows us to separate between different theoretical mechanisms for potential gains: a horizontal dimension (access to a wider set of differentiated varieties), a vertical dimension (access to higher quality varieties), and a price dimension (cheaper access to otherwise available varieties). Around two weeks after recruitment, firms are contacted over WhatsApp by a mystery shopper, played by a trained surveyor. Firms are not aware that the customer is part of the survey team, but are expecting to be contacted, as we explain to them at the end of the baseline survey that we will put them in touch with customers. This exercise enables us to capture firms' sourcing capabilities and pricing in a natural, incentive-compatible way, shortly after treatment exposure.

The mystery shopper explains that they would like to purchase a certain high quality product for an event (see Appendix F.4 for the full protocol). Each product is defined by five horizontal criteria that are largely unrelated to quality, such as color, sleeve style, and presence of a graphic (see Appendix Figure A5 for two examples).[19] The primary outcome for this horizontal component, pre-specified in our PAP, is an indicator for whether the firm had a good with at least three of the five criteria.[20] The average order size was USD 21.

If the firm has a good with at least three criteria, the mystery shopper proceeds to buy the good (with some exceptions described in Appendix F.4). Then, once the good arrives in our office, two expert tailors assess its quality according to a 50-point scorecard that we

[18] As an early, self-reported measure of how the search and trust treatments shift perceptions, Appendix Table A2 reports one-week treatment effects on access and trust outcomes. All treatments significantly raise self-reported access to Turkish-made goods. The trust treatments jointly increase self-reported supplier trustworthiness, although it is individually significant only for the Search+AS+MH group. Because these measures are taken after just one week, changes in trust likely reflect the researcher-controlled components of the intervention rather than organic relationship formation yet.

[19] We created a bank of 28 products in total and chose products that were fairly general (e.g., a red polo). For each product, we checked whether each supplier had advertised the specific image in their group, and, if so, excluded it from firms that were or would have been matched with that supplier.

[20] Our definition of having a good was that the firm said that they could deliver it to our office within two weeks. First, this matches the natural cadence of retail trade for the firms without physical stores (and is also common for firms with a physical store). Second, our goal is to measure whether the treatments increase *access* to foreign goods. Third, as discussed later, a secondary goal of the mystery shopping was to provide firms with an opportunity to experiment with the new suppliers.

developed (see Appendix F.5 for the scorecard). To validate the quality measure, we gave the surveyors conducting the consumer survey a subset of these goods to present and elicit willingness to pay (WTP). We show a binscatter of the relationship between the quality score and WTP in Panel (a) of Figure 4. There is a clear positive relationship, although it becomes flat in the left tail, reflecting the fact that beyond a certain point consumers simply view goods as "low quality". In Panel (b), we classify goods as "high quality" or "low quality" (defined as whether a good is above or below the median quality score of its product type), and plot the CDF of consumer WTP separately by quality. The high quality CDF is shifted to the right of the low quality CDF, with an average premium of 35%. The outcomes are exactly these two measures: the high quality indicator and the raw 50-point quality score. The advantage of the binary outcome is that it is not vulnerable to a long left tail of quality scores that, as we saw in Figure 4, are not meaningful in terms of WTP.

We also attempt to infer whether the good was manufactured in Türkiye. As we showed in Figure A3, there is a large premium for Turkish-made goods since it is a strong signal of quality. Thus, while the other two vertical outcomes measure quality directly, in practice quality can be difficult for consumers to observe and so product origin plays an important role in consumer WTP.[21] For most goods, we record this information from the label, and the outcome is 1 if the label says "Made in Türkiye" and 0 if it says it was made elsewhere.[22]

Finally, the mystery shopping also had a secondary goal of providing treated firms with an opportunity to experiment with the study suppliers. Nothing in the procedure makes this explicit, the goods are not specific to the study suppliers, and the mystery shopper does not know the treatment status of the firm, but if the firm was considering making an order, then the mystery shopper reduces the risk that they will be unable to find a buyer. We mystery shop at *all* firms, so this secondary goal does not confound our analysis.

Followup Survey We conducted a 30-minute followup survey with similar questions to the baseline survey between February and April 2024, around 3 months after a firm is recruited to the study. We successfully surveyed 90% of the sample (1671 firms). The followup rate is very similar and not significantly different across the four treated groups, but is 5 percentage points higher and statistically significantly different in the pure control

[21] We pre-specified this outcome, but did not attach it to either the horizontal or vertical dimensions. Since the consumer preference for Turkish-made goods reflects a preference for quality, it seems more fitting to include it under the vertical dimension.

[22] For the few goods where this was not possible, the expert tailors independently evaluated origin based on design and construction cues. If they both determine the good was made in Türkiye, we impute 1; otherwise, we impute 0. In Table A4, we show that the results are not sensitive to how we treat these goods.

group.[23] The main outcomes are questions about the number and location of the firms' suppliers, their profits and sales, and their e-commerce use.

Mobile Money To complement the survey data, we also use real-time, transaction-level data from the largest mobile money provider in Senegal, Wave Mobile Money, made available for this study. This data contains the value, timestamp, sender, and receiver for the universe of transfers between the phone numbers of study firms and the phone numbers of study suppliers. This data complements survey-based measures and has several advantages: (1) we can see transaction profiles over time, (2) it continues after the followup survey, (3) it captures real transactions (i.e., it is not self-reported).

While this data cannot capture the full universe of transactions, we expect that a non-trivial share takes place through this provider, for two reasons. First, as described in Section 4.1, we hired a set of non-study firms to make mystery orders from the suppliers prior to the intervention, and every supplier accepted payment through this provider. Second, in the baseline survey, 86% of firms reported regularly using this provider to pay suppliers for remote transactions, with 35% exclusively so.[24] However, larger wholesale orders are more often conducted through formal channels such as bank transfers or international remittance services (e.g., Western Union). This distinction is reflected in the baseline survey data: wholesalers are significantly more likely than retailers to use such methods—12% versus 4% of them, respectively.

5.2 Empirical Methodology

Our primary empirical method, pre-specified in our PAP, is the following OLS specification

$$y_i = \alpha + \sum_{j=1}^{4} \beta_j T_{ji} + \delta y_{0i} + \gamma_s + \rho' X_i + \varepsilon_i, \tag{2}$$

where y_i is the outcome for firm i and T_{ji} for $j = \{1, 2, 3, 4\}$ are indicators for treatment arms Search Only, Search + Adverse Selection, Search + Moral Hazard, and Search + Adverse Selection + Moral Hazard. y_{0i} is the outcome measured at baseline, if available. γ_s are stratum fixed effects. X_i are firm-level covariates selected by Double Lasso (Belloni

[23] We present Lee (2009) bounds for the main survey outcomes in Appendix Table A11. The results are similar, in line with the differential attrition being relatively small and a higher followup rate in control than treatment (i.e., the lower bounds trim the left tail of the control distribution, which is typically bounded).

[24] For comparison, 37% reported using the second-most-common provider, with 2% of them exclusively. Firms using this mobile money provider are more likely to be female-owned, sell online, be relatively new, earn lower profits, and be less engaged in e-commerce at baseline.

et al. (2014)).[25] We also report a similar regression that pools the four treated groups.

Inference Our primary method of inference is randomisation inference, as recommended by Athey and Imbens (2017) and Young (2019). In particular, we compute two-sided p-values for the sharp null of zero treatment effect using 2,000 permutations of the t-statistic. We report conventional heteroskedasticity-robust standard errors in parentheses, but we do not use these for inference directly. As each regression estimates four coefficients, we also calculate Romano-Wolf multiple-testing adjusted p-values in square brackets (Romano and Wolf, 2005). Since there are three combinations of trust treatments, to test the null that none of these has any effect, we report the p-value of a joint test that all four coefficients are equal (computed by permuting the F-statistic).

Indexes To account for multiple hypothesis testing across outcomes, for any table that presents more than one outcome corresponding to the same family of outcomes, we also report the results on an index that aggregates the outcomes using the standardised inverse-variance weighted method of Anderson (2008). Since the disaggregated regressions may include different covariates y_{0i} and X_i, before indexing we first residualise each outcome using the covariates that were included in their respective regressions.

Quantile Regression Because prior evidence suggests that some of our outcomes, notably profit and sales, may have thick tails and have important distributional treatment effects (Meager, 2022), we also included in our PAP that we may use quantile regression to examine distributional treatment effects. For these, we follow the same specification as above, except that we omit the stratum fixed effects (γ_s) and the vector of Double-Lasso-selected covariates (X_i) as quantile regressions are much more demanding and the covariate selection procedure in Belloni et al. (2014) is designed for linear models.

6 Results

In Section 6.1, we present results on access to foreign goods, as measured by our mystery shopping activity. In Section 6.2, we present results on supplier relationships. In Section 6.3, we present results on profit and sales. In most tables, we show the pooled regression in Panel A, and then disaggregate across the treated groups in Panel B, with standard errors in parentheses, Romano-Wolf multiple-testing adjusted p-values in square brackets, and the p-value of a joint test that all coefficients are equal at the bottom.

[25]This means that, prior to each regression, we run lasso to predict y_i and each T_i and include the union of selected covariates. In practice, this selects few covariates and thus makes little difference. Nonetheless, in Appendix D, we reproduce the main tables using only treatment indicators. The results are very similar.

6.1 Access to Foreign Goods

As described in Section 5, we designed a mystery shopping exercise to obtain a revealed preference of access to foreign goods, and to separate potential gains across horizontal, vertical and price-based mechanisms. Table 1 reports the outcomes of this exercise.

Horizontal In Column 1 , we report the main horizontal outcome, which is an indicator for whether the firm had a product with at least three horizontal criteria.[26] Pooling the treatments together, treated firms are 7.1 percentage points more likely to find a suitable good ($p = 0.010$). This is a 23.1% increase from the control mean of 30.7%. In Panel B, we see that the effect is broadly similar in all four treatment groups and we do not reject the joint null that they are all equal, which implies that the effect is not different for the groups with the trust treatments. The results thus show that relaxing the Search friction by connecting firms with suppliers abroad via social media led to a sizable increase in the set of varieties that they can provide to real customers.

Vertical The outcome in Column 2 is an indicator for whether the product is "High Quality", defined as whether the product's quality score is greater than its product-group median.[27] Pooling the treatments together, treated firms are 13.1 percentage points more likely to be high quality ($p = 0.058$), a 30.4% increase from the control mean of 43.1%. In Panel B, we see that the coefficient is positive and similar in all four treatment groups.

The outcome in Column 3 is the raw quality score out of 50. Here, the effect is both small and insignificant. As we noted in Section 5, this outcome is vulnerable to a long left tail having an outsize influence. This is indeed what happens: in Appendix Figure A6 we plot the CDF of quality score by treatment status, and we see that, while the treatment CDF is to the right of the control CDF from the 30th percentile onwards (as reflected in the binary outcome), it also contains a handful of very low scoring goods that skew the mean.

[26] Our PAP specified this indicator as the main outcome. For robustness, we use the raw number of criteria as an outcome in Appendix Table A3. The pattern of the results is the same. We also disaggregate this outcome into an extensive margin effect (agreeing to sell a product at all) versus an intensive margin effect (how many criteria the product matched), and find that the effect comes mostly through the intensive margin.

[27] The sample size is lower for the vertical and price outcomes because the mystery shopper only proceeded to the negotiation phase (to measure price) if the firm had a good matching at least 3 horizontal criteria, and then only purchased the good (to measure quality) under some additional conditions described in Appendix F.4. To investigate whether this introduces selection bias, we conduct two analyses in Appendix Table A5. First, we define versions of the vertical outcomes that are unaffected by this issue: one that is 1 if the firm had a suitable and high quality good (and 0 otherwise) and another that is 1 if the firm had a suitable good from Turkey (and 0 otherwise). We find positive and significant effects on both of these outcomes. Second, to ensure differences do not come from comparing different products, we replicate Columns 2-6 of Table 1 but also include fixed effects for each of the 28 products. The qualitative takeaways are the same. Finally, we use PDS Lasso in all regressions, so any observables strongly correlated with either the outcome or the treatments would likely already be controlled for.

The outcome in Column 4 is an indicator for whether the product was made in Türkiye. Pooling the treatments together, treated firms are 16.2 percentage points more likely to supply a good made in Türkiye, a 34.0% increase from the control mean of 47.7% ($p = 0.017$). In Panel B, we see that the effect is positive and similar for all four treatments.

Finally, to account for multiple hypothesis testing across outcomes, we aggregate these three outcomes into a vertical index using the standardised inverse-variance weighting method in Anderson (2008). The pooled coefficient is 0.412 standard deviations ($p = 0.005$), and similar across the four treatment groups. The results thus show that relaxing the search friction by connecting firms with suppliers abroad via social media led to a sizable increase in their access to higher quality goods.

Price The outcome in Column 6 is the price charged to the mystery shopper. The coefficient is positive, small, and statistically insignificant, suggesting no meaningful price effects in either direction. While this could reflect two offsetting effects if the observed increase in quality causes prices to rise by an amount that offsets a decrease through a price-based mechanism, it at least suggests that a price-based mechanism is not so large enough as to overcome this.

Summary: Relaxing Search Frictions Expands Access to Foreign Varieties Putting together these results, we find that all of the treated groups saw large and significant increases in access to foreign goods. Treated firms are able to sell a wider set of varieties and higher quality varieties. The fact that we find no detectable effects on the price suggests that gains from relaxing the frictions come more from improving access to otherwise unavailable varieties than from lowering prices for current varieties.

Across all of these outcomes, the pattern is consistent: the results are driven by relaxing the search friction. We therefore conclude that finding a supplier abroad is costly, and that social media can alleviate this friction. This does not necessarily mean that the trust frictions do not exist: these are small orders that have a customer already in place, so for many firms the risk may simply be quite low. However, for firms to realise the gains from trade implied by these results, they will have to eventually make larger orders and develop these new connections into relationships, which we look at next.

6.2 Building Supplier Relationships

6.2.1 Survey Data

In our follow-up survey, conducted after 3 months, we asked firms how many regular suppliers they had, and where those regular suppliers were based. We defined a regular supplier as any supplier from whom the firm had ordered at least twice in the past three months, and intended to continue the relationship. We analyse these outcomes in Table 2. In Column 1, the outcome is an indicator for whether the firm has a regular supplier in Türkiye. In Column 2, the outcome is the number of regular suppliers in Türkiye. In both cases, pooling all four treated groups shows that treatment caused firms to develop new relationships with suppliers in Türkiye. The pooled coefficients are 3.7 percentage points (an increase of 22.2% relative to control mean of 16.7%) with $p = 0.088$, and 0.083 suppliers (an increase of 37.4% relative to the control mean of 0.222) with $p = 0.023$.

Disaggregating by treatment arm, for both outcomes, the coefficient for Search + Adverse Selection + Moral Hazard is substantially larger: this group sees a 7.5 percentage point increase in the likelihood of having a supplier in Türkiye, and a 0.188 increase in the number of suppliers in Türkiye, both of which are highly significant (including after adjusting for multiple hypothesis testing). The effect is also larger for the Search + Adverse Selection group. Formally, we can reject the null that all four coefficients are equal. This suggests that relaxing trust frictions, and perhaps in particular relaxing them jointly (consistent with the complementarity in the theory), increased the likelihood that these new connections developed into relationships.

In the second part of Table 2, we provide evidence on whether these new relationships complement or substitute for existing relationships. In Column 3, the outcome is the total number of suppliers; in Column 4, the outcome is the number of suppliers in Senegal. The general direction looks closer to a world of substitutes: the coefficients on the total number of suppliers are close to zero, and the coefficients on the number of suppliers in Senegal, while noisy, are of similar magnitude (but opposite sign) to the coefficients on the number of suppliers in Türkiye. Finally, Column 5 shows an indicator for whether the firm said that they have ended a relationship with a regular supplier in the past 3 months. The coefficients are generally positive, which is also suggestive of substitutes.

To assess how the fourth WhatsApp group (for the adverse selection treatment) influenced buyer perceptions and behavior, we included targeted questions in the follow-up survey. A majority (53%) reported that this group provided them with meaningful value. Specifically, 24% stated that the group increased their trust in the suppliers or helped them obtain useful information about them. Another 12% indicated that the group facilitated new connections, and an equal share (12%) reported that they had privately contacted other firms from the group to learn more about the study suppliers. In addition, 24% of firms reported knowing other firms who had placed orders with the study suppliers (compared to 13% in the absence of this treatment)—evidence pointing to the group's role in disseminating information about peer purchasing behavior.

6.2.2 Mobile Money Data

As discussed in Section 5, we use data made available for this research from the largest mobile money provider in Senegal to directly observe transactions between study firms and study suppliers. Before turning to formal regression results, we first present graphical evidence from the raw data. In Figure 5, we plot cumulative order value from study suppliers as a function of time in the study aggregated across the four treatment groups. In the initial month, the total value ordered is very similar across the four treatment groups. This is the period when the mystery shopping is taking place, so limited differences appears in line with the results in Section 6.1.[28] Outside of this period, differences start to open up, with all three trust groups ordering more than Search Only, especially the Search + Adverse Selection + Moral Hazard group, which is in line with the survey results.

We formally test these patterns in Table 3, where the omitted category is Search Only. The outcome in Column 1 is the total value transacted with study suppliers. The coefficients are positive and large—ranging from a 30% increase under AS to a 90% increase under AS+MH—-but are imprecisely estimated and not statistically significant. We then decompose this total value into the number of orders (defined as the number of weeks with a positive transaction value) and the average order value (defined as the value transacted in weeks with positive transaction values). We find no effect on the number of orders, and positive, large, and significant effects on the average order value. The mobile money data therefore suggests that the new matches translated into more meaningful relationships in the trust groups. This looks closer to an effect on the intensive rather than the extensive margin, though it is worth bearing in mind that this dataset does not capture all trade. In interpreting the magnitudes, recall that these are unconditional averages, that relationships are concentrated among a subset of firms, and that larger transactions tend to occur off-platform.

The next three columns repeat the same exercise, but exclude the first four weeks to remove entirely the mystery shopping period. The general pattern is similar.

[28] We do not know if a given transaction corresponds to fulfilling a mystery shopper's order. However, we can do a back-of-the-envelope calculation of aggregate values. There are 1,500 treated firms, the mean value of a mystery shopping order was USD 21, the share of firms that we purchased from was 0.8*0.37, and the mean markup from the baseline survey was 0.5. Thus, if every treated firm that sold a mystery shopping product sourced it from a study supplier (a very strong assumption) and paid with this method (another strong assumption), we would expect a total value of around USD 4,700, as compared to around USD 7,500 in the first month of the mobile money data. The mystery shopping therefore likely accounts for an important share of purchases in the first month, which is not unexpected since an explicit secondary goal of the mystery shopping was to subsidize experimentation. It need not be surprising that some firms ordered from study suppliers for the mystery shopping but not again; the model explicitly assumes that small orders provide limited learning about supplier type.

6.2.3 Summary: Trust Treatments Convert Matches into Supplier Relationships

Putting together the results from Sections 6.2.1 and 6.2.2, we find that trust treatments made it more likely for these new connections to develop into meaningful relationships. In both datasets, the largest effects come from the Search + Adverse Selection + Moral Hazard group. This is consistent with the implication from the model that the trust treatments are complements, although we are not powered to formally test for a complementarity.

6.3 Profit and Sales

6.3.1 Average Treatment Effects

To see whether improved access to foreign goods and new relationships flow through to profits, we report the results on profits and sales in Table 4. For profit, we use the summary survey question from De Mel et al. (2009) for the last 30 days. For sales, we use a similar summary question.

Columns 1-2 show the results on raw profit and sales. We find large and significant effects. For profit, the pooled coefficient is 82.4 USD ($p = 0.025$), or a 43.8% increase from the control mean. For sales, the pooled coefficient is 245.2 USD ($p = 0.051$), or a 40.2% increase from the control mean. When we disaggregate the four treated groups in Panel B, we see that (for both outcomes), while the effect is positive in all four groups, it is much larger and only significant in the Search + Adverse Selection + Moral Hazard group (including after adjusting for multiple hypothesis testing). Formally, we can reject the null that the four coefficients are equal. The same pattern holds when we index the two outcomes. This aligns with Table 2, which shows this group has significantly more suppliers in Türkiye.

To limit the influence of outliers, in Columns 4-5 we report the results when we winsorize the outcomes at the top 1%. The coefficients decrease in magnitude by around half, but the same pattern remains: the coefficient on the Search + Adverse Selection + Moral Hazard group is large and significant, including after adjusting for multiple hypothesis testing. We also report Poisson regressions, which are less sensitive to skewness, in Appendix Table A8. The pattern is the same.

6.3.2 Distributional Treatment Effects

It is now well established in the literature studying small firms in lower-income countries that profit and sales tend to be thick-tailed, and that these tails can have large effects on the coefficients in OLS regressions (Meager, 2022).[29] Thus, as discussed in Section 5.2 and specified in our PAP, we use quantile regression to examine distributional effects.

Quantile Treatment Effects In Figure 6 Panel (a), we show the quantile treatment effects for profit for percentiles 5-95. Across all four groups, the coefficients are small and generally insignificant up to roughly the middle of the distribution. However, starting from around the upper third, the Search + Adverse Selection + Moral Hazard group coefficient becomes large and significant. Expressed relative to the control quantiles, these effects from roughly the 65th to 95th percentiles are close to constant, on the order of 15–20%. The coefficients for Search Only and Search + Adverse Selection are also large at the 95th percentile, but are very noisy. The results for sales in Panel (b) are similar.[30] We relate these distributional findings to the OLS estimates and the model in Section 6.3.3.

Threshold Regression As an alternative way to analyse distributional treatment effects, we also run OLS regressions on indicators that are 1 if the outcome is greater than t, for a range of t. This is unaffected by high variance in the tails as all that matters is whether the outcome is above the threshold t. The results, reported in Appendix Figure A7, are similar to the quantile regressions: large and positive effects at the upper end of the distribution for the Search + Adverse Selection + Moral Hazard group, and some suggestive evidence of positive effects for the other groups at the very top.

6.3.3 Interpreting Profit Effects

The pattern of the profit results is similar to the pattern in Section 6.2. However, a few points warrant further discussion. First, several features of the data indicate that the conclusions are not driven by a handful of outliers that happen to be in the treatment group: (1) the quantile regressions show gains starting at around the 65th or 75th percentile, beyond what would be expected from a few extreme cases, (2) the threshold regressions are by construction robust to a small number of observations having outsize influence, (3) the p-values in Table 4 highlight that the patterns we observe are very unlikely under the sharp null, and (4) a placebo using baseline profit (Appendix Figure A9) shows no such pattern.

[29] To verify that treatment effects are not driven by measurement error in the tails, we called back all firms whose profit was more than 5 times that at baseline and exceeded a threshold at endline, and we asked them to confirm their previous response. Out of 13 such firms, 12 confirmed that their previous response was correct.

[30] While quantile treatment effects are only interpretable as the effect for firms at a given quantile under rank preservation, we think that it is likely that these are treatment effects for firms that are larger at baseline. First, in Appendix Figure A8 we show treatment effects on the transition matrix between baseline and endline profit: the largest coefficients are in the northeast region, suggesting that treatment makes large firms more likely to remain large. Firms in the 95th percentile of baseline profit have, on average, one employee (compared with none among other firms); 30% report traveling abroad in the past five years; they are also more likely to be male-owned, to operate physical stores, and to be wholesalers (see Section 6.5).

Second, the pattern of gains across the distribution is broadly consistent with the model and suggests a natural way to interpret the magnitudes in light of the large OLS estimates. In the model, there is a productivity cutoff $\bar{z}$ above which firms engage in search for a supplier abroad. For firms with $z > \bar{z}$, the combined treatment generates (i) a level effect from a new match created by the search treatment, (ii) an effect that scales with z from the trust treatments, so that treatment effects rise with size and converge to a roughly constant percentage gain.[31] For firms with $z < \bar{z}$, revealed preference implies that the value of a new match is less than the search cost s, and our assumption that learning is limited at small order sizes further dampens the gains, so effects can be small (or even zero) and difficult to detect.

The empirical distributional effects line up with this logic. Assuming rank preservation for simplicity, firms in roughly the bottom half of the profit distribution show little to no detectable gains. Firms in the 65th-95th percentile show increases in profit on the order of 15% to 20% of the corresponding control quantile. These are economically meaningful but smaller than the OLS point estimate (though within the 95% confidence interval). Finally, firms in the extreme right tail (around the 95th-99th percentiles) exhibit very large but noisy gains that have a large influence on the OLS point estimate; as we show in Section 7, the model cannot fully quantitatively match the size of these tail effects. Taken together, the most prudent interpretation is that the intervention generates (i) little detectable profit impacts in the bottom half of the distribution, (ii) economically important profit increases in the upper third of the distribution, (iii) possibly very large, but rare and imprecisely estimated, effects at the very top.

Third, the profit effects appear before the mobile money data in Figure 5 diverge. We view this timing pattern as plausible in light of how firms transact. Larger orders with new suppliers in Türkiye that would cause changes in profits tend to take place off-platform, especially among wholesalers (who are more likely to use other payment methods like bank transfers, and, as we show in Section 6.5 drive profit effects). Ultimately, Table 2 shows that the Search + AS + MH group had significantly more regular suppliers in Türkiye after 3 months, and the profit results suggest that these new relationships were profitable. The subsequent divergence in the mobile money data, which is completely independent of our

[31] The search effect is constant in levels because the sequential search equilibrium implies that the expected gain from a new match must equal s. This is true regardless of the quality of the current match, because, as discussed in Section 7, we interpret the baseline equilibrium as the long-run steady state of the model, in which the firm has already extracted its surplus and thus earns just enough to make sure it does not want to leave. This means that all firms with a given z earn their outside option, which is the expected value of searching again, which is equal to $U_0(\bar{\psi}; z)$, where $\bar{\psi}$ is the cutoff draw and is characterized by the property that the expected gain from one more draw equals the expected cost, s. The trust effect scales with z as the dynamic program is homothetic in z: aside from the limited learning at small order sizes, there are no scale effects and thus the gains to improving trust are constant in percentage terms (but increasing in level terms).

survey-based profit measure, therefore provides complementary evidence that the these treated built and then exploited new trading relationships.[32]

6.3.4 Summary: Profit Gains from Reducing Trust Frictions

In Table 4, we saw large average effects on profit and sales. As in Section 6.2, the effects are concentrated in the trust groups, and in particular in the group with both trust treatments. This suggests that trust frictions matter for firm profits, and that alleviating them using social media can unlock meaningful gains.

6.4 Firms Prefer Social Media to Formal E-commerce Platforms

Our final set of outcomes examines a light-touch Alibaba training sub-treatment intended to probe one specific mechanism: whether very limited use of formal B2B platforms reflects platform complexity/know-how. As described above, the brief training was offered only once and only to firms that did not receive any of the main treatments, so our main analysis excludes firms in the main treatments (though results are similar if we include them).

We report the results in Table 5. The training has a first-stage: treated firms are 6.5 percentage points more likely to have heard of Alibaba, 11.3 percentage points more likely to have searched for goods on the platform, and 8.7 percentage points more likely to have compared prices on Alibaba with prices from their regular supplier. However, we detect no increase in actual purchases: the coefficient is 1.4 percentage points, with standard errors tight enough to rule out modest to large effects. Taken together, these estimates suggest that a lack of basic familiarity or simple usability is unlikely to be the primary binding constraint. At the same time, this design was not intended to—and cannot—disentangle alternative mechanisms such as language barriers, broader technological constraints (payments, logistics, dispute resolution), or social/relational factors (direct messaging, negotiation, supplier access). This points toward deeper frictions that social media may relax but current B2B platforms like Alibaba do not.

6.5 Heterogeneous Treatment Effects: Higher Profit Gains for Wholesalers

We pre-specified a set of heterogeneity analyses to explore how treatment effects vary with firm characteristics, and we report detailed tables in Appendix C. Several patterns emerge.

First, we compare retailers and wholesalers. We define wholesalers as the 33% of the sample for whom some (or all) of their sales are wholesale to other firms. While we find no con-

[32] The firms that drive the early profit gains need not be the same firms that later transact more heavily via mobile money, and the frequency of larger restocking orders may differ from that of mobile money payments. We therefore do not expect a one-to-one mapping between the timing of profit and mobile money effects.

sistent heterogeneity across mystery shopping or relationship outcomes, the profit gains appear concentrated among wholesalers. This aligns with the distributional treatment effects in Section 6.3. One possible explanation is that wholesalers, who typically buy and sell in bulk, stand to gain more immediately from access to new suppliers. Alternatively, the effects may reflect reduced travel costs for sourcing, given that wholesalers are more likely to travel abroad for procurement. While we cannot definitively distinguish between these mechanisms, Appendix Table A9 provides suggestive evidence that wholesalers in the treatment group are less likely to have traveled internationally for business in the three months between the intervention and the follow-up survey. The estimated effects are large and mostly negative, with a few significant coefficients, though wide confidence intervals suggest they should be interpreted with caution.

Second, we examine whether treatment effects differ between online-only firms and those with a physical store. The horizontal mystery shopping effects (e.g., custom order responsiveness) are more pronounced among online-only firms, likely reflecting their greater flexibility in adapting to buyer requests. However, we find no systematic differences in vertical mystery shopping or relationship outcomes. Interestingly, the profit effects are instead concentrated among firms with physical stores, many (45%) of which are wholesalers. This suggests that while online-only firms may be more responsive to initial contact, firms with physical stores may be better positioned to convert new relationships into sustained profit gains. Since online-only firms are disproportionately women-operated (63% against 25% for physical stores), we observe a similar pattern when disaggregating by the gender of the firm owner.

Finally, we find no meaningful heterogeneity in treatment effects based on prior importing experience, initial exposure to Turkish-made goods, or existing membership in a WhatsApp group with a supplier in Türkiye. This suggests that the intervention may be useful for firms regardless of their baseline access to international sourcing networks.

7 Model Estimation

In this section, we use the results of the experiment to estimate the model from Section 3. We have two main goals. First, we want to estimate the parameters governing the frictions, as these are important inputs to quantitative trade models but few estimates are available in the literature. Second, since our treatments are only one particular way of alleviating the frictions, we want to evaluate the gains from trade available if the frictions were to be counterfactually alleviated in different combinations, to different magnitudes, or through different interventions.

7.1 Estimation Details

Modifications to the Model In order to make the model estimable, we need to make a few modifications. First, we let the revenue function r be such that the firm faces and internalises a constant elasticity residual demand curve for goods, $zQ^{-1/\sigma} = P$, with $\sigma > 1$.[33] The aggregate good Q is a linear combination of inputs from either a new supplier abroad that they are matched with, q_f, or their existing suppliers (who may also be foreign), q_e. For simplicity, we let firms' existing suppliers be summarized by constant price p_e. Timing is such that the firm chooses (q_f, q_e), observes whether the order q_f is high quality and therefore whether $Q = q_f + q_e$ or $Q = q_e$, and then sells to their downstream buyers. The firm's stage game payoff is therefore the following,

$$(1 - \mu_t(1-\lambda))z(\psi q_{ft} + q_{et})^{\frac{\sigma-1}{\sigma}} + \mu_t(1-\lambda)zq_{et}^{\frac{\sigma-1}{\sigma}} - p_{ft}q_{ft} - p_e q_{et}, \tag{3}$$

where $z > 0$ is firm-specific productivity and ψ is match-specific productivity. We can see from inspection that the firm will always choose $q_{et} > 0$, but may choose $q_{ft} = 0$ if either beliefs μ_t, or the prices $\{p_{ft}\}_t^\infty$ required to incentivise the supplier, are large.[34]

Second, we need distributions for the heterogeneity. For the match-specific productivity, ψ, we normalize the supplier's base cost, c, to 1 and then let ψ be distributed lognormal with parameters (ψ_μ, ψ_σ), which we will estimate. We draw the firm-specific productivity, z, from the empirical distribution that rationalises the control group's profits (described in more detail below).

Third, we need to take a stand on the baseline equilibrium. The sequential search process implies that there exists a cutoff value $\bar{z}$ such that firms with $z > \bar{z}$ will search until they find a supplier abroad with sufficiently high match-specific productivity, where $\bar{z}$ is a function of parameters to be estimated. We interpret the baseline equilibrium as the long-run steady state of the model, meaning that all firms with $z > \bar{z}$ have found a satisfactory supplier abroad and are thus in the mature phase of the relationship, which means they earn $(1-\delta)\bar{U}(z)$ every period. For such firms, we then set p_e to be the constant price that delivers exactly $(1-\delta)\bar{U}(z)$. For all other firms, p_e represents the cost of buying from a local

[33] Since our experiment was not large enough to generate general equilibrium effects, we do not explicitly model competition between firms in the estimation or counterfactuals. Our counterfactuals remain valid so long as the shock does not affect very large numbers of firms. We do not expect that introducing general equilibrium effects would meaningfully affect the relative gains to alleviating the frictions (e.g., adverse selection vs moral hazard), but it would likely lower the absolute returns.

[34] As discussed in Section 3, to prevent a discontinuity in learning when going from $q_{ft} = 0$ to any positive value, we assume that for q_{ft} below a small $\underline{q}$, posteriors are given by $\mu_{t+1} = \mu_{t+1}^{\text{Bayes}} + (\mu_t - \mu_{t+1}^{\text{Bayes}}) \cdot (1 - q_{ft}/\underline{q})^p$, with $p \in (1, 2)$. Making learning a continuous function of q_{ft} ensures the problem satisfies appropriate regularity conditions, and generates similar behavior to a minimum order size without introducing non-convexities into the principal's choice set.

supplier in Dakar, which we calibrate using the baseline survey. A firm's outside option, $\bar{U}$, is then defined by the value of purchasing at p_e forever,

$$\bar{U}(z) = \left(\max_{q_e} z q_e^{\frac{\sigma-1}{\sigma}} - p_e(z) q_e \right) / (1-\delta).$$

Parameters Our goal is to estimate the parameters governing the three frictions: the search cost, s, the share of bad types, μ, and the moral hazard effort cost, ξ. As the experiment has four treatment effects per outcome, we include two additional, related parameters to better match the combinations of treatments. First, we include the discount factor, δ, which is important for dynamic incentives and thus has been a target parameter in previous studies of dynamic moral hazard (e.g., Einav et al. (2015)). We interpret one period as one month. Second, we include the probability of low effort generating high quality goods, λ. This parameter affects the speed of learning and, indirectly, the strength of the interaction between adverse selection and moral hazard: when learning is slow, moral hazard becomes less consequential. Finally, we also need to estimate the parameters governing the lognormal distribution of match-specific productivity, (ψ_μ, ψ_σ), which are related to the search friction as they govern the expected returns to searching. We thus estimate these seven parameters, and calibrate the remaining two, (σ, p_l). We discuss this calibration in Appendix Table A10.

Moments Since the ultimate goal is to use the model to extrapolate from the treatments to consider counterfactuals where we vary the extent of the frictions, we estimate the model by simulating the impact of treatment in the model and then matching the reduced form treatment effects. We select the four treatment effects on likelihood of having a supplier in Türkiye as these are the cleanest empirical expression of the extensive margin in the model, the three treatment effects on mobile money order value post mystery shopping as these capture the intensive margin (as we do not know the exact share of transactions covered by the mobile money data, we express these as percentages changes), and the four treatment effects on winsorized profit as this is the most welfare-relevant empirical object that we measure (we select the winsorized version to reduce the influence of the large tail).

We simulate the three treatments in the model as follows. We implement search as the firm being matched at zero cost to a supplier abroad with match-specific productivity equal to the maximum of three draws from the distribution (to capture the idea that the firm is matched to three suppliers in the experiment). We implement adverse selection as the firm receiving one high quality signal realisation without having to purchase anything, meaning that they update as a Bayesian and thus begin the relationship with belief that the supplier is a bad-type of $\mu_1 = \mu_0\lambda/(1-\mu_0(1-\lambda)) < \mu_0$. This intends to capture the recommendation call from the adverse selection treatment that explicitly told the firm about one positive order experience.[35] We implement moral hazard as treated firms solving for the optimal dynamic path under a joint-punishment strategy among study firms, as opposed to an individual grim-trigger strategy. That is, treated firms know that the suppliers stands to lose δNV_{t+1} instead of δV_{t+1} if low-quality is realised, where we calibrate $N = 5$ as the empirical ratio of the number of firms that made orders from study suppliers to the number of study suppliers.[36] The DICC for these firms can be rewritten as $(1-\lambda)\delta V_{t+1} \geq \tilde{\xi} c q_t$, where $\tilde{\xi} \equiv \xi/N$, so this treatment is equivalent to lowering ξ by a factor of N.

Solving and Estimating the Model Dynamic models of moral hazard are non-stationary in general and solving them numerically is non-trivial. We use the method of Marcet and Marimon (2019), which involves rewriting the problem as an as-if planner's problem with Pareto weights that evolve endogenously to summarize past utility promises.[37] This permits a recursive formulation of the original Lagrangean with the Pareto weights as additional co-states, which—under some regularity conditions—we can then solve by iterating on the value function.

For estimation, we use Simulated Method of Moments (SMM) with a weighting matrix equal to the inverse of the variance-covariance matrix of the empirical moments, and with 50 randomized initial points for the numerical minimization algorithm. For each guess of the parameters, we do the following two-step procedure. First, we calculate the value of ψ_μ and the empirical distribution of z that rationalise the control group's share of firms with a supplier in Türkiye and profit distribution, respectively.[38] Second, we use these to draw from the (z, ψ) distribution and calculate the moments. We compute standard errors by bootstrapping the estimation procedure 200 times and calculating the standard deviation

[35] We focus on the direct information provision, as opposed to any social learning in the discussion WhatsApp group, as we believe this approach is more transparent. Modeling the WhatsApp group would introduce several additional assumptions (and parameters), such as those governing firms' incentives to share information and firms' beliefs about when other firms will make orders.

[36] This implicitly assumes that firms treat other firms as identical to themselves (i.e., V_{t+1} is the same). This simplifies the computation significantly, as otherwise we would need to repeatedly solve for a fixed point in V_{t+1} (for all t), and can be reasonably interpreted as firms extrapolating from their own experience.

[37] The Pareto weights are equal to the sum of the Lagrange multipliers from the incentives constraints in earlier periods. Intuitively, if the DICC binds in period 0, which implies that the agent must be delivered a certain amount of utils in the future, then the agent's Pareto weight increases in proportion to the multiplier to ensure that the "planner" in the future delivers these utils. This allows the problem to be written recursively because the endogenous Pareto weights (which become state variables) fully summarizes previous constraints.

[38] Specifically, the model implies that profits for firm i are given by $\pi_i = z_i^\sigma \frac{(\sigma-1)^{\sigma-1}}{\sigma^\sigma} p_{ei}^{1-\sigma}$. Since the definition of baseline described above implies a (parameter-dependent) mapping from z to p_e, we can define a bijective function $\pi(z) = z^\sigma \frac{(\sigma-1)^{\sigma-1}}{\sigma^\sigma} p_e(z)^{1-\sigma}$. Then, we can construct a distribution of z by drawing π_i from the control group's empirical distribution and inverting this function to obtain z_i.

of the estimates. We provide a more detailed description of the procedure to solve and estimate the model in Appendix E.

7.2 Parameter Estimates

We present the estimated parameters in Table 6.[39] The value of $\hat{\mu} = 0.57$ suggests an important adverse selection problem as over half of suppliers are bad types. The value of $\hat{\xi} = 0.91$ is close to a standard model of moral hazard (which would have $\xi = 1$, i.e., low effort production is costless), although how much moral hazard actually constrains the firm depends on how severely the DICCs bind.[40] The search cost of $s = 94$ USD is somewhat large, although it has a very high standard error.

While there is a large literature linking institutional quality and trade outcomes, there are relatively few papers that explicitly estimate micro-level structural parameters that can be readily plugged in to quantitative models, and fewer still that exploit experimental or quasi-experimental variation to do so. We therefore view our parameter estimates as contributing to this literature, both in terms of providing estimates from a new setting and of using experimental variation to validate previous findings. The closest estimates to our μ and ξ are provided by Monarch and Schmidt-Eisenlohr (2016), who use the dynamic patterns of US imports to back out the implied values of similar parameters for a range of countries. In their estimates, μ varies from 0.56 to 0.91, placing our estimate toward the lower end of this range, and ξ varies from 0.63 to 0.98, placing our estimate toward the upper end, consistent with their finding that this parameter is negatively correlated with local institutional quality. We can also compare the endogenous implications of the estimated parameters. In a similar setting to ours, though with a different model, the estimates in Startz (2025) imply that Nigerian traders that order from abroad pay a price premium to incentivize suppliers of around 15% on average. At the estimated parameter values in our model, calculating a similar object—discounted total incentive premium, i.e., $\sum_t \delta^t (p_t - c) q_t / \sum_t \delta^t c q_t$—yields values of around 12%.

[39] We plot the targeted moments in Panel A of Appendix Figure A10. The model is able to match the treatment effects reasonably well. It cannot match the scale of the complementarity in the profit moments, likely because it is unable to generate the extreme right tail, but it is close in the survey relationship moments. As described in Section 6.3.3, the model generates distributional results that are broadly comparable to the experimental ones: in Panel B, we show that quantile treatment effects computed in the model, which are not directly targeted, look similar to the experimental ones (modulo standard errors at the higher percentiles).

[40] The moral hazard problem also appears in the monthly discount factor $\hat{\delta}$ being quite low at 0.78—this happens because it is hard for dynamic moral hazard to have meaningful quantitative impacts if there is a large amount of surplus in the relationship and the players are very patient. While this value is low if interpreted purely as impatience, it is not unreasonable as a composite that embeds an exogenous breakup probability, especially in a volatile informal trading environment.

7.3 Counterfactuals

We now use the estimated model to evaluate counterfactuals that represent how different market institutions might reduce trust frictions. First, consider a more formal social commerce platform or online matching service where a trusted central actor pre-screens or endorses suppliers based on public reputations on social media platforms, mobile money transaction history, or a verification of any physical establishments. We first restrict attention to environments in which this institution does not (yet) alter suppliers' incentives to exert effort, so that it operates purely by changing the composition of suppliers (adverse selection) rather than their behavior (moral hazard). Since these low-cost signals are unlikely to be perfectly informative about supplier types, suppose that this pre-screening results in a pool of suppliers available to a user of the platform that is characterized by $\hat{\mu}/2$ instead of $\hat{\mu}$. This is different to our intervention because it affects the overall distribution of suppliers available to a firm, which changes the firm's endogenous search behavior; in our experiment, the improvements are only among a fixed set of suppliers.

We show the results in Figure 7, Panel (a). The effect is very large in percentage terms, an increase in lifetime discounted profits of 180%. In absolute terms, the implied gains (around USD 1,900) are comparable to fixed costs associated with alternative solutions, such as traveling internationally to source supplies. The change in search behavior is an important contributor to this, and can be seen in Figure 7, Panel (b), where we plot firm lifetime profits, $U(z)$, over the distribution of firm productivity, z, at baseline and in the counterfactual. On the extensive margin, the cutoff determining which firms search, $\bar{z}$, shifts left, meaning that more firms engage in trade. On the intensive margin, even firms already importing increase their reservation utility and may optimally search again.

Second, suppose this platform goes beyond pre-screening and implements a review system that also affects suppliers' incentives. While a review system also provides information directly, for the sake of comparison suppose that its additional impact in this counterfactual is to reduce the moral hazard parameter by half, holding fixed the pre-screening described above. The combined effect is much larger, in the order of a 260% increase in lifetime discounted profits (Figure 7). A large part of this reflects complementarity between the two channels, since strengthening incentives to a similar degree *on its own* has a much smaller impact of only 27%.

The mechanism is that, when μ is relatively high (adverse selection is severe), moral hazard is not the binding constraint—firms are already reluctant to order from suppliers abroad because of adverse selection, so a steep learning-induced quantity schedule itself provides dynamic incentives. The incentive constraint thus binds infrequently in practice and improving incentives alone has limited bite. On the platform, however, pre-screening removes some of the bad types, so firms are more optimistic about any given new match, and the DICC becomes the binding constraint more often. Consistent with this, the average Lagrange multiplier on the DICC is around four times larger when adverse selection is lowered. This helps explain why practical solutions that address both adverse selection and moral hazard simultaneously (such as review systems) can be especially powerful.

8 Conclusion

We study the extent to which search and trust frictions limit the ability of small firms to source inputs from foreign markets—and whether social media can help reduce these barriers. Finding and developing relationships with suppliers is a first-order issue, and thus understanding these frictions—as well as how they interact—is important. Moreover, the rapid recent growth of firms using social media to find and evaluate suppliers makes it especially timely to understand the role that these new digital tools play in alleviating information frictions.

Our results have two main implications. First, search and trust frictions are both substantial barriers for trade. Our mystery shopping exercise shows that lowering search costs by connecting firms to new suppliers abroad allows them to provide both a wider set of varieties and higher quality varieties to real customers. However, using both survey and mobile payments data, we find that these connections only develop into meaningful and lasting relationships when we also alleviate the trust frictions. This, in turn, leads to higher profits for some firms. Beyond the experiment, our structural exercise suggests that adverse selection may be the more important trust friction, though both are important and there is scope for large complementarities to alleviating them jointly. While our experiment focuses on international trade, our results likely apply more generally when buyers and sellers are spatially distant or where alternative ways to mitigate the frictions are unavailable.

Second, the fact that we find these effects through interventions that deliberately mimic how firms use social media in the real world suggests that the rapid growth of trade through social media should be at least partially understood as a new way for firms to mitigate search and trust frictions. These technologies are changing the structure of supply chains in ways that may complement or substitute for existing modes of sourcing: retailers can develop new international connections, and wholesalers may be able to reduce costly travel or use the increased information to gain more value from it. Moreover, these changes are here to stay, as smartphone access and mobile connectivity continue to grow, and as social media companies themselves introduce e-commerce features to their products.

Yet, despite substantial demand for online commerce, firms in our setting appear to rarely use formal B2B platforms like Alibaba. Our e-commerce training had no effect on purchasing behavior, suggesting that the constraint is not just a lack of knowledge. The stronger appeal of social media may reflect deeper differences in how it reduces frictions in this setting in ways that formal platforms do not—for example, through greater familiarity, flexibility, or trust. We view understanding these differences as a promising direction for future research.

Tables

Table 1: Access to Foreign Goods

	Horizontal	Vertical				Price
	Has Product $\geq$ 3 Criteria	High Qual Dummy	Qual Score (/50)	Made in Turkey	Index	Price (USD)
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Pooled						
Treatment	0.071***	0.131*	-0.391	0.162**	0.412***	0.710
	(0.027)	(0.067)	(0.562)	(0.067)	(0.135)	(0.699)
Panel B: Individual Treatments						
Search Only	0.107***	0.175**	0.259	0.138	0.426**	1.360
	(0.035)	(0.083)	(0.675)	(0.082)	(0.166)	(0.896)
	[0.010]	[0.126]	[0.906]	[0.183]	[0.048]	[0.369]
Search + Adverse Selection	0.041	0.159*	-0.027	0.130	0.398**	0.445
	(0.034)	(0.087)	(0.858)	(0.086)	(0.169)	(0.833)
	[0.244]	[0.170]	[0.970]	[0.183]	[0.048]	[0.813]
Search + Moral Hazard	0.083**	0.099	-1.179	0.183**	0.399**	1.064
	(0.034)	(0.082)	(0.789)	(0.081)	(0.162)	(0.866)
	[0.045]	[0.379]	[0.385]	[0.068]	[0.048]	[0.465]
Search + AS + MH	0.050	0.083	-0.717	0.211**	0.424**	-0.025
	(0.034)	(0.084)	(0.814)	(0.081)	(0.171)	(0.818)
	[0.244]	[0.379]	[0.719]	[0.037]	[0.048]	[0.973]
Control Mean	0.307	0.431	43.064	0.477	0.000	19.990
% Increase (Pooled)	23.1%	30.4%	-0.9%	34.0%	N/A	3.6%
All Coefs Equal p-val	0.198	0.619	0.321	0.668	0.996	0.317
Adjusted R^2	0.06	0.04	0.34	0.09	0.02	0.40
N	1859	359	359	361	359	642

Note: p-values are computed using randomisation inference. Specifically, we compute the randomised randomisation-t p-value from Young (2019) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. We also report Romano-Wolf multiple-testing adjusted p-values in square brackets (Romano and Wolf, 2005). To test the null that the trust treatments have no effect, we report at the bottom of the table the p-value for an F-test that all coefficients are equal, computed by permuting the F-statistic. Panel A shows the coefficient from a regression on an indicator that pools all treated groups. Panel B shows the coefficients corresponding to treatment indicators for each of the four treatment arms. All regressions include any covariates selected by Double Lasso (Belloni, Chernozhukov, and Hansen, 2014), as well as stratum fixed effects and the outcome measured at baseline (if available).

Column 1 is an indicator that is one if the firm has a good that matches at least 3 horizontal criteria. Column 2 is an indicator for whether the good's quality score is above the median product-group quality score. Column 3 is the raw quality score. Column 4 is an indicator for whether the good is made in Turkey, primarily inferred based on the label. See the text for full details of how this outcome is constructed. Column 5 is the Anderson (2008) index combining the vertical outcomes to account for multiple hypothesis testing across outcomes. Column 6 is the price in USD, which is only measured conditional on the firm finding a good matching at least three horizontal criteria.

Table 2: Supplier Relationships (Followup Survey)

	Regular Suppliers in Türkiye		Previous Suppliers		
	Any Sup in Türkiye	Num Sup in Türkiye	Num Sup Total	Num Sup in Senegal	Ended with Sup
	(1)	(2)	(3)	(4)	(5)
Panel A: Pooled					
Treatment	0.037*	0.083**	-0.102	-0.130	0.062***
	(0.021)	(0.035)	(0.164)	(0.170)	(0.021)
Panel B: Individual Treatments					
Search Only	0.024	0.081	-0.101	-0.111	0.050*
	(0.027)	(0.051)	(0.216)	(0.222)	(0.028)
	[0.576]	[0.263]	[0.885]	[0.924]	[0.103]
Search + Adverse Selection	0.049*	0.063	-0.171	-0.245	0.066**
	(0.029)	(0.047)	(0.218)	(0.226)	(0.028)
	[0.205]	[0.321]	[0.836]	[0.636]	[0.049]
Search + Moral Hazard	0.003	0.003	-0.118	-0.064	0.076**
	(0.026)	(0.041)	(0.201)	(0.212)	(0.029)
	[0.916]	[0.936]	[0.885]	[0.924]	[0.038]
Search + AS + MH	0.075**	0.188***	-0.014	-0.096	0.053*
	(0.028)	(0.057)	(0.222)	(0.225)	(0.028)
	[0.035]	[0.003]	[0.951]	[0.924]	[0.103]
Control Mean	0.167	0.222	3.700	3.213	0.135
% Increase (Pooled)	22.2%	37.4%	-2.8%	-4.0%	45.9%
All Coefs Equal p-val	0.075	0.013	0.919	0.861	0.826
Adjusted R^2	0.14	0.12	0.32	0.26	0.05
N	1680	1680	1681	1681	1671

Note: p-values are computed using randomisation inference. Specifically, we compute the randomised randomisation-t p-value from Young (2019) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. We also report Romano-Wolf multiple-testing adjusted p-values in square brackets (Romano and Wolf, 2005). To test the null that the trust treatments have no effect, we report at the bottom of the table the p-value for an F-test that all coefficients are equal, computed by permuting the F-statistic. Panel A shows the coefficient from a regression on an indicator that pools all treated groups. Panel B shows the coefficients corresponding to treatment indicators for each of the four treatment arms. All regressions include any covariates selected by Double Lasso (Belloni, Chernozhukov, and Hansen, 2014), as well as stratum fixed effects and the outcome measured at baseline (if available).

Column 1 is 1 if the firm says that they have a regular supplier in Türkiye. Column 2 is the number of regular suppliers in Türkiye. Column 3 is the total number of regular suppliers. Column 4 is the number of regular suppliers in Senegal. Column 5 is 1 if the firm has ended a relationship with a regular supplier in the past 3 months. A regular supplier is defined as a supplier from whom the firm has made two or more orders with the intention to continue the relationship.

Table 3: Order Value (Mobile Money Data)

	Entire Period			Post Mystery Shopping		
	Total Order Value (USD)	Num Weeks Ordered	Avg Order Value (USD)	Total Order Value (USD)	Num Weeks Ordered	Avg Order Value (USD)
	(1)	(2)	(3)	(4)	(5)	(6)
Search + Adverse Selection	4.18	-0.03	23.33**	3.12	-0.06	5.62
	(5.28)	(0.11)	(10.68)	(4.44)	(0.10)	(22.31)
	[0.910]	[0.993]	[0.092]	[0.784]	[0.920]	[0.891]
Search + Moral Hazard	4.79	0.01	17.18*	5.60	0.03	11.16
	(6.31)	(0.12)	(8.57)	(5.60)	(0.11)	(10.59)
	[0.910]	[0.993]	[0.099]	[0.718]	[0.958]	[0.790]
Search + AS + MH	12.94	0.00	44.50	16.10	0.03	40.45
	(19.15)	(0.15)	(28.92)	(19.10)	(0.14)	(20.44)
	[0.910]	[0.993]	[0.209]	[0.784]	[0.958]	[0.457]
Search Only Mean	14.31	0.36	39.24	7.92	0.23	33.95
% Increase (AS)	29.2%	-7.8%	59.4%	39.4%	-24.4%	16.6%
% Increase (MH)	33.5%	3.0%	43.8%	70.7%	13.8%	32.9%
% Increase (AS+MH)	90.4%	0.4%	113.4%	203.2%	12.6%	119.1%
All Coefs Zero p-val	0.879	0.995	0.088	0.737	0.878	0.516
Adjusted R^2	0.01	0.04	0.03	0.00	0.03	0.09
N	1500	1500	538	1500	1500	351

Note: p-values are computed using randomisation inference. Specifically, we compute the randomised randomisation-t p-value from Young (2019) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. We also report Romano-Wolf multiple-testing adjusted p-values in square brackets (Romano and Wolf, 2005). To test the null that the trust treatments have no effect, we report at the bottom of the table the p-value for an F-test that all coefficients are equal, computed by permuting the F-statistic. This table shows the coefficients corresponding to treatment indicators for each of the three treatment groups with trust treatments, where Search Only is the omitted category. All regressions include any covariates selected by Double Lasso (Belloni, Chernozhukov, and Hansen, 2014), as well as stratum fixed effects and the outcome measured at baseline (if available).

Column 1 is the total value of transfers to study suppliers. Column 2 is the total number of weeks in which a firm made a transfer to a study supplier. Column 3 is the mean value transferred to study suppliers in weeks with a positive transaction value. Column 4 is the total value of transfers to study suppliers, excluding the first four weeks. Column 5 is the total number of weeks in which a firm made a transfer to a study supplier, excluding the first four weeks. Column 6 is the mean value transferred to study suppliers in weeks with a positive transaction value, excluding the first four weeks. Mystery shopping generally took place during within 2-3 weeks of the firm entering the study. The data covers up to 22 months after being added to the study. Columns 3 and 6 are at the firm-week level, with standard errors clustered by firm. All values are in USD.

Table 4: Profit and Sales

	Raw			Winsorized (1%)		
	Profit (USD)	Sales (USD)	Index	Profit (USD)	Sales (USD)	Index
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Pooled						
Treatment	82.4**	245.2*	0.314**	45.5**	121.5	0.174**
	(31.6)	(108.4)	(0.107)	(21.2)	(78.6)	(0.076)
Panel B: Individual Treatments						
Search Only	31.7	311*	.197*	22.1	178	.21*
	(28.8)	(177)	(.114)	(25.4)	(112)	(.108)
	[0.561]	[0.183]	[0.187]	[0.650]	[0.270]	[0.124]
Search + Adverse Selection	43.5	80.8	.193	30.8	73.1	.124
	(37.7)	(114)	(.129)	(30.7)	(100)	(.102)
	[0.561]	[0.729]	[0.235]	[0.650]	[0.695]	[0.387]
Search + Moral Hazard	15.8	-24	.036	8.21	-25.3	6.2e-03
	(25.1)	(106)	(.085)	(23.2)	(87)	(.078)
	[0.562]	[0.829]	[0.695]	[0.741]	[0.786]	[0.943]
Search + AS + MH	254***	636**	.88***	128***	269**	.371***
	(89.3)	(267)	(.315)	(39.2)	(124)	(.137)
	[0.004]	[0.031]	[0.005]	[0.002]	[0.084]	[0.018]
Control Mean	188.3	609.5	0.000	188.3	609.5	0.000
% Increase (Pooled)	43.8%	40.2%	N/A	24.2%	19.9%	N/A
All Coefs Equal p-val	0.051	0.041	0.041	0.024	0.054	0.027
Adjusted R^2	0.13	0.12	-0.01	0.25	0.29	-0.02
N	1351	1378	1298	1351	1378	1298

Note: p-values are computed using randomisation inference. Specifically, we compute the randomised randomisation-t p-value from Young (2019) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. We also report Romano-Wolf multiple-testing adjusted p-values in square brackets (Romano and Wolf, 2005). To test the null that the trust treatments have no effect, we report at the bottom of the table the p-value for an F-test that all coefficients are equal, computed by permuting the F-statistic. Panel A shows the coefficient from a regression on an indicator that pools all treated groups. Panel B shows the coefficients corresponding to treatment indicators for each of the four treatment arms. All regressions include any covariates selected by Double Lasso (Belloni, Chernozhukov, and Hansen, 2014), as well as stratum fixed effects and the outcome measured at baseline (if available).

Column 1 is total profit from the past 30 days in USD. Column 2 is total sales from the past 30 days in USD. Column 3 is the Anderson (2008) index combining the previous two columns to account for multiple hypothesis testing across outcomes. Column 4 is total profit from the past 30 days in USD, winsorizing the top 1%. Column 5 is total sales from the past 30 days in USD, winsorizing the top 1%. Column 6 is the Anderson (2008) index combining the previous two columns to account for multiple hypothesis testing across outcomes. Profit is measured using the survey question from De Mel, McKenzie, and Woodruff (2009). Sales is measured using a similar survey question.

Table 5: Effect of Alibaba Training

	Heard of Alibaba (1)	Searched on Alibaba (2)	Compared Prices with Supplier (3)	Bought on Alibaba (4)
e-Commerce Treatment	0.065**	0.113**	0.087*	0.014
	(0.025)	(0.052)	(0.049)	(0.035)
Control Mean	0.908	0.423	0.319	0.135
% Increase	7.2%	26.8%	27.4%	10.7%
Adjusted R^2	0.07	0.09	0.14	0.22
N	340	340	340	340

Note: p-values are computed using randomisation inference. Specifically, we compute the randomised randomisation-t p-value from Young (2019) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. Conventional robust standard errors are reported in parentheses. This table shows the effect of the Alibaba training treatment on Alibaba usage.

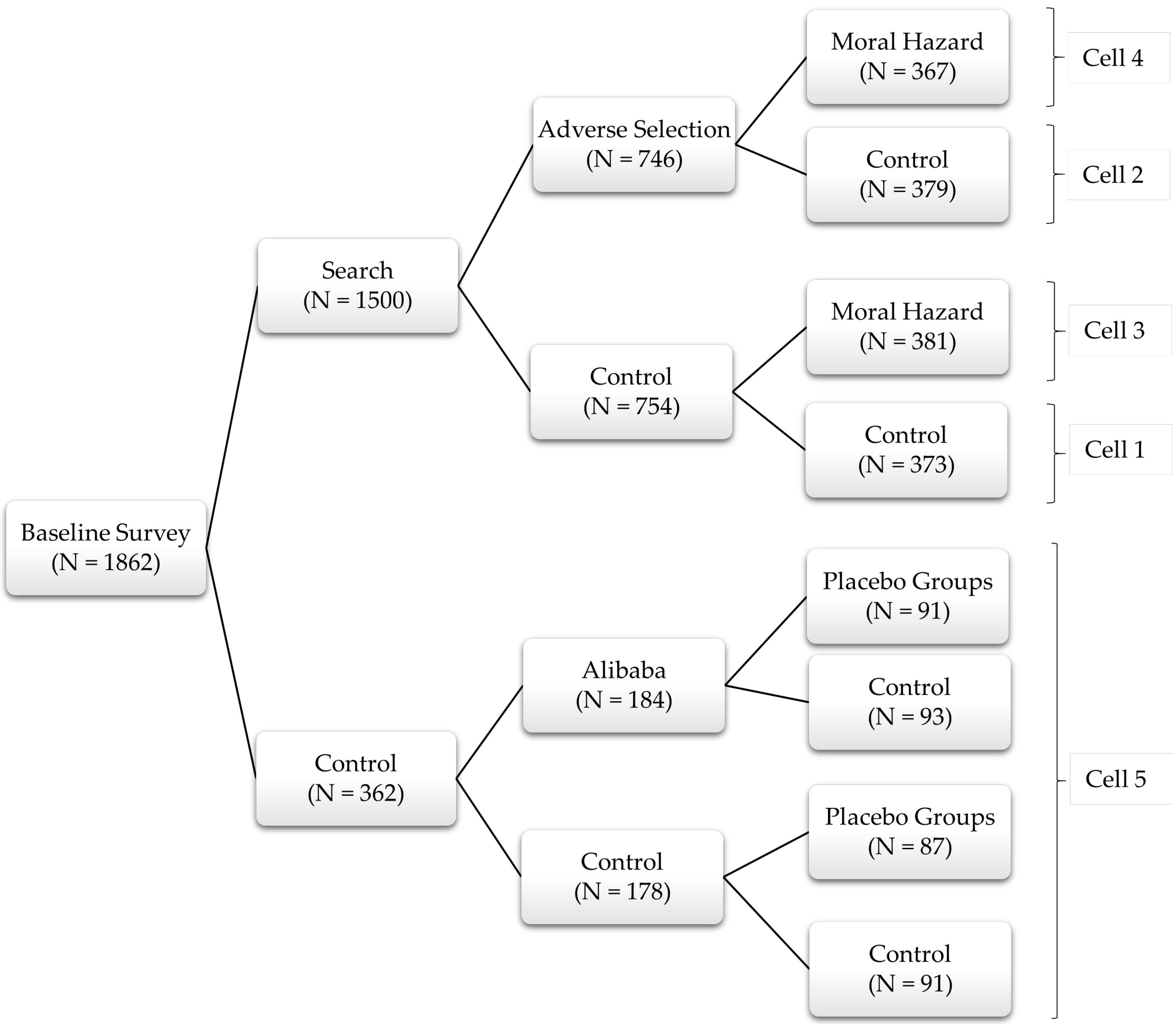
Table 6: Estimated Structural Parameters

Description	Parameter	Estimate
Share of bad types	μ	0.57 (0.09)
Moral hazard cost saving	ξ	0.91 (0.16)
Sequential search cost (USD)	s	94 (156)
Match productivity (mean)	ψ_μ	−6.26 (0.89)
Match productivity (std dev)	ψ_σ	1.66 (0.44)
Discount factor	δ	0.78 (0.11)
Bad type quality probability	λ	0.30 (0.05)

Notes: This table shows the estimated structural parameters. We estimate the parameters using Simulated Method of Moments, where we simulate the treatment in the model and match the model-implied treatment effects to the experimental treatment effects on profit, whether the firm has a supplier in Turkey, and orders from mobile money. We weight the moments by the inverse of the empirical variance-covariance matrix, and run the optimiser on 50 randomized starting points. We present bootstrapped standard errors in parentheses, which we calculate as the standard deviation of the results of the estimation procedure run on 200 sets of treatment effects obtained by bootstrapping the experimental data. The parameters for the Lognormal match-specific productivity distribution are for the underlying Normal. Please see Appendix E for more detail on the procedure to solve and estimate the model.

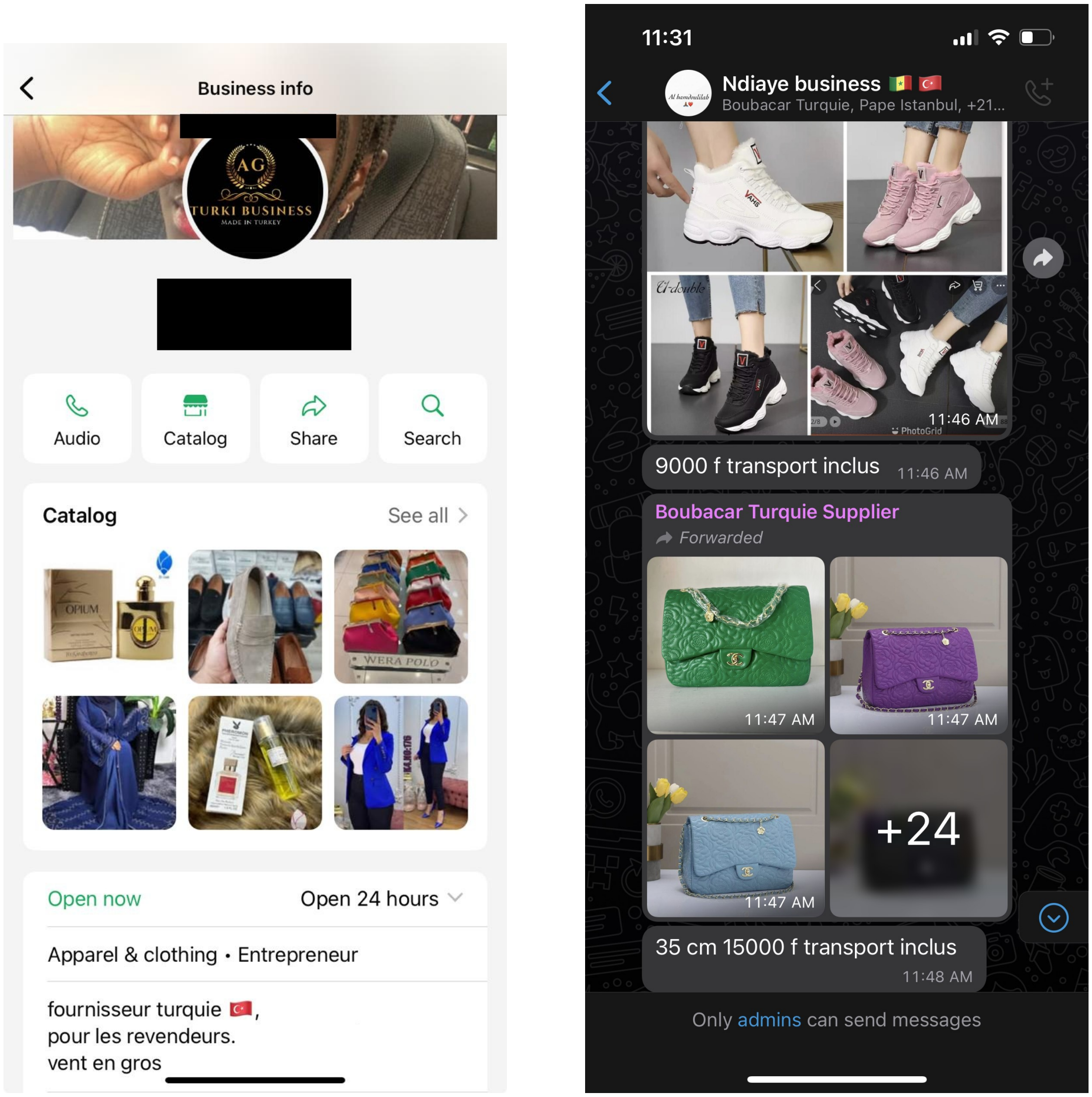
Figures

Figure 1: Experimental Design Tree



Note: This figure presents the design tree of the experiment, which identifies three frictions—search, adverse selection, and moral hazard—via randomized variation in information and messaging. Treatments are described in Section 4.1. Among recruited firms, 80% of them are randomly assigned to the Search treatment, in which they are matched via WhatsApp groups to three suppliers of Turkish-made goods. Within the Search group, 50% are cross-randomized into the Adverse Selection treatment, where they are placed in peer groups to share supplier experiences and receive seeded information from independent recommenders. Separately, 50% of firms in the Search group are cross-randomized into the Moral Hazard treatment, in which firms are told that suppliers are rated and subject to removal, increasing perceived incentives for supplier effort. Two additional sub-treatments—Alibaba training and placebo peer groups—are cross-randomized within the pure control group to test secondary hypotheses within the control group. The first provides a brief Alibaba training to assess whether limited platform knowledge constrains B2B e-commerce adoption. The second creates peer WhatsApp groups without supplier matches to test whether connecting firms alone drives outcomes, serving as a placebo for the Adverse Selection treatment. Neither sub-treatment yields meaningful effects.

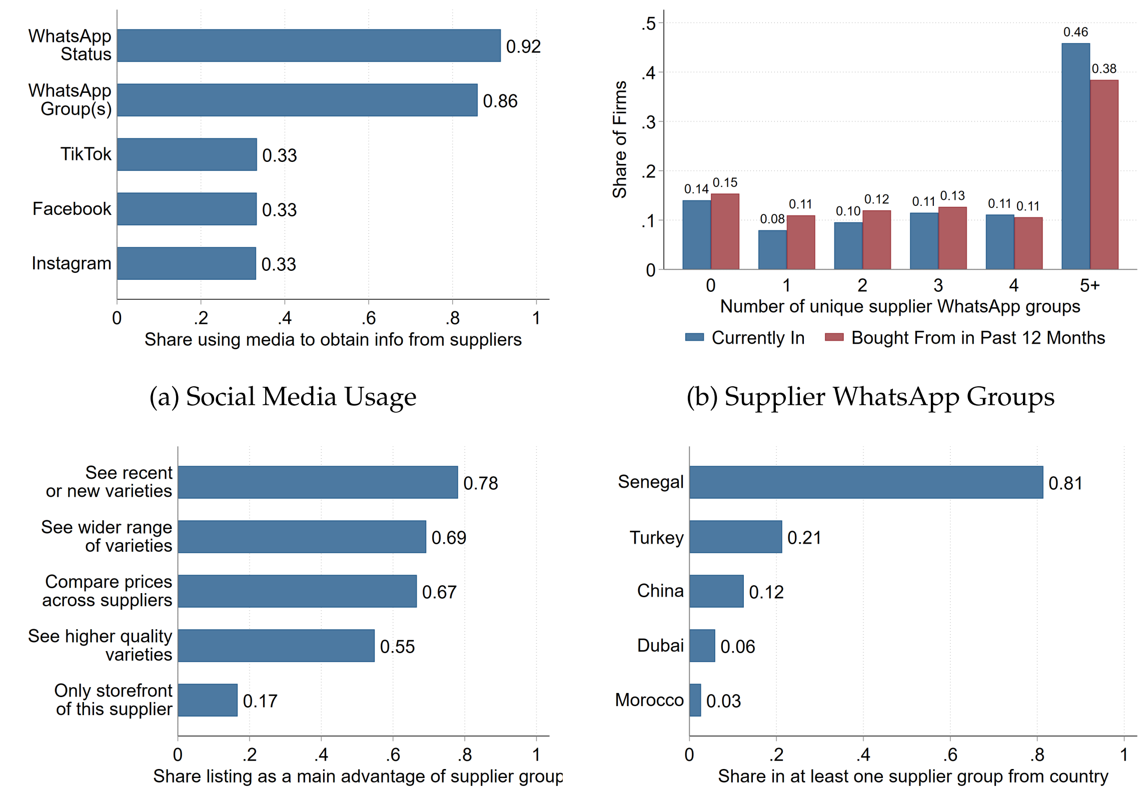
Figure 2: Supplier WhatsApp Profile



(a) WhatsApp Business Profile (b) Supplier WhatsApp Group

Note: These pictures show two common ways that suppliers use WhatsApp for social commerce, as described in Section 2.3. Panel (a) shows a typical WhatsApp Business profile, which allows suppliers to present key business information, including a catalog of goods. WhatsApp Business includes several features to facilitate commerce, such as the ability to run targeted advertisements on Facebook or Instagram that click through to this profile, and tools to automate and manage responses to customer inquiries. Panel (b) shows a typical supplier WhatsApp group. These groups function as virtual storefronts: the supplier regularly posts photos or videos showcasing products and can use the group to advertise promotions or new varieties. Clients—usually 50 to 100 per group—observe the posts but can rarely interact publicly; instead, they inquire or negotiate privately with the supplier via direct messages. Such groups could help reduce search costs by giving buyers access to a steady stream of updated product information and facilitating seamless communication. They might also reduce trust frictions to a lesser extent, as visible group membership and repeated engagement raise the reputational costs of cheating. Many suppliers operate both a supplier WhatsApp group and a WhatsApp Business profile, using the WhatsApp Business profile as the public-facing storefront and the supplier WhatsApp group as a way to engage with regular customers.

Figure 3: Firm Social Media Usage to obtain Information about Suppliers



(a) Social Media Usage

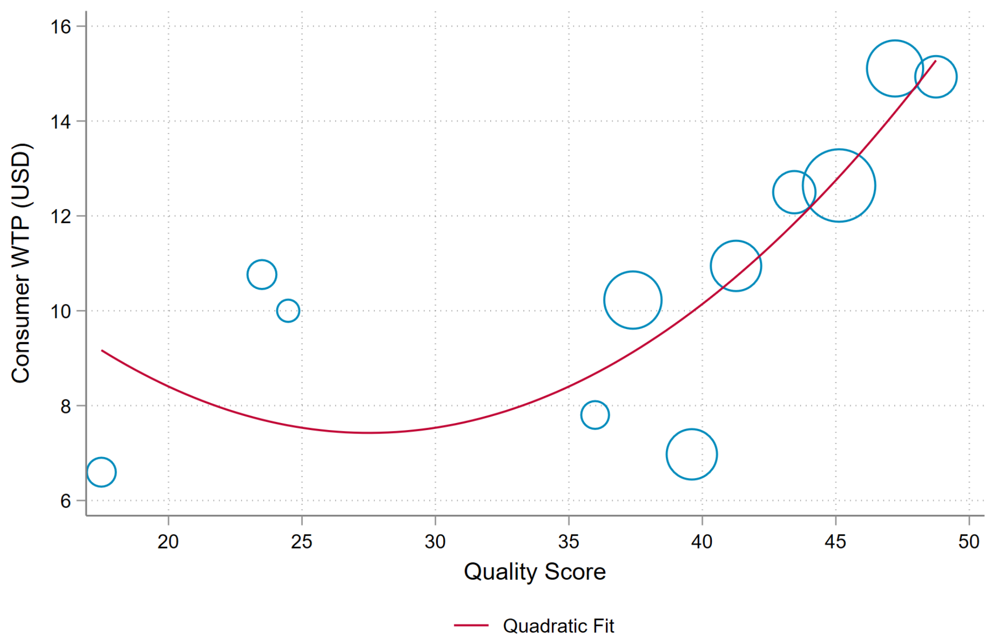
(b) Supplier WhatsApp Groups

(c) Advantages of Supplier Groups

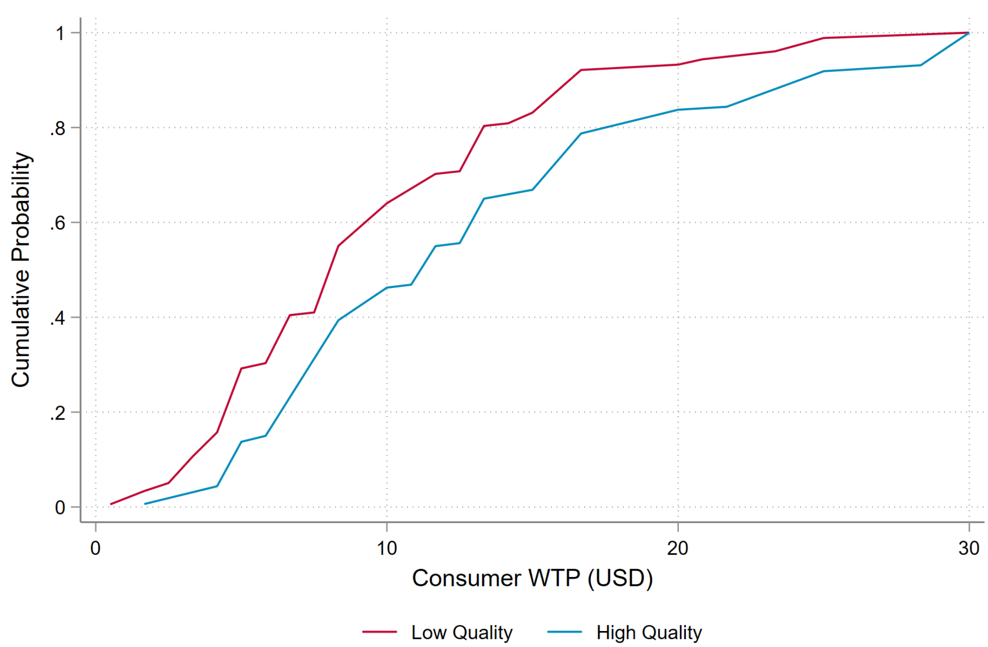
(d) Supplier Groups, by Supplier Country

Note: This figure shows a number of statistics about how firms in our sample use social media to obtain information about suppliers. All data is from our baseline survey with 1,862 firms. Panel (a) shows the results of a question asking firms to select all social media that they use to obtain information about suppliers for their business. Panel (b) shows the distribution of the number of supplier WhatsApp groups a firm is in at the time of the baseline survey, as well as the distribution of the number of such groups that the firm has directly made at least one purchase from in the past 12 months. Supplier WhatsApp groups are defined as WhatsApp groups in which the primary purpose is for suppliers to advertise their wares to downstream clients. Panel (c) shows the results of a question asking firms that use supplier WhatsApp groups to select all reasons why they find these groups useful. Panel (d) shows the share of firms who are in at least one supplier WhatsApp group where the supplier located and based in the country listed.

Figure 4: Consumer Willingness to Pay for Quality



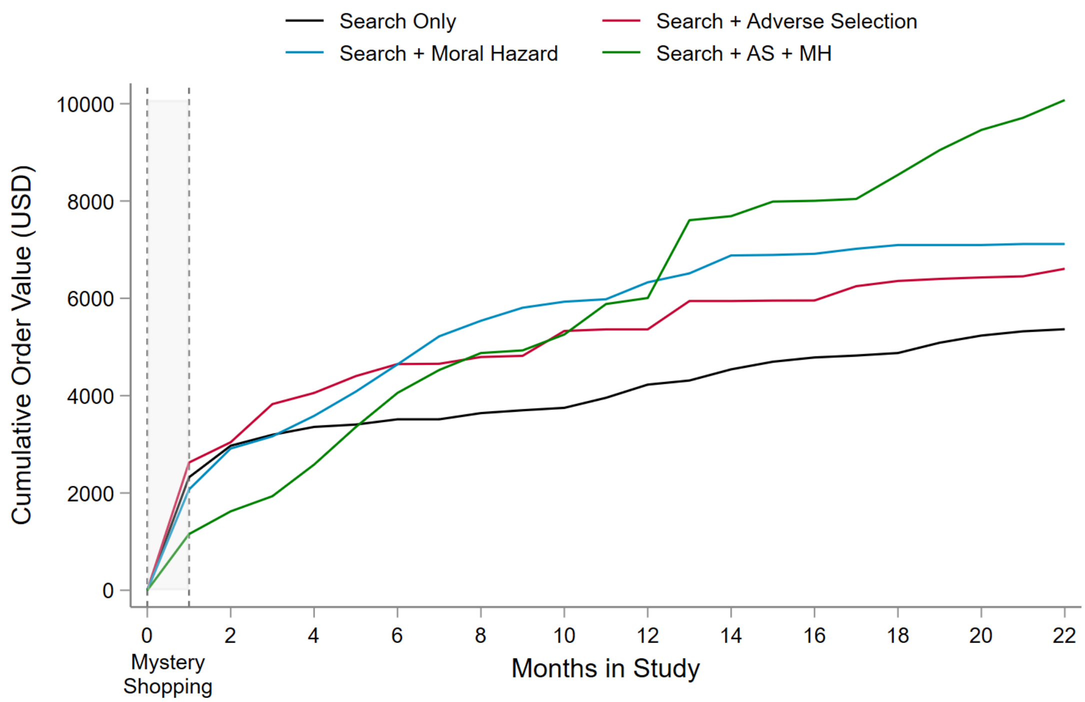
(a) Raw Score



(b) CDF of WTP by High or Low Quality

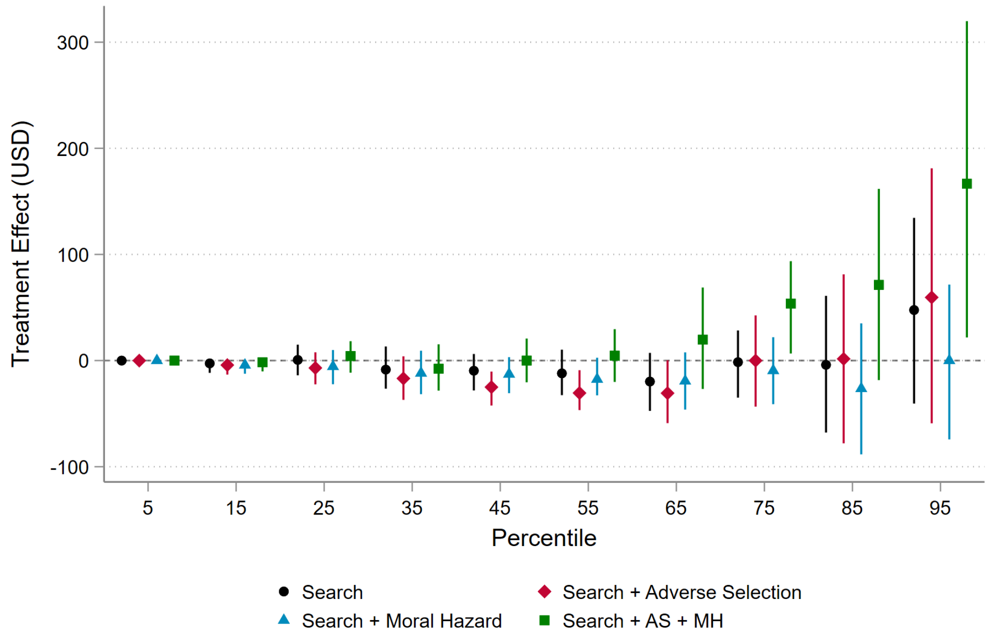
Note: Panel (a) shows a binscatter of consumer willingness to pay for garments (as measured by the consumer survey) against the quality score of the garments. The size of each bubble is proportional to the number of observations. Panel (b) shows the CDF of consumer willingness to pay separately based on whether the garment met our definition of high quality. We truncate willingness to pay at 30 USD to avoid unnecessarily stretching the x-axis. See the main text for full details on the consumer survey and variable construction.

Figure 5: Cumulative Order Value (Mobile Money Data)

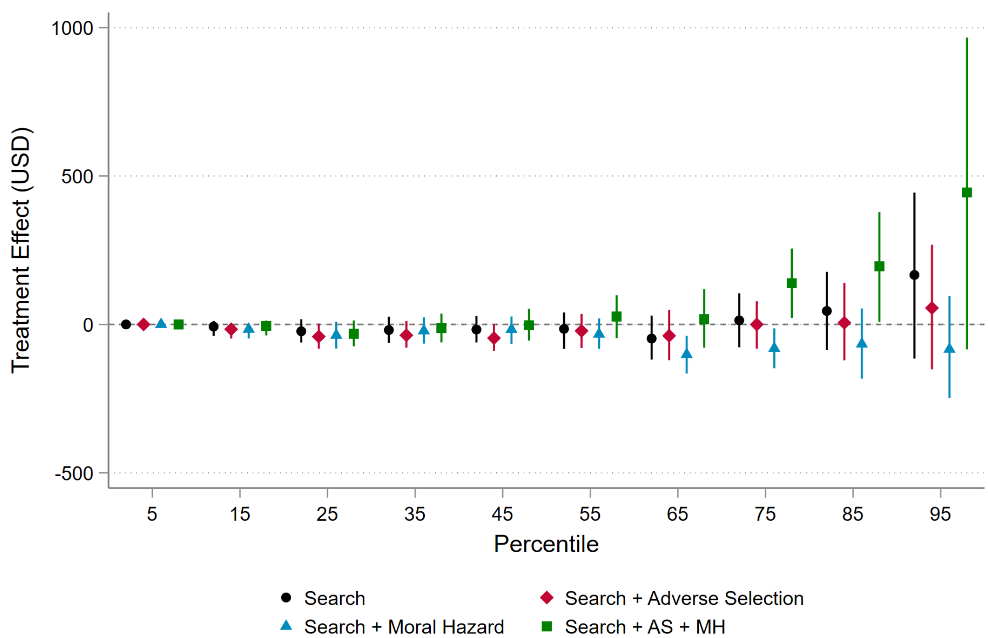


Note: This figure shows the total value flowing from study firms to study suppliers in the mobile money data, in each treatment group, as a function of number of months since the firm entered the study. As the group sizes are slightly different due to integer indivisibility issues, we normalize group-level totals by N_j/N_{avg}, where N_j is the number of units in treatment group j and N_{avg} is the average number of units across all four treatment groups. Pure control is omitted as they were not connected to any study suppliers.

Figure 6: Quantile Treatment Effects



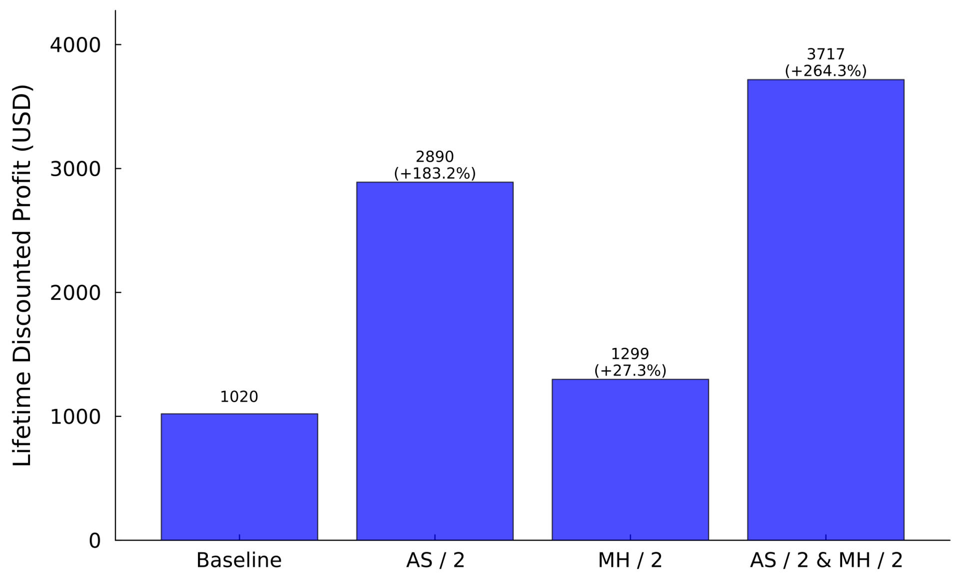
(a) Profit



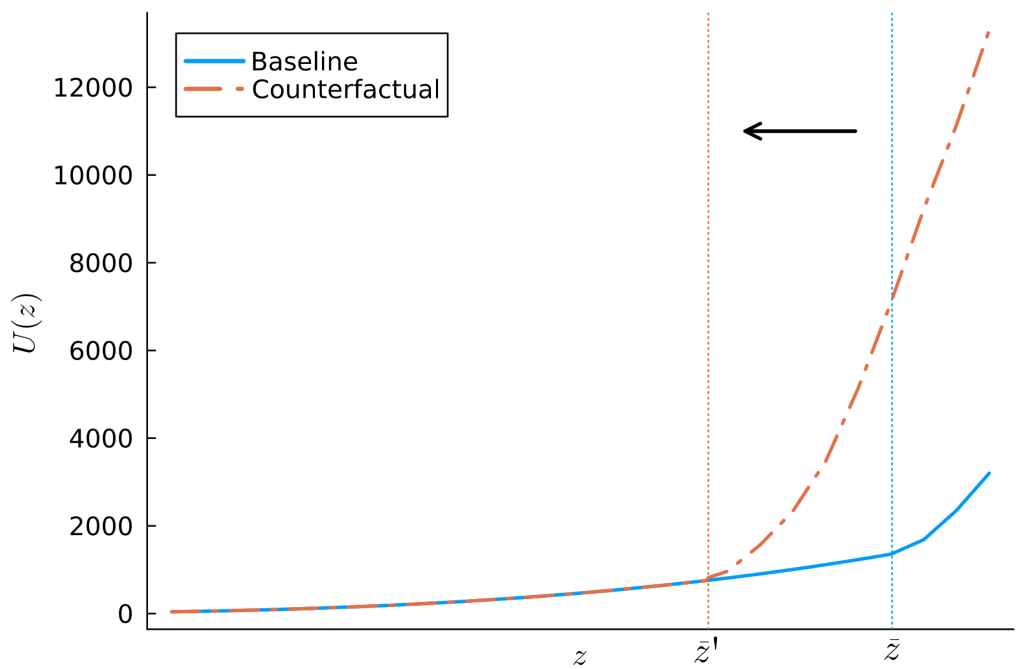
(b) Sales

Note: This figure shows the coefficients from quantile regressions of profit and sales on the four treatment groups. All quantile regressions include the outcome measured at baseline (if available), but otherwise do not include any covariates. We plot 95% confidence intervals constructed using randomisation inference, defined as the set of sharp nulls that do not reject at the 5% level.

Figure 7: Counterfactual Analysis



(a) Counterfactuals



(b) Importing productivity cutoff, $\bar{z}$, shifts left when adverse selection is lowered

Note: Panel (a) shows counterfactuals alleviating combinations of the frictions. The first column is the baseline scenario. The second column reduces the share of bad types, μ_0, which governs the size of adverse selection, by half. The third column reduces ξ, the parameter governing how much a supplier can reduce the marginal cost by if they choose low effort, which scales moral hazard, by half. Finally, the last column reduces both μ_0 and ξ by half each. All values are expressed in the expected present discounted value of lifetime profits for the firm. Panel (b) shows firm discounted profits, $U(z)$, as a function of productivity, z, in both the Baseline and the AS / 2 scenarios. The vertical lines show the productivity cutoff, $\bar{z}$, in the two scenarios.

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Supplemental Appendix

Appendix A - Additional Tables and Figures

Tables

Table A1: Balance Table

Variable	Control (1)	Search (2)	Search AS (3)	Search MH (4)	Search AS MH (5)	p(2) (6)	p(3) (7)	p(4) (8)	p(5) (9)	Joint (10)
Female	0.49	0.51	0.49	0.53	0.49	0.71	0.9	0.29	0.84	0.68
Business Age (Years)	4.85	4.74	5.2	4.72	5.21	0.67	0.32	0.87	0.3	0.47
# Supp Groups	5.43	5.96	5.22	5.38	5.78	0.31	0.59	0.88	0.51	0.52
In Supp Group Turkey	0.22	0.21	0.22	0.21	0.21	0.77	0.85	0.83	0.87	0.99
Uses FB for Suppliers	0.32	0.31	0.35	0.31	0.37	0.8	0.34	0.63	0.13	0.21
Share Cust Turkey	0.43	0.45	0.45	0.44	0.45	0.48	0.45	0.67	0.53	0.96
Travelled Business (5y)	0.09	0.08	0.11	0.09	0.11	0.47	0.31	0.96	0.42	0.42
Profit USD (30d)	221	221	262	195	236	0.92	0.25	0.47	0.59	0.47
Bought Alibaba Ever	0.16	0.13	0.15	0.18	0.13	0.15	0.48	0.64	0.17	0.22
# Customers (3d)	9.46	8.93	9.09	8.61	10.07	0.43	0.47	0.28	0.44	0.39
# Suppliers (12m)	5.52	5.82	5.41	5.18	5.56	0.39	0.75	0.27	0.91	0.49
# Suppliers Turkey (12m)	0.42	0.33	0.4	0.32	0.39	0.22	0.84	0.2	0.71	0.51
Avg Order Size (USD)	463	468	499	421	527	0.99	0.76	0.65	0.51	0.85
Modal Good Price (USD)	17.67	17.03	17.66	17.61	17.41	0.5	1	0.9	0.78	0.95
Pays Supp Mobile Money	0.87	0.84	0.87	0.86	0.88	0.24	0.96	0.53	0.8	0.6
N	362	373	379	381	367					

Note: Columns 1–5 report means by treatment cell. Columns 6–9 report randomization-inference p-values for each arm relative to control (2vs1, 3vs1, 4vs1, 5vs1), computed following the regression specification in (2). Column 10 reports the RI p-value from a joint test all four coefficients are zero. Finally, we run a multinomial logit of the four treatment groups on all variables in the table; a joint test that all coefficients are zero has RI p-value of 0.958.

Table A2: One Week Survey

	Indicator for Answers 1 or 2 (out of 5)	
	Find Products From Turkey	Trust Supplier
	(1)	(2)
Panel A: Pooled		
Treatment	0.110***	
	(0.029)	
Panel B: Individual Treatments		
Search Only	0.078**	
	(0.038)	
	[0.047]	
Search + Adverse Selection	0.140***	0.009
	(0.037)	(0.036)
	[0.000]	[0.791]
Search + Moral Hazard	0.087**	-0.025
	(0.037)	(0.036)
	[0.044]	[0.694]
Search + AS + MH	0.136***	0.075**
	(0.038)	(0.038)
	[0.000]	[0.105]
Control Mean	0.453	0.303
% Increase (Pooled)	24.3%	6.6%
All Coefs Equal p-val	0.241	N/A
All Coefs Zero p-val	N/A	0.054
Adjusted R^2	0.06	0.06
N	1636	1201

Note: p-values are computed using randomisation inference. Specifically, we compute the randomised randomisation-t p-value from Young (2019) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. We also report Romano-Wolf multiple-testing adjusted p-values in square brackets (Romano and Wolf, 2005). To test the null that the trust treatments have no effect, we report at the bottom of the table the p-value for an F-test that all coefficients are equal, computed by permuting the F-statistic. Panel A shows the coefficient from a regression on an indicator that pools all treated groups. Panel B shows the coefficients corresponding to treatment indicators for each of the four treatment arms. All regressions include any covariates selected by Double Lasso (Belloni, Chernozhukov, and Hansen, 2014), as well as stratum fixed effects and the outcome measured at baseline (if available).

Column 1 is an indicator for whether they can find Turkish products. Column 2 is 1 if they reported a trust level 1 or 2, 0 if they reported a trust level of 3 or 4 or 5, and missing otherwise. Because control firms were not connected to any suppliers, they were not asked to provide ratings. Accordingly, Panel B, Column 2, omits the control group. Panel A, Column 2, reports estimates comparing the pooled trust treatments (AS, MH, and AS+MH) to the Search Only group.

Table A3: Horizontal Outcomes (Detailed)

	Extensive vs Intensive Margin		Number of Criteria	
	Agree Search	Find Product Conditional	Num Criteria Unconditional	Num Criteria Conditional
	(1)	(2)	(3)	(4)
Panel A: Pooled				
Treatment	0.015	0.103***	0.281**	0.368***
	(0.027)	(0.034)	(0.112)	(0.139)
Panel B: Individual Treatments				
Search Only	0.020	0.143***	0.531***	0.701***
	(0.034)	(0.043)	(0.148)	(0.178)
	[0.888]	[0.004]	[0.002]	[0.001]
Search + Adverse Selection	0.004	0.069	0.143	0.191
	(0.034)	(0.043)	(0.144)	(0.182)
	[0.993]	[0.116]	[0.503]	[0.301]
Search + Moral Hazard	0.037	0.103**	0.318**	0.326*
	(0.033)	(0.043)	(0.143)	(0.174)
	[0.626]	[0.040]	[0.071]	[0.149]
Search + AS + MH	-0.003	0.096**	0.132	0.249
	(0.034)	(0.044)	(0.142)	(0.179)
	[0.993]	[0.047]	[0.503]	[0.283]
Control Mean	0.673	0.453	1.421	2.094
% Increase (Pooled)	2.2%	22.7%	19.8%	17.6%
All Coefs Equal p-val	0.597	0.369	0.025	0.020
Adjusted R^2	0.06	0.04	0.04	0.03
N	1859	1274	1859	1274

Note: p-values are computed using randomisation inference. Specifically, we compute the randomised randomisation-t p-value from Young (2019) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. We also report Romano-Wolf multiple-testing adjusted p-values in square brackets (Romano and Wolf, 2005). To test the null that the trust treatments have no effect, we report at the bottom of the table the p-value for an F-test that all coefficients are equal, computed by permuting the F-statistic. Panel A shows the coefficient from a regression on an indicator that pools all treated groups. Panel B shows the coefficients corresponding to treatment indicators for each of the four treatment arms. All regressions include any covariates selected by Double Lasso (Belloni, Chernozhukov, and Hansen, 2014), as well as stratum fixed effects and the outcome measured at baseline (if available).

Column 1 is 1 if the firm agrees to sell or search for the product. Column 2 is 1 if the firm has a suitable product, conditional on agreeing to sell or search for the product. Column 3 is the number of horizontal criteria of the product, and is 0 if the firm either did not agree to sell or search for a product or agreed but never sent any product. Column 4 is the number of horizontal criteria of the product, conditional on agreeing to sell or search for a product, and is 0 if the firm agreed but never sent any product.

Table A4: Vertical Outcomes (Detailed)

	From Turkey		
	Made in Turkey (Label)	Made in Turkey (Tailor Judgement)	Made in Turkey (Label + Tailors)
	(1)	(2)	(3)
Panel A: Pooled			
Treatment	0.210**	0.099	0.159**
	(0.081)	(0.081)	(0.072)
Panel B: Individual Treatments			
Search Only	0.169*	0.134	0.138
	(0.096)	(0.095)	(0.086)
	[0.141]	[0.437]	[0.195]
Search + Adverse Selection	0.172*	0.010	0.107
	(0.100)	(0.102)	(0.089)
	[0.141]	[0.927]	[0.221]
Search + Moral Hazard	0.241**	0.132	0.192**
	(0.094)	(0.095)	(0.084)
	[0.038]	[0.437]	[0.087]
Search + AS + MH	0.271***	0.096	0.189**
	(0.097)	(0.097)	(0.084)
	[0.028]	[0.515]	[0.087]
Control Mean	0.489	0.581	0.544
% Increase (Pooled)	42.9%	17.0%	29.2%
All Coefs Equal p-val	0.538	0.489	0.625
Adjusted R^2	0.10	-0.00	0.07
N	255	287	330

Note: p-values are computed using randomisation inference. Specifically, we compute the randomised randomisation-t p-value from Young (2019) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. We also report Romano-Wolf multiple-testing adjusted p-values in square brackets (Romano and Wolf, 2005). To test the null that the trust treatments have no effect, we report at the bottom of the table the p-value for an F-test that all coefficients are equal, computed by permuting the F-statistic. Panel A shows the coefficient from a regression on an indicator that pools all treated groups. Panel B shows the coefficients corresponding to treatment indicators for each of the four treatment arms. All regressions include any covariates selected by Double Lasso (Belloni, Chernozhukov, and Hansen, 2014), as well as stratum fixed effects and the outcome measured at baseline (if available).

Column 1 is 1 if the label says , 0 if the label says for X other than Turkey, and missing otherwise. Column 2 is 1 if both tailors independently determined that the product was made in Turkey, and is 0 if both tailors independently determined that the product was not made in Turkey. It is missing if the tailors disagreed. For shoes, as there was only one expert shoemaker, we take their opinion directly. Column 3 is an indicator that combines the label and tailor measures of whether the good was made in Turkey. It is equal to the label measure where available, and the tailor measure otherwise.

Table A5: Access to Foreign Goods (Selection and Robustness Analysis)

	Combined		Horiz	Vertical				Price
	Sold HQ	Sold Turkish	Has Product	HQ Dummy	Q Score (/50)	Made in Turkey	Index	Price (USD)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Pooled								
Treatment	0.037*	0.064***	0.065**	0.111	-0.767	0.123*	0.308**	0.590
	(0.021)	(0.021)	(0.027)	(0.069)	(0.633)	(0.063)	(0.137)	(0.703)
Panel B: Individual Treatments								
Search Only	0.060**	0.070**	0.106***	0.150*	-0.378	0.104	0.398**	1.054
	(0.028)	(0.029)	(0.035)	(0.085)	(0.706)	(0.076)	(0.159)	(0.867)
	[0.111]	[0.046]	[0.011]	[0.250]	[0.819]	[0.303]	[0.056]	[0.541]
Search + Adverse Selection	0.029	0.040	0.029	0.153*	-0.088	0.081	0.358**	0.829
	(0.027)	(0.028)	(0.034)	(0.088)	(0.948)	(0.082)	(0.169)	(0.816)
	[0.543]	[0.142]	[0.396]	[0.250]	[0.930]	[0.316]	[0.096]	[0.605]
Search + Moral Hazard	0.030	0.068**	0.077**	0.092	-1.314	0.146**	0.266	0.655
	(0.027)	(0.028)	(0.035)	(0.084)	(0.868)	(0.073)	(0.167)	(0.874)
	[0.543]	[0.046]	[0.071]	[0.443]	[0.381]	[0.156]	[0.209]	[0.660]
Search + AS + MH	0.029	0.077***	0.047	0.062	-1.109	0.148**	0.210	-0.214
	(0.026)	(0.029)	(0.035)	(0.085)	(0.846)	(0.076)	(0.181)	(0.824)
	[0.543]	[0.031]	[0.301]	[0.447]	[0.424]	[0.156]	[0.261]	[0.778]
Control Mean	0.097	0.105	0.307	0.431	43.064	0.477	0.000	19.990
% Increase (Pooled)	38.1%	61.0%	21.2%	25.8%	-1.8%	25.8%	N/A	3.0%
All Coefs Equal p-val	0.673	0.627	0.114	0.654	0.502	0.772	0.659	0.356
Adjusted R^2	0.02	0.04	0.07	0.07	0.35	0.25	0.01	0.45
N	1438	1438	1859	359	359	361	356	642

Note: p-values are computed using randomisation inference. Specifically, we compute the randomised randomisation-t p-value from Young (2019) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. We also report Romano-Wolf multiple-testing adjusted p-values in square brackets (Romano and Wolf, 2005). To test the null that the trust treatments have no effect, we report at the bottom of the table the p-value for an F-test that all coefficients are equal, computed by permuting the F-statistic. Panel A shows the coefficient from a regression on an indicator that pools all treated groups. Panel B shows the coefficients corresponding to treatment indicators for each of the four treatment arms. All regressions include any covariates selected by Double Lasso (Belloni, Chernozhukov, and Hansen, 2014), as well as stratum fixed effects and the outcome measured at baseline (if available).

Column 1 is 1 if the firm both sold a suitable good and it was high quality (following the definition in Column 2), and 0 otherwise (regression excludes cases where firm was randomly selected for non-purchase). Column 2 is 1 if the firm both sold a suitable good and it was from Turkey (following the definition in Column 4), and 0 otherwise (regression excludes cases where firm was randomly selected for non-purchase). Column 3 is an indicator that is one if the firm has a good that matches at least 3 horizontal criteria, and is missing if the firm never replied to the mystery shopper or was otherwise unreachable. Column 4 is an indicator for whether the good's quality score is above the median product-group quality score. Column 5 is the raw quality score. Column 6 is an indicator for whether the good is made in Turkey, primarily inferred based on the label. See the text for full details of how this outcome is constructed. Column 7 is the Anderson (2008) index combining the vertical outcomes to account for multiple hypothesis testing across outcomes. Column 8 is the price in USD, which is only measured conditional on the firm finding a good matching at least three horizontal criteria.All columns from 3 onwards include fixed effects for each of the 28 products.

Table A6: Supplier Relationships (Further Results on Substitution)

	Reg Supp in China		Media for Suppliers			Forward Media	
	Any Supp in China	Num Supp in China	Uses Facebook	Uses TikTok	Uses Instagram	Fwd Photo for Search	Fwd Photo for Price
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: Pooled							
Treatment	-0.025	-0.014	-0.084***	-0.025	-0.027	-0.054**	-0.097***
	(0.016)	(0.035)	(0.027)	(0.025)	(0.026)	(0.027)	(0.030)
Panel B: Individual Treatments							
Search Only	-0.030	-0.023	-0.090***	-0.047	-0.037	-0.054	-0.103***
	(0.019)	(0.040)	(0.033)	(0.032)	(0.032)	(0.036)	(0.039)
	[0.335]	[0.929]	[0.024]	[0.386]	[0.592]	[0.230]	[0.019]
Search + Adverse Selection	-0.029	-0.028	-0.063*	-0.006	-0.038	-0.033	-0.088**
	(0.019)	(0.045)	(0.034)	(0.032)	(0.033)	(0.035)	(0.038)
	[0.335]	[0.929]	[0.076]	[0.924]	[0.592]	[0.337]	[0.035]
Search + Moral Hazard	-0.018	0.009	-0.080**	-0.039	-0.016	-0.063*	-0.081**
	(0.020)	(0.044)	(0.033)	(0.032)	(0.033)	(0.035)	(0.038)
	[0.424]	[0.946]	[0.037]	[0.467]	[0.836]	[0.204]	[0.035]
Search + AS + MH	-0.023	-0.013	-0.103***	-0.011	-0.012	-0.058*	-0.109***
	(0.020)	(0.045)	(0.033)	(0.033)	(0.033)	(0.035)	(0.039)
	[0.424]	[0.946]	[0.007]	[0.924]	[0.836]	[0.230]	[0.018]
Control Mean	0.099	0.167	0.328	0.290	0.279	0.716	0.659
% Increase (Pooled)	-25.3%	-8.4%	-25.6%	-8.6%	-9.7%	-7.5%	-14.7%
All Coefs Equal p-val	0.910	0.840	0.630	0.515	0.774	0.837	0.889
Adjusted R^2	0.12	0.36	0.09	0.12	0.12	0.04	0.03
N	1680	1680	1671	1671	1671	1671	1565

Note: p-values are computed using randomisation inference. Specifically, we compute the randomised randomisation-t p-value from Young (2019) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. We also report Romano-Wolf multiple-testing adjusted p-values in square brackets (Romano and Wolf, 2005). To test the null that the trust treatments have no effect, we report at the bottom of the table the p-value for an F-test that all coefficients are equal, computed by permuting the F-statistic. Panel A shows the coefficient from a regression on an indicator that pools all treated groups. Panel B shows the coefficients corresponding to treatment indicators for each of the four treatment arms. All regressions include any covariates selected by Double Lasso (Belloni, Chernozhukov, and Hansen, 2014), as well as stratum fixed effects and the outcome measured at baseline (if available).

Column 1 is 1 if the firm says that they have a regular supplier in China. Column 2 is 1 if the firm says that they have a regular supplier in China. Column 3 is 1 if the firm says that they use Facebook to learn about suppliers. Column 4 is 1 if the firm says that they use TikTok to learn about suppliers. Column 5 is 1 if the firm says that they use Instagram to learn about suppliers. Column 6 is 1 if the firm says that they have forwarded a photo or video from a supplier group to a regular supplier to try to obtain a similar product in the past 3 months. Column 7 is 1 if the firm says that they have forwarded a photo or video from a supplier group to a regular supplier to try to obtain a better price in the past 3 months. A regular supplier is defined as a supplier from whom the firm has made two or more orders with an intention of continuing the relationship.

Table A7: Mobile Money Order Value (Poisson Regressions)

	Entire Period			Post Mystery Shopping		
	Total Order Value (USD)	Num Weeks Ordered	Avg Order Value (USD)	Total Order Value (USD)	Num Weeks Ordered	Avg Order Value (USD)
	(1)	(2)	(3)	(4)	(5)	(6)
Search + Adverse Selection	0.22	-0.06	0.32	0.28	-0.29	0.49
	(0.32)	(0.33)	(0.20)	(0.45)	(0.50)	(0.32)
	[0.959]	[0.997]	[0.568]	[0.817]	[0.929]	[0.603]
Search + Moral Hazard	0.28	0.06	0.23	0.52	0.16	0.31
	(0.35)	(0.32)	(0.17)	(0.47)	(0.42)	(0.21)
	[0.959]	[0.997]	[0.568]	[0.817]	[0.932]	[0.603]
Search + AS + MH	0.63	0.01	0.65	1.10	0.09	0.91
	(0.73)	(0.41)	(0.39)	(0.85)	(0.57)	(0.37)
	[0.959]	[0.997]	[0.568]	[0.817]	[0.932]	[0.284]
Search Only Mean	14.31	0.36	39.24	7.92	0.23	33.95
% Increase (AS)	125.0%	94.5%	138.4%	132.3%	74.7%	163.2%
% Increase (MH)	132.6%	105.7%	125.3%	168.9%	117.0%	137.0%
% Increase (AS+MH)	188.3%	101.1%	190.9%	299.3%	109.8%	249.1%
All Coefs Zero p-val	0.955	0.993	0.426	0.870	0.895	0.360
Adjusted R^2	0.01	0.04	0.05	0.02	0.00	0.12
N	1500	1500	538	1500	1500	351

Note: p-values are computed using randomisation inference. Specifically, we compute the randomised randomisation-t p-value from Young (2019) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. We also report Romano-Wolf multiple-testing adjusted p-values in square brackets (Romano and Wolf, 2005). To test the null that the trust treatments have no effect, we report at the bottom of the table the p-value for an F-test that all coefficients are equal, computed by permuting the F-statistic. This table shows the coefficients corresponding to treatment indicators for each of the three treatment groups with trust treatments, where Search Only is the omitted category. All regressions include any covariates selected by Double Lasso (Belloni, Chernozhukov, and Hansen, 2014), as well as stratum fixed effects and the outcome measured at baseline (if available).

Column 1 is the total value of transfers to study suppliers. Column 2 is the total number of weeks in which a firm made a transfer to a study supplier. Column 3 is the mean value transferred to study suppliers in weeks with a positive transaction value. Column 4 is the total value of transfers to study suppliers, excluding the mystery shopping period. Column 5 is the total number of weeks in which a firm made a transfer to a study supplier, excluding the mystery shopping period. Column 6 is the mean value transferred to study suppliers in weeks with a positive transaction value, excluding the mystery shopping period. All regressions are estimated using Poisson Pseudo Maximum Likelihood (PPML). Mystery shopping generally took place during within 2-3 weeks of the firm entering the study. The data covers up to 22 months after being added to the study. Columns 3and 6are at the firm-week level, with standard errors clustered by firm. All values are in USD.

Table A8: Profit and Sales (Poisson Regression)

	Profit		Sales	
	Profit 30 Days (USD)	Profit 30 Days Winsorized 1% (USD)	Sales 30 Days (USD)	Sales 30 Days Winsorized 1% (USD)
	(1)	(2)	(3)	(4)
Panel A: Pooled				
Treatment	0.307*	0.188	0.217	0.118
	(0.132)	(0.112)	(0.142)	(0.126)
Panel B: Individual Treatments				
Search Only	0.088	0.074	0.315	0.198
	(0.147)	(0.133)	(0.210)	(0.158)
	[0.916]	[0.826]	[0.431]	[0.511]
Search + Adverse Selection	0.097	0.119	-0.041	-0.026
	(0.189)	(0.162)	(0.167)	(0.159)
	[0.916]	[0.826]	[0.830]	[0.876]
Search + Moral Hazard	-0.047	-0.009	-0.151	-0.147
	(0.138)	(0.143)	(0.174)	(0.143)
	[0.916]	[0.950]	[0.683]	[0.546]
Search + AS + MH	0.865***	0.480***	0.475**	0.260
	(0.214)	(0.142)	(0.212)	(0.162)
	[0.001]	[0.007]	[0.159]	[0.396]
Control Mean	188.3	188.3	609.5	609.5
% Increase (Pooled)	35.9%	20.7%	24.2%	12.5%
All Coefs Equal p-val	0.009	0.020	0.067	0.057
Adjusted R^2	0.34	0.38	0.46	0.48
N	1351	1351	1378	1378

Note: p-values are computed using randomisation inference. Specifically, we compute the randomised randomisation-t p-value from Young (2019) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. We also report Romano-Wolf multiple-testing adjusted p-values in square brackets (Romano and Wolf, 2005). To test the null that the trust treatments have no effect, we report at the bottom of the table the p-value for an F-test that all coefficients are equal, computed by permuting the F-statistic. Panel A shows the coefficient from a regression on an indicator that pools all treated groups. Panel B shows the coefficients corresponding to treatment indicators for each of the four treatment arms. All regressions include any covariates selected by Double Lasso (Belloni, Chernozhukov, and Hansen, 2014), as well as stratum fixed effects and the outcome measured at baseline (if available).

Column 1 is total profit from the past 30 days in USD. Column 2 is total profit from the past 30 days in USD, winsorizing the top 1%. Column 3 is total sales from the past 30 days in USD. Column 4 is total sales from the past 30 days in USD, winsorizing the top 1%. Profit is measured using the survey question from De Mel, McKenzie, and Woodruff (2009). Sales is measured using a similar survey question.

Table A9: Travel

	Wholesalers			Retailers		
	Any Travel	Travel China	Travel Turkey	Any Travel	Travel China	Travel Turkey
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Pooled						
Treatment	-0.033	-0.035	-0.020	-0.004	-0.021*	0.003
	(0.036)	(0.030)	(0.019)	(0.014)	(0.011)	(0.005)
Panel B: Individual Treatments						
Search Only	-0.001	-0.009	-0.020	-0.005	-0.022	-0.004
	(0.049)	(0.039)	(0.026)	(0.017)	(0.013)	(0.004)
	[0.980]	[0.936]	[0.856]	[0.984]	[0.272]	[0.692]
Search + Adverse Selection	0.007	-0.011	-0.012	-0.001	-0.025**	0.004
	(0.041)	(0.034)	(0.022)	(0.018)	(0.012)	(0.008)
	[0.980]	[0.936]	[0.856]	[0.997]	[0.126]	[0.813]
Search + Moral Hazard	-0.051	-0.050*	-0.014	-0.008	-0.015	0.010
	(0.038)	(0.030)	(0.020)	(0.017)	(0.014)	(0.008)
	[0.425]	[0.272]	[0.856]	[0.977]	[0.298]	[0.586]
Search + AS + MH	-0.052	-0.043	-0.016	-0.001	-0.022	0.001
	(0.037)	(0.032)	(0.020)	(0.020)	(0.014)	(0.006)
	[0.425]	[0.399]	[0.856]	[0.997]	[0.272]	[0.865]
Control Mean	0.130	0.090	0.040	0.041	0.033	0.004
% Increase (Pooled)	-25.4%	-38.9%	-50.0%	-9.8%	-63.6%	75.0%
All Coefs Equal p-val	0.168	0.248	0.987	0.986	0.866	0.151
Adjusted R^2	0.18	0.28	0.09	0.04	0.07	0.01
N	546	546	546	1125	1125	1125

Note: p-values are computed using randomisation inference. Specifically, we compute the randomised randomisation-t p-value from Young (2019) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. We also report Romano-Wolf multiple-testing adjusted p-values in square brackets (Romano and Wolf, 2005). To test the null that the trust treatments have no effect, we report at the bottom of the table the p-value for an F-test that all coefficients are equal, computed by permuting the F-statistic. Panel A shows the coefficient from a regression on an indicator that pools all treated groups. Panel B shows the coefficients corresponding to treatment indicators for each of the four treatment arms. All regressions include any covariates selected by Double Lasso (Belloni, Chernozhukov, and Hansen, 2014), as well as stratum fixed effects and the outcome measured at baseline (if available).

Column 1 is 1 if the firm travelled internationally for business in the past 3 months. Column 2 is 1 if the firm travelled for business to China in the past 3 months. Column 3 is 1 if the firm travelled for business to Turkey in the past 3 months. Column 4 is 1 if the firm travelled internationally for business in the past 3 months. Column 5 is 1 if the firm travelled for business to China in the past 3 months. Column 6 is 1 if the firm travelled for business to Turkey in the past 3 months. Travel is 1 if either the firm owner or someone closely involved with the firm travelled internationally for firm-specific business in the past 3 months.

Table A10: Calibrated Parameters

Parameter	Value	Origin
σ	3.02	Average markup from baseline survey.
p_l	12.29	Average input price from baseline survey among firms without a foreign supplier.

Note: This table presents the values of the parameters that are calibrated in the model.

Table A11: Lee Bounds on Main Survey Outcomes

Outcome	Treatment	Coef	SE	Lee LB	LB SE	Lee UB	UB SE
Any Sup Turkey	Pooled	0.039*	0.021	0.030	0.029	0.089***	0.032
Any Sup Turkey	S Only	0.024	0.027	0.013	0.035	0.062*	0.037
Any Sup Turkey	S+AS	0.051*	0.029	0.044	0.036	0.093**	0.040
Any Sup Turkey	S+MH	0.006	0.027	–0.005	0.034	0.061	0.038
Any Sup Turkey	S+AS+MH	0.075***	0.029	0.068*	0.035	0.143***	0.042
Num Sup Turkey	Pooled	0.085**	0.035	0.073	0.046	0.171***	0.047
Num Sup Turkey	S Only	0.080	0.051	0.064	0.062	0.164***	0.059
Num Sup Turkey	S+AS	0.069	0.046	0.059	0.056	0.138**	0.062
Num Sup Turkey	S+MH	0.002	0.040	–0.012	0.051	0.098**	0.049
Num Sup Turkey	S+AS+MH	0.193***	0.058	0.185***	0.066	0.278***	0.078
Profit (W. 1%)	Pooled	42.346**	21.414	21.431	22.939	69.399*	40.765
Profit (W. 1%)	S Only	19.276	26.129	–16.096	112.056	10.220	30.194
Profit (W. 1%)	S+AS	30.947	31.180	14.094	33.439	57.659	54.219
Profit (W. 1%)	S+MH	–4.981	24.780	–31.444	27.323	26.762	49.715
Profit (W. 1%)	S+AS+MH	131.630***	44.353	103.253**	46.973	189.377***	46.504
Sales (W. 1%)	Pooled	133.128	83.904	60.980	88.451	300.616***	75.961
Sales (W. 1%)	S Only	185.777	123.823	113.197	127.067	328.926**	143.754
Sales (W. 1%)	S+AS	70.475	109.672	16.607	116.684	250.794**	111.430
Sales (W. 1%)	S+MH	–59.735	92.478	–141.587	99.540	108.860	132.227
Sales (W. 1%)	S+AS+MH	349.621**	148.693	268.666*	158.297	558.849***	146.320

Note: This table shows Lee (2009) bounds for the main outcome variables from the survey. Columns 1 and 2 present the main coefficient and standard error. Column 3 presents the Lee lower bound and Column 4 presents its standard error. Column 5 presents the Lee upper bound and Column 6 presents its standard error. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. As Lee bounds are based on bivariate regressions and do not easily permit covariates, we only include the baseline value of the outcome: we use it directly to tighten the bounds if it is binary, and otherwise we convert it to binary by creating an indicator for above or below the median.

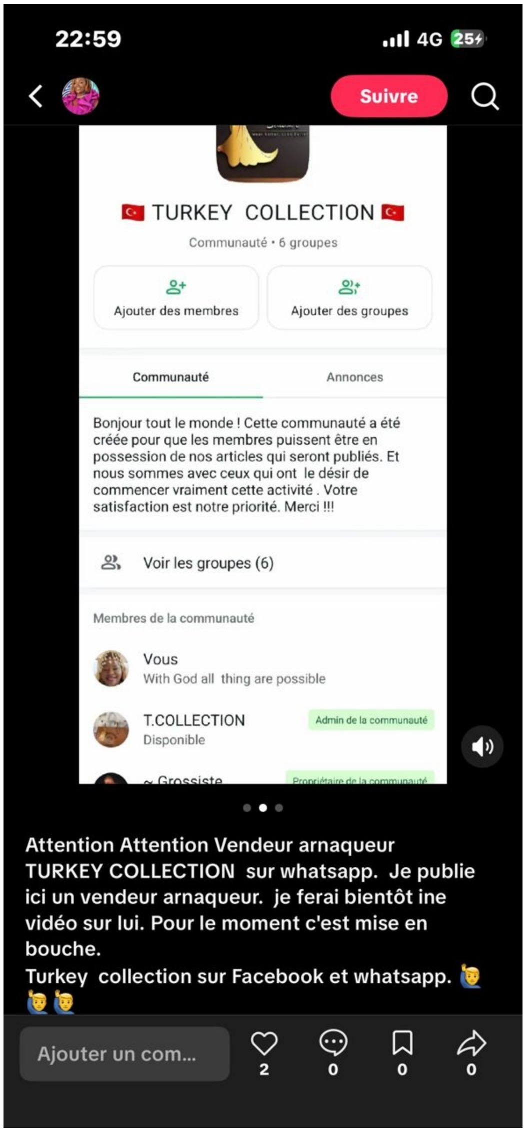
Table A12: Effect of Placebo Firm Groups

	Horizontal Dummy	Vertical Dummy	Num Supp In Turkey	Profit Sales Index (1%)
	(1)	(2)	(3)	(4)
Placebo Group Treatment	-0.034	-0.013	0.013	0.012
	(0.054)	(0.125)	(0.063)	(0.123)
Control Mean	0.374	0.438	0.216	-0.006
% Increase	-9.2%	-3.0%	6.0%	N/A
Adjusted R^2	-0.00	-0.02	-0.00	-0.00
N	311	65	342	266

Note: p-values are computed using randomisation inference. Specifically, we compute the randomised randomisation-t p-value from Young (2019) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. Conventional robust standard errors are reported in parentheses. This table shows the effect of the placebo firm discussion groups on the main outcomes (see Section 4.1 for details). The omitted category is the half of the pure control group that did not receive as this treatment. The outcome in Column 1 is the same as Column 1 of Table 1. The outcome in Column 2 is the same as Column 2 of Table 1. The outcome in Column 3 is the same as Column 2 of Table 2. The outcome in Column 4 is the same as Column 6 of Table 4.

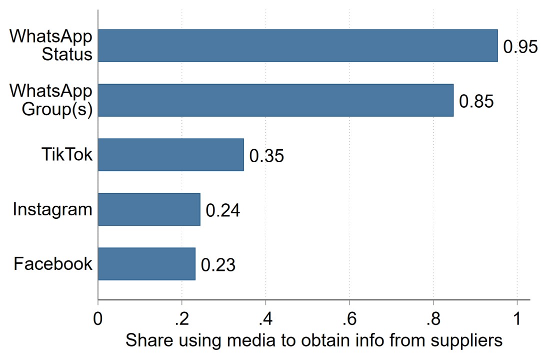
Figures

Figure A1: Examples of Group Punishment / Warning about Suppliers

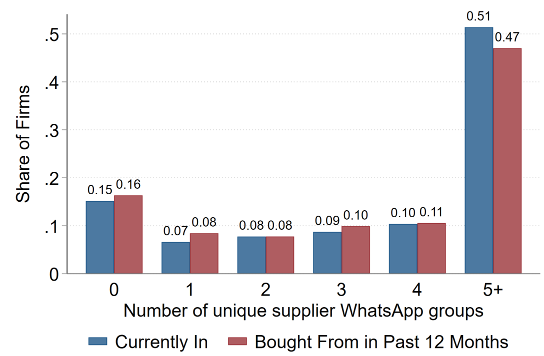


Note: This figure shows two examples of firms broadcasting on social media to alert other firms about bad experiences with particular suppliers in Turkey. In the left hand side image, the text states that this supplier is a scammer and that the firm will make and post a video with details, but is posting in the meantime to alert others about the supplier. In the right hand side image, the post explains that this firm made an order of 8 wigs (worth around USD 240), but, after paying, nothing has happened, and the supplier has repeatedly obfuscated.

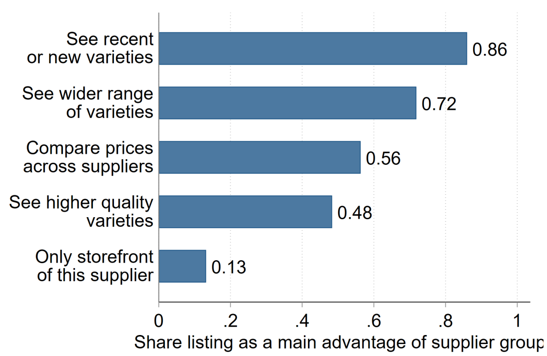
Figure A2: Social Media Usage (Physical Store Only)



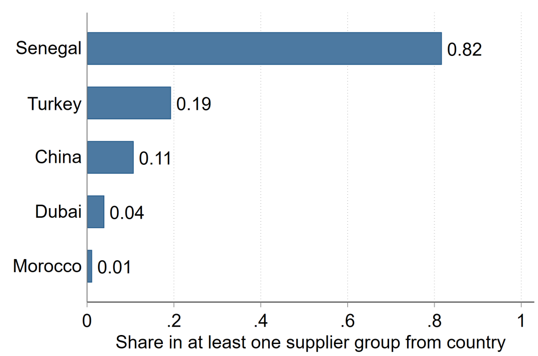
(a) Social Media Usage



(b) Supplier WhatsApp Groups



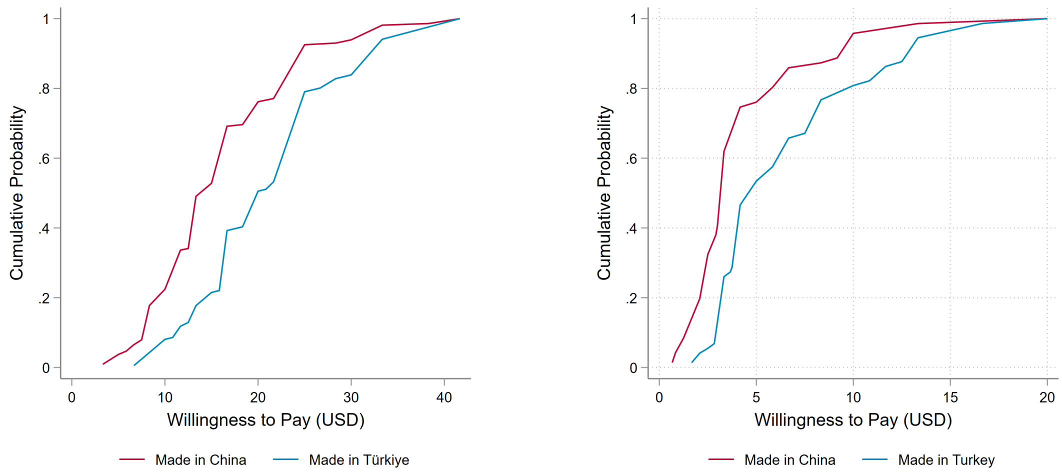
(c) Advantages of Supplier Groups



(d) Supplier Groups, by Supplier Origin

Note: This figure shows a number of statistics about how firms in our sample use social media to obtain information about suppliers. It is the same as Figure 3, but instead calculates statistics only for the 607 firms that have physical stores. Panel (a) shows the results of a question asking firms to select all social media that they use to obtain information about suppliers for their business. Panel (b) shows the distribution of the number of supplier WhatsApp groups a firm is in at the time of the baseline survey, as well as the distribution of the number of such groups that the firm has directly made at least one purchase from in the past 12 months. Supplier WhatsApp groups are defined as WhatsApp groups in which the primary purpose is for suppliers to advertise their wares to downstream clients. Panel (c) shows the results of a question asking firms that use supplier WhatsApp groups to select all reasons why they find these groups useful. Panel (d) shows the share of firms who are in at least one supplier WhatsApp group where the supplier located and based in the country listed.

Figure A3: Willingness to Pay by Product Origin



(a) Consumer Survey ($N = 400$) (b) Firm Survey ($N = 144$)

Note: This figure shows the results of two exercises in which surveyors showed various images of garments to respondants, randomising whether they stated that the good was made in Türkiye or made in China, and elicited willingness to pay for the garments. Panel (a) shows the CDF of WTP in the consumer survey, separately by whether the surveyor stated the good was made in Turkey or China. The distribution is truncated at 40 USD for ease of readability. Panel (b) shows the CDF of WTP for a small, separate survey of firms (for a different set of goods), with distribution truncated at 20 USD for ease of readability.

Figure A4: Business Cards for Moral Hazard Treatment

(a) Treatment (b) Control

Note: This figure shows the business cards used in the Moral Hazard treatment. In the treatment version, the card prominently displays a phone number for firms to rate suppliers or report problems, reinforcing the message that supplier performance will be monitored and can have consequences. In the control version, the card contains the same phone number but refers only to general questions about the study, with no mention of ratings or suppliers.

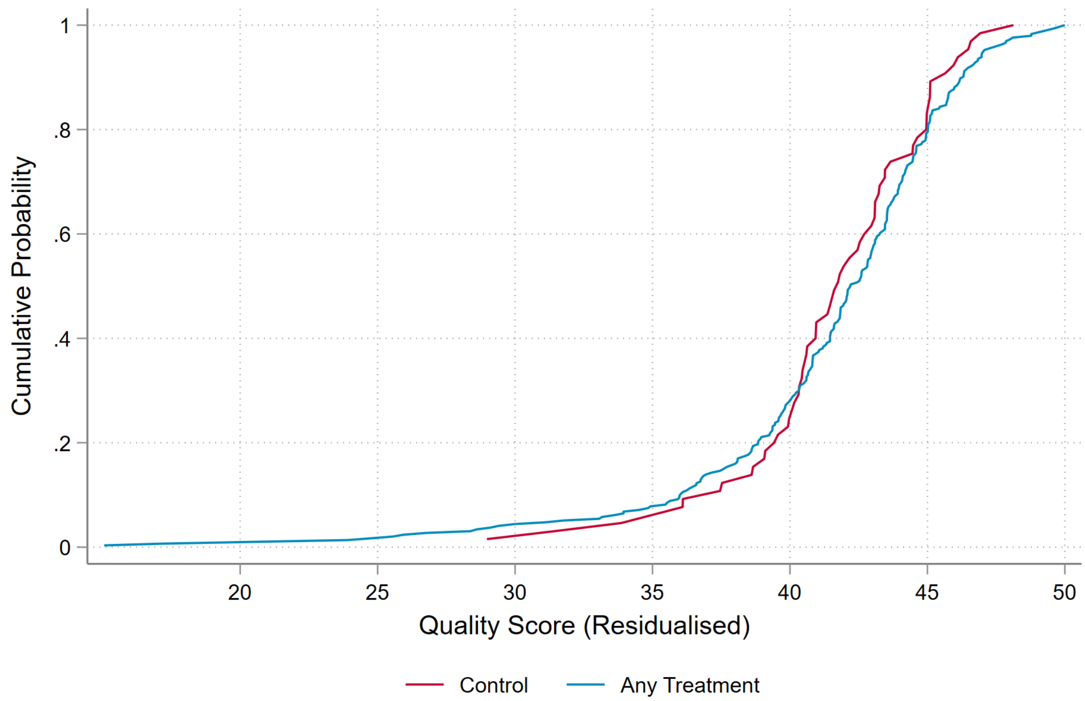
Figure A5: Mystery Shopping Goods (Examples)

1. *Red*
2. *One colour*
3. *No large logo/print*
4. *3 black buttons*
5. *Size L*

1. *Black*
2. *Long sleeves*
3. *Knee length*
4. *Turtleneck*
5. *Ribbed Cotton*

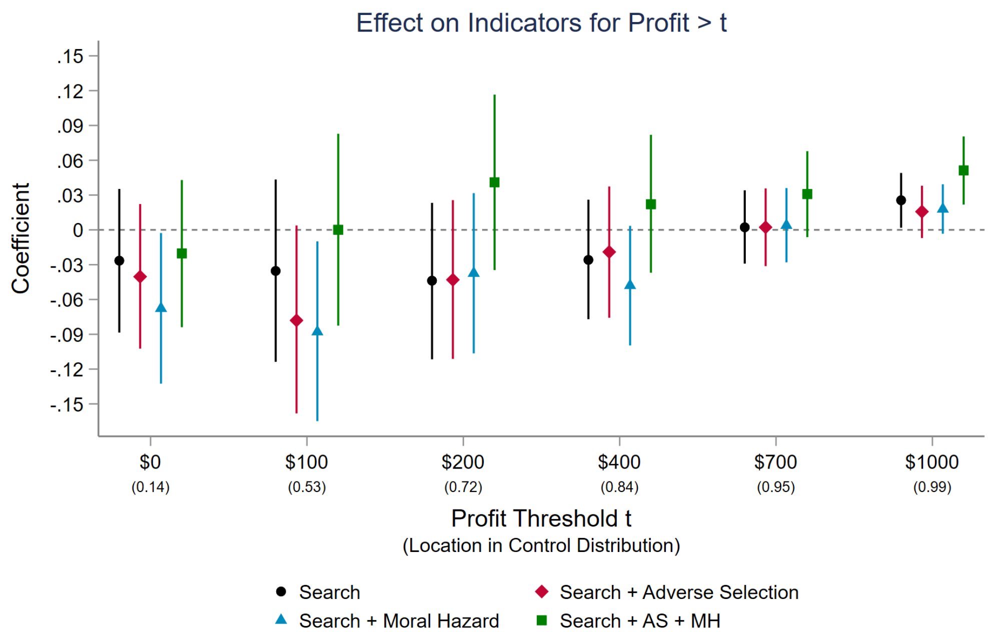
Note: Examples of goods requested in the mystery shopping exercise. In total, there were 28 different goods.

Figure A6: Quality Score Distribution

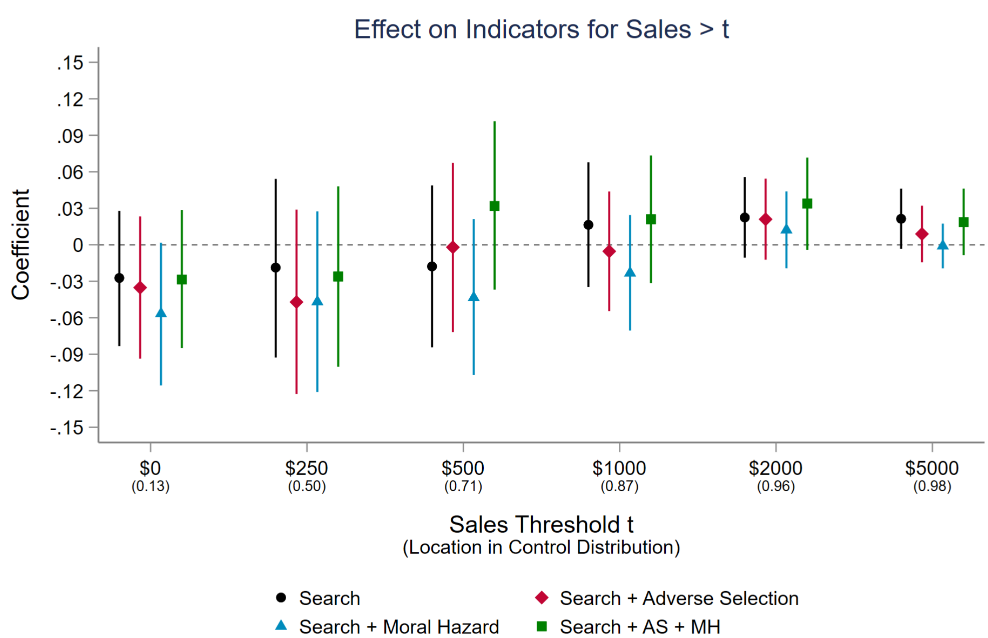


Note: This figure shows CDF of the quality score separately by treatment status, with all four treatment groups (Search Only, Search + Adverse Selection, Search + Moral Hazard, Search + AS + MH) pooled for visual ease. To be consistent with the regression in the table, we first residualise quality using stratum fixed effects and any covariates selected by PDS Lasso in the regression.

Figure A7: Threshold Regressions for Profit and Sales



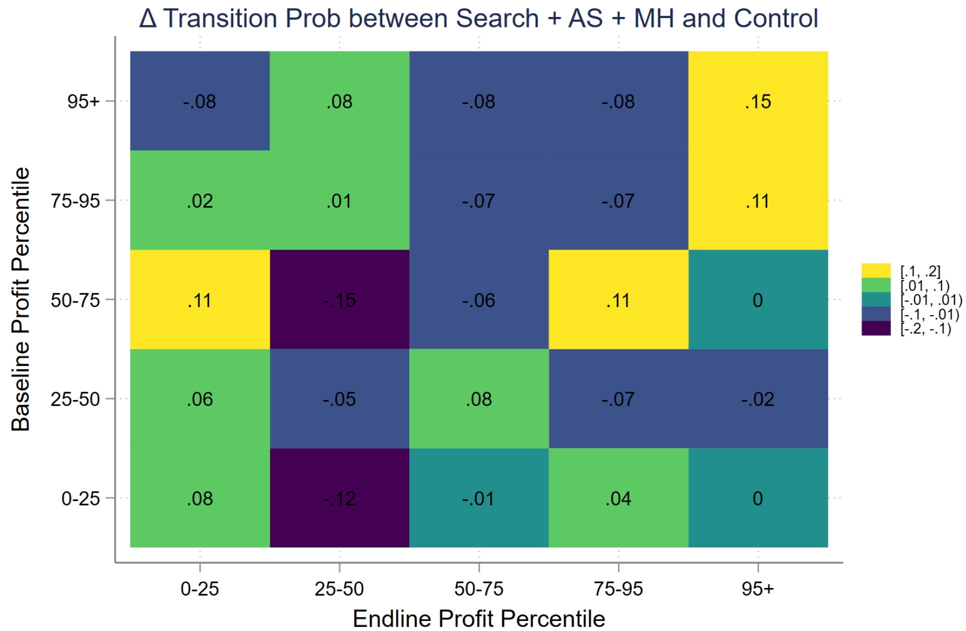
(a) Profit



(b) Sales

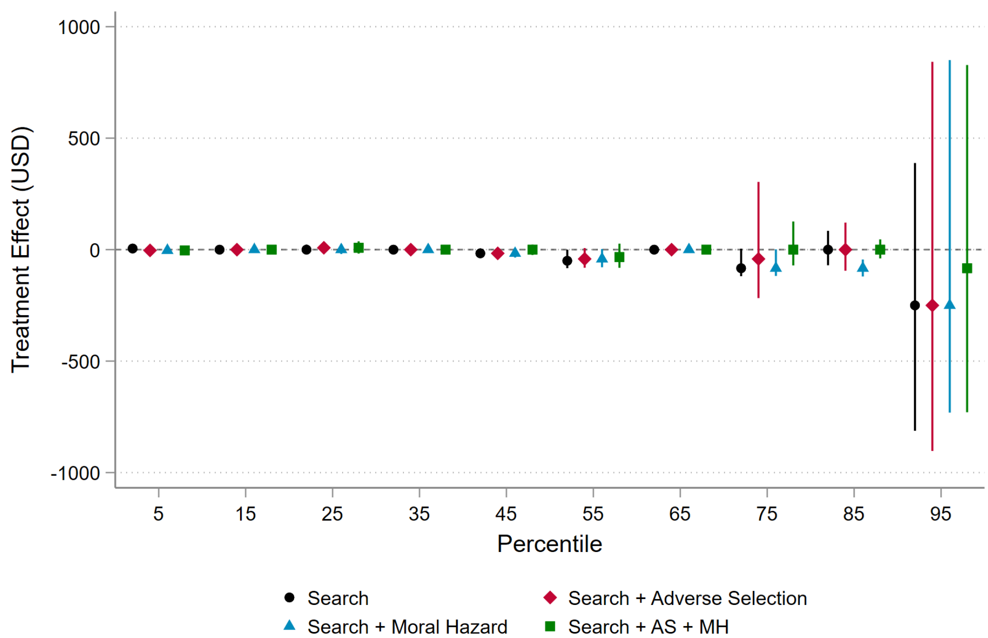
Note: This figure shows the coefficients from regressions of indicators for whether profit and sales are above some threshold t, for a range of t. All regressions include covariates selected by Double Lasso (Belloni, Chernozhukov, and Hansen, 2014), as well as the outcome measured at baseline (if available). The numbers in parentheses show the percentiles at which t is located in the distribution of the pure control group. We plot 95% confidence intervals constructed using randomisation inference, defined as the set of sharp nulls that do not reject at the 5% level, using the procedure in Young (2024).

Figure A8: Treatment Effects on Transition Matrix



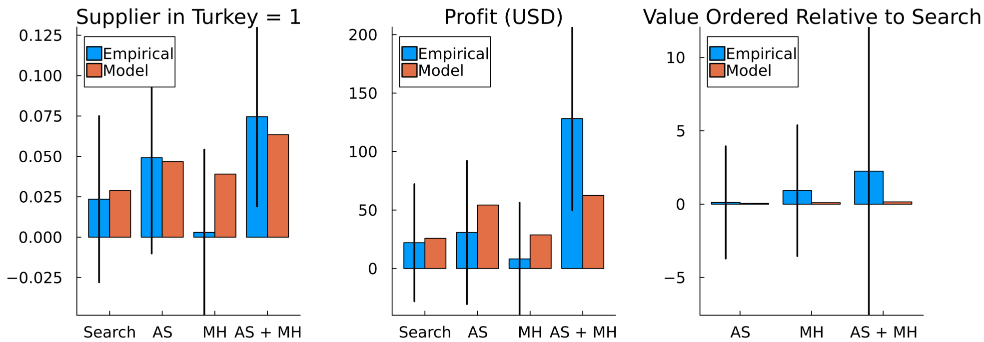
Note: This figure shows the treatment effects for the Search + Adverse Selection + Moral Hazard treatment on indicators for whether a firm transitioned between different percentile profit buckets between baseline and endline.

Figure A9: Placebo Check: Quantile Treatment Effects on Baseline Profit

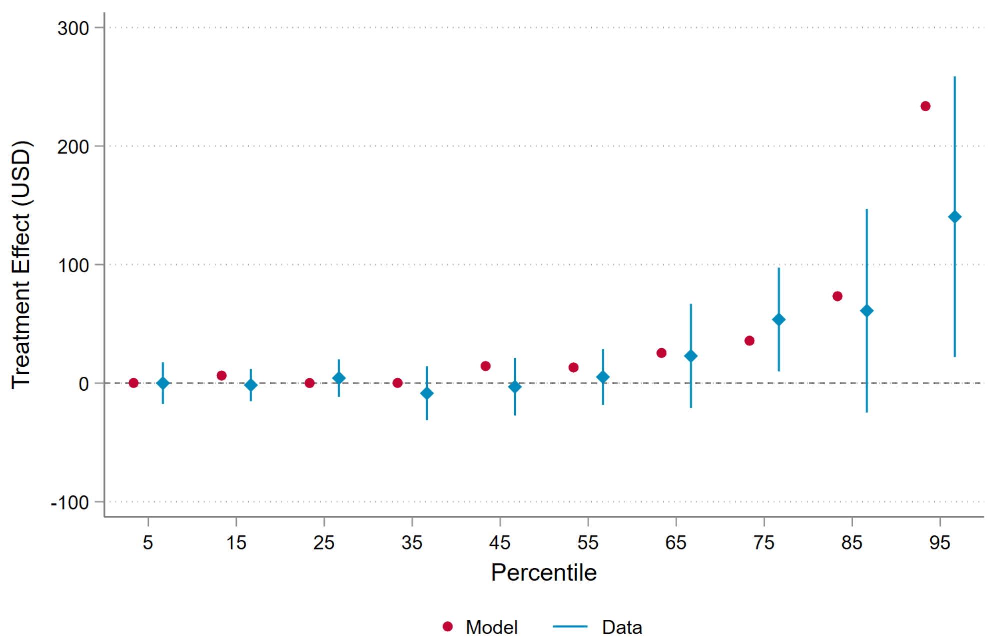


Note: This figure shows the coefficients from quantile regressions of baseline profit on the four treatment groups, intended as a placebo test. As the only covariate included in the main quantile regressions is the outcome measured at baseline, which is itself the outcome here, we do not include any covariates. We plot 95% confidence intervals constructed using randomisation inference, defined as the set of sharp nulls that do not reject at the 5% level.

Figure A10: Model Fit



(a) Fit on Targeted Moments



(b) Fit on Quantile Treatment Effects

Note: Panel A shows the 11 targeted moments in the structural estimation, with confidence internals on the empirical moments (computed by bootstrapping the experimental treatment effects) shown to highlight the relative precisions. We weight moments by the inverse of the variance-covariance matrix of the treatment effects. Panel B compares quantile treatment effects between the model and the data, which were not explicitly targeted moments, for the group with both trust treatment. The Data moments are the same as in Figure 6.

Appendix B - Mathematical Appendix

In this Appendix, we prove further detail on the model. In the main text, we defined the contract as a sequence of quantities, q_t, and prices, p_t. For the purposes of stating and characterizing the solution, it is easier to define the contract as a sequence of quantities, q_t, and transfers, τ_t, such that the firm orders quantity q_t and pays transfer τ_t. This is purely notational and does not affect the solution, as the price can always be backed out as $p_t = \tau_t/q_t$.

Further Details on the Optimal Contract

Assumption 1. *(No trade with bad types)*

$$\max_q \lambda r(q) - (1-\xi)cq < \max_q r(q) - p_l q$$

This assumption states that if a firm knew that the supplier was a bad type, they would prefer to order from the local supplier.

Proposition 1. *It is not optimal to offer a menu of contracts that separates good and bad suppliers.*

Proof. Suppose the contrary, and consider the state of the world where the supplier is a bad type. Since the menu fully separates the types, the firm's posterior is then that the supplier is a bad type with probability 1. Because the supplier has limited liability, the maximum that the firm can earn under any such contract is the full surplus, i.e., $(\max_q \lambda r(q) - (1-\xi)q)/(1-\delta)$. Assumption 1 implies that the firm can always do better than this, because at the very least they can buy from a local supplier in every period. Since the contract is relational, the firm would thus renege before sending the first transfer. Thus, the expected payoff to the bad type from accepting the revealing contract is 0. So long as the contract recommended to the good type involves positive quantity, the bad type can always earn a positive expected payoff by accepting the good type's contract, because limited liability ensures that $\tau_t \geq cq_t > (1-\xi)cq_t$ for all t. Therefore, the bad type would not accept the contract that reveals their type, which is a contradiction. $\square$

The original program is as follows:

$$L = \min_{\{\rho_t\},\{\eta_t\},\{\gamma_t\}} \max_{\{q_t\},\{\tau_t\}} \sum_{t=0}^{\infty} \delta^t \left(1 - \mu_0(1-\lambda^t)\right) \left((1-\mu_t(1-\lambda)r(q_t) - \tau_t + \delta\mu_t(1-\lambda)\bar{U}\right)$$
$$+ \sum_{t=0}^{\infty} \delta^t \rho_t \left[\sum_{\tau=t+1}^{\infty} \delta^{\tau-t}(R_\tau - cq_\tau) - \xi c q_t \right]$$
$$+ \sum_{t=0}^{\infty} \delta^t (1-\mu_0(1-\lambda^t))\eta_t \left[\sum_{\tau=t}^{\infty} \delta^{\tau-t} \left(1 - \mu_\tau(1-\lambda^{\tau-t})\right) \left((1-\mu_\tau(1-\lambda)r(q_t) - \tau_t + \delta\mu_t(1-\lambda)\bar{U}\right) - \bar{U} \right]$$
$$+ \sum_{t=0}^{\infty} \delta^t \gamma_t \left[\tau_t - cq_t\right]$$

The modified program is as follows.

$$\begin{aligned} W_t(U_t, V_t, \mu_t) = {} & y(q_t, \tau_t, \mu_t) + \rho_t\left(\delta(1-\lambda)V_{t+1} - \xi c q_{t+1}\right) \\ & + \eta_t\left(U_{t+1} - \bar{U}\right) \\ & + \gamma_t\left(\tau_t - cq_t\right) \\ & + \nu_t^b\left(y(q_t,\tau_t,\mu_t) + \delta(1-\mu_t(1-\lambda))U_t + \delta\mu_t(1-\lambda)\bar{U} - U_t\right) \\ & + \nu_t^s\left(\tau_t - cq_t + \delta V_{t+1} - V_t\right) \\ & + \delta(1-\mu_t(1-\lambda))W_{t+1}(U_{t+1}, V_{t+1}, \mu_{t+1}) \end{aligned}$$

The FOCs are as follows

$$(1-\mu_t(1-\lambda))r'(q_t)(1+\nu_t^b) = \rho_t \xi c + \left(\gamma_t + \nu_t^s\right)c \qquad (q_t)$$
$$1 + \nu_t^b = \gamma_t + \nu_t^s \qquad (\tau_t)$$
$$\rho_t(1-\lambda) + \nu_t^s = -(1-\mu_t(1-\lambda))\frac{\partial W_{t+1}}{\partial V_{t+1}} \qquad (V_{t+1})$$
$$\eta_{t+1} + \delta(1-\mu_t(1-\lambda))\nu_t^b = -\delta(1-\mu_t(1-\lambda))\frac{\partial W_{t+1}}{\partial U_{t+1}} \qquad (U_{t+1})$$

Substituting the FOC for τ_t into the FOC for q_t and re-arranging gives

$$r'(q_t) = \frac{1}{1-\mu_t(1-\lambda)}\left(1 + \xi\frac{\rho_t}{1+\nu_t^b}\right)c.$$

The Envelope Condition implies that $\frac{\partial W_{t+1}}{\partial V_{t+1}} = -\nu_{t+1}^s$ and $\frac{\partial W_{t+1}}{\partial U_{t+1}} = -\nu_{t+1}^b$. Combining FOCs 2-4 then gives the following equation relating γ_t and γ_{t+1} that we will make extensive use of in the following proofs.

$$\rho_t(1-\lambda) + 1 - \gamma_t + \mu_t(1-\lambda)\nu_t^b = (1-\gamma_{t+1})(1-\mu_t(1-\lambda)) + \frac{\eta_{t+1}}{\delta}. \tag{4}$$

Remark 1. $\nu_{t+1}^b > \nu_t^b \iff \eta_{t+1} > 0$

Proof. This follows immediately from the FOC for U_{t+1} after substituting in the Envelope Condition. □

The optimal contract is generally not available in closed form, but we state and prove some properties in the following proposition.

Proposition 2. *There exists finite T^* such that:*

1. *The agent earns zero stage profits for all $t < T^*$ (i.e., LL binds for all $t < T^*$).*
2. *The principal earns $(1-\delta)\bar{U}$, that is, zero stage profits net of their outside option, for all $t > T^*$ (i.e., DEC binds for all $t > T^*$).*
3. *q_t is strictly increasing for $t < T^*$.*
4. *ICC binds for at least some $t < T^*$.*

Proof. We prove this through a series of Lemmas, which we state and prove below. For parts 1 and 2, see Lemma 3. For part 3, see Lemma 1. For part 4, see Lemma 5. □

Before formally stating the Lemmas, we first provide an intuitive sketch of the approach. First, we show that q_t must be strictly increasing whenever LL binds. Intuitively, a (weakly) decreasing q_t despite beliefs improving would imply that ICC strongly becomes "more binding" over time. But when LL is binding, the supplier is earning zero stage profits, so the ICC must be getting less binding over time.

Second, we show that if LL in t binds, then DEC in t cannot bind. Intuitively, both parties cannot be earning their outside option at the same time, as belief improving and q_t growing would imply that in other periods one of them must be making a loss.

Third, we show that the problem can be divided into two phases: LL will bind for all early periods and be slack for all late periods. "Backloading" results of this kind are standard in the dynamic moral hazard literature. Backloading happens for two reasons. The first reason is that incentives must be given to the agent at some point, and backloading incentives is efficient because it improves both early and late ICCs (whereas frontloading or even-loading still improves early ICCs but improves late ICCs less). The second reason is that adverse selection means that the "good type" agent is more patient than the principal, as the good type knows their own type. This means that it is always cheaper for the principal to backload payments. As a corollary, the DEC must bind for all late periods, as the principal wants to backload as much as possible, and will continue to do so until the DEC binds.

Finally, we show that ICC must bind for some t in the early section. Intuitively, if it didn't, then the principal would just extend the early period–where they earn all the surplus–for longer. The only reason to ever end this early phase is precisely because an ICC eventually binds (the principal has to pay the agent eventually, and in absence of the ICC would always prefer not to).

Lemma 1. *If LL binds in $t+1$, then $q_{t+1} > q_t$.*

Proof. Suppose that LL binds in $t+1$ and $q_{t+1} \leq q_t$. The FOC for q_t is

$$r'(q_t) = \frac{1}{1-\mu_t(1-\lambda)}\left(1+\xi\frac{\rho_t}{1+\nu_t^b}\right)c.$$

We already know that $\nu_{t+1}^b \geq \nu_t^b$, and that $\mu_{t+1} \leq \mu_t$. Then, the hypothesis $q_{t+1} \leq q_t$ implies that $\rho_{t+1} > \rho_t$. Then, we can write

$$\delta(1-\lambda)V_{t+1} < (1-\lambda)V_{t+1} = (1-\lambda)\left(R_{t+1} - cq_{t+1} + \delta V_{t+2}\right) = \delta(1-\lambda)V_{t+2} = \xi cq_{t+1} \leq \xi cq_t.$$

The first equality is a definition, the second equality follows from the fact that LL binds in $t+1$, the third equality follows from the fact that $\rho_{t+1} > \rho_t$ implies that $\rho_{t+1} > 0$, which implies that ICC binds in $t+1$. The final inequality follows from the hypothesis that $q_{t+1} \leq q_t$. Thus, we have shown that

$$\delta(1-\lambda)V_{t+1} < \xi cq_t.$$

But this implies that ICC in t is violated, which is a contradiction. □

Lemma 2. *If LL in $t+1$ binds, then DEC in $t+1$ is slack.*

Proof. Suppose that DEC in $t+1$ binds. Then, we can write

$$\begin{aligned}
U_t &= (1-\mu_t(1-\lambda))r(q_t) - \tau_t + \delta(1-\mu_t(1-\lambda))U_{t+1} + \delta\mu_t(1-\lambda)\bar{U} \\
&= (1-\mu_t(1-\lambda))r(q_t) - \tau_t + \delta\bar{U} \\
&\leq (1-\mu_t(1-\lambda))r(q_t) - cq_t + \delta\bar{U} \\
&< (1-\mu_{t+1}(1-\lambda))r(q_t) - cq_t + \delta\bar{U} \\
&\leq (1-\mu_{t+1}(1-\lambda))r(q_{t+1}) - cq_{t+1} + \delta\bar{U} \\
&\leq (1-\mu_{t+1}(1-\lambda))r(q_{t+1}) - cq_{t+1} + \delta(1-\mu_{t+1}(1-\lambda))U_{t+2} + \delta\mu_{t+1}(1-\lambda)\bar{U} \\
&= (1-\mu_{t+1}(1-\lambda))r(q_{t+1}) - R_{t+1} + \delta(1-\mu_{t+1}(1-\lambda))U_{t+2} + \delta\mu_{t+1}(1-\lambda)\bar{U} \\
&= U_{t+1} \\
&= \bar{U}.
\end{aligned}$$

The first line is the definition of U_t. The second line follows from DEC binding in $t+1$. The third line follows from LL in t. The fourth line follows from $\mu_{t+1} < \mu_t$. The fifth line follows from Lemma 1. The sixth line follows from DEC in period $t+2$, i.e., $U_{t+2} \geq \bar{U}$. The seventh line follows from the hypothesis that LL binds in $t+1$, i.e., $\tau_t = cq_t$. The eighth line is the definition of U_{t+1}. The ninth line follows from DEC binding in $t+1$.

The above thus establishes that $U_t < \bar{U}$. But this is a violation of DEC in period t, which is a contradiction. □

Corollary 1. *If LL is slack in t, then it is also slack in $t+1$.*

Proof. LL slack in t means $\gamma_t = 0$. Suppose it binds in $t+1$, which means $\gamma_{t+1} > 0$. Lemma 2 then implies that $\eta_{t+1} = 0$. But, Equation (4) gives

$$\gamma_{t+1}(1 - \mu_t(1-\lambda)) = \frac{\eta_{t+1}}{\delta} - \rho_t(1-\lambda) - \mu_t(1-\lambda)(1+\nu_t^b).$$

It must then be that $\eta_{t+1} > 0$, which is a contradiction. □

Corollary 2. *If LL is slack in t, then DEC binds in $t+1$.*

Proof. Corollary 1 implies that $\gamma_t = \gamma_{t+1} = 0$. Then, the FOC implies

$$\frac{\eta_{t+1}}{\delta} = \rho_t(1-\lambda) + \mu_t(1-\lambda)(1+\nu_t^b) \geq \mu_t(1-\lambda)(1+\nu_t^b).$$

The final term is strictly positive, which implies that $\eta_{t+1} > 0$. □

Lemma 3. *If any trade occurs, then there exists finite $T^* \geq 1$ such that (i) LL binds for all $t < T^*$ and is slack for all $t \geq T^*$, and (ii) DEC is slack for all $t < T^*$ and binds for all $t > T^*$.*

Proof. For (i): Corollary 1 shows that $\gamma_t = 0 \implies \gamma_{t+1} = 0$. Thus, if there exists T^* such that $\gamma_t = 0$, then $\gamma_s = 0$ for all $s \geq T^*$. We already know that $\gamma_0 = 1$, so this is not the case for $t = 0$. Suppose that $\gamma_t > 0$ for all t. Then, the supplier earns zero profit, which implies that all ICCs will fail unless $q_t = 0$ for all t, which violates the supposition that trade occurs at some point. Thus, there must be at least one $t \geq 1$ such that LL is slack in t. If there are multiple, define T^* as the earliest such t.

For (ii): Since LL is slack for all $t \geq T^*$, Corollary 2 implies that DEC binds for all $t > T^*$. □

Corollary 3. *$\nu_t^b = 0$ for all $t \leq T^*$, and $\nu_{t+1}^b > 0$ for all $t > T^*$.*

Proof. This follows from Remark 1 and Lemma 3. □

Lemma 4. *$\gamma_{t+1} \leq \gamma_t$, with inequality strict if $\gamma_t \in (0,1)$.*

Proof. If $\gamma_{t+1} = 0$, then this holds trivially. We thus need to establish the claim for $\gamma_{t+1} > 0$. Suppose then that $\gamma_{t+1} > \gamma_t$ with $\gamma_{t+1} > 0$. Then, Equation (4) implies

$$0 > \rho_t(1-\lambda) - \frac{\eta_{t+1}}{\delta} + \mu_t(1-\lambda)(1+\nu_t^b - \gamma_{t+1}).$$

Since $\gamma_{t+1} > 0$, Lemma 2 implies that $\eta_{t+1} = 0$. We are thus left with

$$\begin{aligned} 0 &> \rho_t(1-\lambda) + \mu_t(1-\lambda)(1+\nu_t^b - \gamma_{t+1}) \\ &= \rho_t(1-\lambda) + \mu_t(1-\lambda)(1+\nu_{t+1}^b - \gamma_{t+1}) \\ &= \rho_t(1-\lambda) + \mu_t(1-\lambda)\nu_{t+1}^s \end{aligned}$$

where the first equality follows from the fact that the FOC for U_{t+1} implies that $\eta_{t+1} = 0 \implies \nu_t^b = \nu_{t+1}^b$, and the second equality follows from the FOC for R_{t+1}. But the RHS is weakly positive, so this is a contradiction, which establishes that $\gamma_{t+1} \leq \gamma_t$.

Then, to establish the claim about strict inequality, suppose instead that $\gamma_{t+1} = \gamma_t$. We instead have

$$0 = \rho_t(1-\lambda) + \mu_t(1-\lambda)\nu_{t+1}^s,$$

which is only possible if $\rho_t = \nu_{t+1}^s = 0$. But this implies that $\gamma_{t+1} = 1 + \nu_t^b \geq 1$. If $\gamma_t \in (0,1)$, this implies $\gamma_{t+1} > \gamma_t$, which is a contradiction. □

Lemma 5. *ICC binds for some $t < T^*$.*

Proof. We prove this by iterating forward Equation (4). Since $\eta_t = \nu_t^b = 0$ for all $t < T^*$, the equation can be written

$$\gamma_t = \mu_t(1-\lambda) + \rho_t(1-\lambda) + \gamma_{t+1}(1-\mu_t(1-\lambda)).$$

Starting with $\gamma_0 = 1$ and iterating this until $T^* - 1$, for which $\gamma_{t+1} = 0$, we get

$$1 + (1-\mu_0(1-\lambda^{T^*}))\frac{\eta_{T^*}}{\delta} = \sum_{t=0}^{T^*-1}(1-\mu_0(1-\lambda^t))[\mu_t(1-\lambda) + \rho_t(1-\lambda)].$$

Note that $LHS \geq 1$. The first term on the RHS simplies to

$$\sum_{t=0}^{T^*-1}(1-\mu_0(1-\lambda^t))\mu_t(1-\lambda) = (1-\lambda)\mu_0\sum_{t=0}^{T^*-1}\lambda^t = \mu_0(1-\lambda^{T^*}) < 1.$$

Thus, it cannot be that $\rho_t = 0$ for all $t \leq T^* - 1$, as otherwise $LHS > RHS$. So there must be some $t \leq T^*$ for which the ICC binds. □

Proposition 3. *The period-0 value of the relationship is decreasing in μ_0 and ξ.*

Proof. We first show the case for ξ. Using the Envelope Theorem to differentiate the original infinite horizon Lagrangian with respect to ξ gives

$$\frac{dL}{d\xi} = -\sum_{t=0}^{\infty} \delta^t \rho_t c q_t.$$

Since Proposition 2 established that the DICC binds for at least some t, we know that $\rho_t > 0$ for at least some t. So this is strictly negative except for the trivial case where $q_t = 0$ for all t, in which case it is zero.

Second, for μ_0, it is easier to work with the recursive formulation. Differentiating the period-0 program gives

$$\frac{dW_0}{d\mu_0} = \underbrace{-(1-\lambda)r(q_0)(1+\nu_0^b) - \delta(1-\lambda)\nu_0^b(U_1 - \bar{U}) - \delta(1-\lambda)W_1}_{\equiv \phi_0} + \delta(1-\mu_0(1-\lambda))\frac{d\mu_1}{d\mu_0}\frac{\partial W_1}{\partial \mu_1}.$$

The first three terms are all negative. The only term that we cannot immediately sign is $\partial W_1/\partial \mu_1$. However, since the problem is recursive, we can simply repeatedly lead ϕ_t to obtain

$$\frac{dW_0}{d\mu_0} = \sum_{t=0}^{\infty} \delta^t \prod_{s=0}^{t} (1-\mu_s(1-\lambda))\frac{d\mu_{s+1}}{d\mu_s}\phi_t.$$

This is strictly negative, except in the trivial case where the firm never purchases anything with the supplier abroad. □

Proposition 4. *The buyer's period 0 value satisfies* $\partial^2 U_0(\mu_0, \xi)/\partial \mu_0 \partial \xi > 0$*.*

Proof. This simply comes from the fact that the stage payoff satisfies increasing differences. It is easiest to work directly with the SPFE as we can then use standard contraction mapping arguments. As described in detail in Appendix E, the problem can be rewritten as

$$\begin{aligned} W(\zeta_t^b, \zeta_t^s, \mu_t) = \min_{\eta_t, \rho_t, \gamma_t} \max_{q_t, \tau_t} \quad & (\zeta_t^b + \eta_t) h_0^b(q_t, \tau_t, \mu_t) + \eta_t h_1^b(q_t, \tau_t, \mu_t) \\ & + (\zeta_t^s + \gamma_t) h_0^s(q_t, \tau_t) + \rho_t h_1^s(q_t, \tau_t) + \delta W(\zeta_{t+1}^b, \zeta_{t+1}^s, \mu_{t+1}), \end{aligned}$$

with

$$\begin{aligned} h_0^b(q_t, \tau_t, \mu_t) &\equiv \left[(1-\mu_t(1-\lambda))r(q_t) + \mu_t(1-\lambda)r(0) - \tau_t + \delta\mu_t(1-\lambda)\bar{U}\right] \\ h_1^b(q_t, \tau_t, \mu_t) &\equiv -\bar{U} \\ h_0^s(q_t, \tau_t) &\equiv \tau_t - c q_t \\ h_1^s(q_t, \tau_t) &\equiv -\xi c q_t \end{aligned}$$

subject to

$$\zeta^b_{t+1} = (\zeta^b_t + \eta_t)(1 - \mu_t(1 - \lambda))$$
$$\zeta^s_{t+1} = \zeta^s_t + \rho_t,$$

and the evolution of μ_t.

The stage payoff function satisfies weak increasing differences in $(-\mu, -\xi)$ because the cross-partial of these terms is zero (in particular, the ICC constraint does not contain μ). However, for strictness, note that the payoff function has increasing differences in $(-\mu, q)$, so q^* is increasing in $-\mu$, and the marginal value of lowering ξ is also proportional to q. Then, since ξ does not affect the transition function for μ and the feasible sets are parameter invariant, and since the stage payoff satisfies the regularity conditions for standard minimax theorems, standard contraction-mapping arguments establish that the value function inherits increasing differences in $(-\mu, -\xi)$. Since the value function in period 0 is equal to the buyer's utility as $\zeta^b_0 = 1$ and $\zeta^s_0 = 0$, and the function is continuous, we thus have $\partial U_0(\mu_0, \xi)^2/\partial\mu_0\partial\xi > 0$.

□

Further Details on the Search Process

In this sub-section, we prove (1) that the sequential search process behaves according to the standard results despite the fact that the outside option, $\bar{U}$, enters the utility function directly and (2) that if there is a firm-specific heterogeneity term z, then there exists a cutoff $\bar{z}$ such that only firms with $z > \bar{z}$ engage in search.

The search process works as follows. To match with a random supplier abroad, a firm can pay a one-time search cost $s > 0$. Upon matching, both parties immediately observe a realisation of a match-specific productivity term, $\psi \sim G$, that is fixed over time. We denote the period-0 value of a relationship with a supplier abroad with match-specific productivity ψ as $U_0(\psi, \bar{U})$, where we include $\bar{U}$ to make explicit that U_0 depends on the outside option directly. It does so through two channels. First, the adverse selection problem means that the firm will eventually earn $\bar{U}$ on path if the supplier is a bad type (because a low quality realization will eventually occur), which will cause U_0 to be increasing in $\bar{U}$. Second, the limited commitment problem means that the firm cannot make promises that would result in earning less than $\bar{U}$ (i.e., the DEC), which limits the enforceable surplus–this causes U_0 to be decreasing in $\bar{U}$. Since this is direct dependence on the outside option is non-standard in sequential search models, we prove the standard results here.

Proposition 5. *The optimal policy in the search process is a cutoff rule, where the cutoff value equals the expected utility from the search process.*

Proof. First, note that the search problem is recursive. That is, the entire history of searching is summarized by the highest value of ψ drawn to date, which, we denote with ψ_b. Second, U_0 is monotonic in ψ, strictly so in the region where the firm develops a relationship. As is standard, these together imply that the optimal policy is to search until the firm receives a draw above an endogenous cutoff $\bar{\psi}$.

For a firm whose best current draw is ψ_t, the expected benefit from obtaining one more draw is

$$\int_{\psi}^{\infty}(U_0(\psi,\bar{U})-U_0(\psi_b,\bar{U}))dG(\psi),$$

where $\bar{U}$ is the expected utility of terminating the current relationship and re-searching. The expected cost is s. By definition, the cutoff value of $\bar{\psi}$ equates expected benefits with expected costs,

$$\int_{\bar{\psi}}^{\infty}(U_0(\psi,\bar{U})-U_0(\bar{\psi},\bar{U}))dG(\psi)=s.$$

As the stopping time is geometrically distributed, the expected number of draws until the firm obtains a realisation above $\bar{\psi}$ is $1/(1-G(\bar{\psi}))$. This means that the expected utility from engaging in search, $\bar{U}$, is

$$\bar{U}=\frac{\int_{\bar{\psi}}^{\infty}U_0(\psi,\bar{U})dG(\psi)}{1-G(\bar{\psi})}-\frac{s}{1-G(\bar{\psi})}.$$

Re-arranging, we can obtain

$$\begin{aligned}s&=\int_{\bar{\psi}}^{\infty}U_0(\psi,\bar{U})dG(\psi)-\bar{U}\int_{\bar{\psi}}^{\infty}dG(\psi)\\ s&=\int_{\bar{\psi}}^{\infty}(U_0(\psi,\bar{U})-\bar{U})dG(\psi),\end{aligned}$$

which, when compared with the definition of the cutoff, implies that $\bar{U}=U_0(\bar{\psi},\bar{U})$. Thus, despite the fact that $\bar{U}$ enters the utility function directly, it does not change the fundamental structure of the search process. □

Proposition 6. *The cutoff $\bar{\psi}$ is decreasing in s.*

Proof. It is easier to eliminate the $\bar{\psi}$ in the integral limit by rewriting the cutoff rule definition as follows

$$\int_{-\infty}^{\infty}\max\{U_0(\psi,\bar{U})-\bar{U},0\}dG(\psi)=s,$$

where $\bar{U}=U_0(\bar{\psi},\bar{U})$. Note that

$$\frac{ds}{d\bar{U}}=\int_{-\infty}^{\infty}\left(\frac{\partial U_0(\psi,\bar{U})}{\partial\bar{U}}-1\right)1\{U_0(\psi,\bar{U})-\bar{U}>0\}dG(\psi).$$

In standard sequential search models, the first term in parentheses is zero, so this is clearly negative. Here, we will show that this new term is always strictly smaller than 1, which is sufficient to establish the claim.

We know that the U_0 is given by the following dynamic program (and we know that a solution exists for the particular draw of ψ under consideration as we only need to consider the cases where $U_0(\psi, \bar{U}) - \bar{U} > 0$),

$$
\begin{aligned}
L = \min_{\{\rho_t\},\{\eta_t\},\{\gamma_t\}} \max_{\{q_t\},\{\tau_t\}} &\sum_{t=0}^{\infty} \delta^t \left(1 - \mu_0(1 - \lambda^t)\right) \left((1 - \mu_t(1 - \lambda)r(\psi q_t) - \tau_t + \delta\mu_t(1 - \lambda)\bar{U}\right) \\
&+ \sum_{t=0}^{\infty} \delta^t \rho_t \left[\sum_{\tau=t+1}^{\infty} \delta^{\tau-t}(R_\tau - cq_\tau) - \xi c q_t\right] \\
&+ \sum_{t=0}^{\infty} \delta^t (1 - \mu_0(1 - \lambda^t))\eta_t \left[\sum_{\tau=t}^{\infty} \delta^{\tau-t} \left(1 - \mu_\tau(1 - \lambda^{\tau-t})\right) \left((1 - \mu_\tau(1 - \lambda)r(\psi q_t) - \tau_t + \delta\mu_\tau(1 - \lambda)\bar{U}\right) - \bar{U}\right] \\
&+ \sum_{t=0}^{\infty} \delta^t \gamma_t \left[\tau_t - cq_t\right].
\end{aligned}
$$

The Envelope Theorem implies that $\frac{dL}{d\bar{U}}$ is given by

$$
\sum_{t=0}^{\infty} \delta^t \left(1 - \mu_0(1 - \lambda^t)\right) \delta\mu_t(1-\lambda) + \sum_{t=0}^{\infty} \delta^t (1-\mu_0(1-\lambda^t))\eta_t \left[\sum_{\tau=t}^{\infty} \delta^{\tau-t} \left(1 - \mu_\tau(1 - \lambda^{\tau-t})\right) \delta\mu_\tau(1 - \lambda) - 1\right].
$$

Since Bayesian updating implies that we can write $\mu_t = \mu_0\lambda^t/(1 - \mu_0(1 - \lambda^t))$, we can rewrite the first term as $\mu_0\delta(1 - \lambda)\sum_{t=0}^{\infty} \delta^t\lambda^t$, or $\frac{\mu_0\delta(1-\lambda)}{1-\delta\lambda}$, which is strictly less than 1. Intuitively, this is a (discounted) sum of probabilities and thus cannot exceed 1–indeed, it is easy to see that if we set δ to 1 then this simplifies to just μ_0. The first term is therefore strictly less than 1. The second term is negative, because the first term in square brackets is strictly less than 1 by the exact same argument. Since this argument did not rely on the value of ψ, we have that $dL/d\bar{U} < 1$ for all ψ for which $U_0(\psi, \bar{U}) - \bar{U} > 0$, which implies that

$$
\frac{ds}{d\bar{U}} = \int_{-\infty}^{\infty} \left(\frac{\partial U_0(\psi, \bar{U})}{\partial \bar{U}} - 1\right) 1\{U_0(\psi, \bar{U}) - \bar{U} > 0\} dG(\psi) < 0.
$$

□

Note that a firm-specific multiplicative utility shifter that takes the form is equivalent to firms having heterogeneous search costs s/z as we simply divide through the cutoff definition equation by z. This proposition then implies that the cutoff utility is increasing in z, and thus that if there is a "global" outside option (from not searching at all) U_l, then there will exist some $\bar{z}$ such that the cutoff $\bar{U}(\bar{z})$ exactly equals U_l. In particular, the cutoff is defined such that,

$$
\int_{-\infty}^{\infty} \max\{U_0(\psi, U_l) - U_l, 0\} dG(\psi) = s/\bar{z}.
$$

Lemma 6. *The cutoff $\bar{z}$ is increasing in μ_0 and ξ.*

Proof. Since Proposition 3 establishes that U_0 is decreasing in μ_0 and ξ, the LHS of the equation defining $\bar{z}$ is decreasing in μ_0 and ξ, which means that $\bar{z}$ must be increasing in μ_0 and ξ to ensure that the RHS decreases. □

Appendix C - Heterogeneity Analysis

Table C1: Heterogeneity by Retailer vs Wholesaler

	Horizontal Dummy (1)	High Quality Dummy (2)	Any Supp in Türkiye (3)	Profit Sales Index (1%) (4)
Search Only	0.147***	0.158	0.021	0.004
	(0.043)	(0.097)	(0.034)	(0.104)
Search + AS	0.031	0.221**	0.053	0.013
	(0.042)	(0.105)	(0.036)	(0.121)
Search + MH	0.097**	0.119	-0.006	0.004
	(0.042)	(0.100)	(0.033)	(0.091)
Search + AS + MH	0.005	0.058	0.054	0.024
	(0.043)	(0.110)	(0.037)	(0.103)
S Only * Wholesaler	-0.105	-0.071	-0.011	0.469
	(0.075)	(0.194)	(0.067)	(0.324)
S + AS * Wholesaler	0.061	-0.179	0.001	0.255
	(0.073)	(0.193)	(0.067)	(0.323)
S + MH * Wholesaler	0.013	-0.139	0.016	-0.144
	(0.075)	(0.190)	(0.066)	(0.231)
S + AS + MH * Wholesaler	0.142*	0.005	0.023	1.209***
	(0.073)	(0.187)	(0.068)	(0.437)
All Inter Zero p-val	0.023	0.783	0.989	0.034
Adjusted R^2	0.01	-0.01	0.00	0.05
N	1859	359	1680	1298

Note: This table shows the main results with treatment interacted with an indicator for whether the firm sells at least partially wholesale, defined as having a positive share of sales that are wholesale to other firms. p-values are computed using randomisation inference. Specifically, we compute the randomised randomisation-t p-value from Young (2019) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. The outcome in Column 1 is the same as Column 1 of Table 1. The outcome in Column 2 is the same as Column 2 of Table 1. The outcome in Column 3 is the same as Column 1 of Table 2. The outcome in Column 4 is the same as Column 6 of Table 4.

Table C2: Heterogeneity by Physical Store vs Online Only

	Horizontal Dummy (1)	High Quality Dummy (2)	Any Supp in Türkiye (3)	Profit Sales Index (1%) (4)
Search Only	0.176***	0.113	0.040	-0.058
	(0.044)	(0.097)	(0.038)	(0.076)
Search + AS	0.091**	0.139	0.065*	-0.017
	(0.043)	(0.104)	(0.039)	(0.098)
Search + MH	0.161***	0.074	0.005	-0.053
	(0.043)	(0.097)	(0.036)	(0.060)
Search + AS + MH	0.098**	0.033	0.081**	-0.026
	(0.044)	(0.104)	(0.040)	(0.066)
S Only * Physical Store	-0.181***	0.090	-0.066	0.653*
	(0.073)	(0.200)	(0.056)	(0.337)
S + AS * Physical Store	-0.135*	0.048	-0.035	0.519
	(0.071)	(0.194)	(0.060)	(0.376)
S + MH * Physical Store	-0.208***	-0.079	-0.036	0.165
	(0.072)	(0.209)	(0.056)	(0.285)
S + AS + MH * Physical Store	-0.120*	0.098	-0.031	1.582***
	(0.072)	(0.185)	(0.061)	(0.497)
All Inter Zero p-val	0.042	0.924	0.836	0.028
Adjusted R^2	0.03	-0.01	0.02	0.12
N	1859	359	1680	1298

Note: This table shows the main results with treatment interacted with an indicator for whether the firm had a physical store at baseline. p-values are computed using randomisation inference. Specifically, we compute the randomised randomisation-t p-value from Young (2019) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. The outcome in Column 1 is the same as Column 1 of Table 1. The outcome in Column 2 is the same as Column 2 of Table 1. The outcome in Column 3 is the same as Column 1 of Table 2. The outcome in Column 4 is the same as Column 6 of Table 4.

Table C3: Heterogeneity by Direct Import Experience

	Horizontal Dummy (1)	High Quality Dummy (2)	Any Supp in Türkiye (3)	Profit Sales Index (1%) (4)
Search Only	0.126***	0.114	0.050	0.074
	(0.044)	(0.104)	(0.030)	(0.121)
Search + AS	0.059	0.120	0.070**	0.008
	(0.043)	(0.106)	(0.031)	(0.102)
Search + MH	0.085**	0.045	0.024	-0.135*
	(0.043)	(0.106)	(0.028)	(0.075)
Search + AS + MH	0.053	0.023	0.072**	0.416**
	(0.043)	(0.108)	(0.031)	(0.192)
S Only * Experience	-0.033	0.076	-0.104	0.180
	(0.074)	(0.177)	(0.068)	(0.278)
S + AS * Experience	-0.030	0.110	-0.053	0.246
	(0.073)	(0.190)	(0.069)	(0.329)
S + MH * Experience	0.042	0.091	-0.080	0.285
	(0.074)	(0.178)	(0.067)	(0.230)
S + AS + MH * Experience	0.006	0.114	-0.014	0.319
	(0.074)	(0.179)	(0.072)	(0.430)
All Inter Zero p-val	0.840	0.968	0.515	0.762
Adjusted R^2	0.00	-0.01	0.05	0.02
N	1859	359	1680	1298

Note: This table shows the main results with treatment interacted with an indicator for whether the firm has direct online importing experience, defined as having bought goods from another country online directly at least once in the past 12 months. p-values are computed using randomisation inference. Specifically, we compute the randomised randomisation-t p-value from Young (2019) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. The outcome in Column 1 is the same as Column 1 of Table 1. The outcome in Column 2 is the same as Column 2 of Table 1. The outcome in Column 3 is the same as Column 1 of Table 2. The outcome in Column 4 is the same as Column 6 of Table 4.

Table C4: Heterogeneity by Baseline Turkish Share

	Horizontal Dummy	High Quality Dummy	Any Supp in Türkiye	Profit Sales Index (1%)
	(1)	(2)	(3)	(4)
Search Only	0.059	0.180	0.024	0.038
	(0.056)	(0.145)	(0.043)	(0.173)
Search + AS	-0.010	0.121	0.052	-0.033
	(0.054)	(0.155)	(0.044)	(0.176)
Search + MH	0.026	0.200	-0.020	-0.168
	(0.054)	(0.150)	(0.038)	(0.125)
Search + AS + MH	0.026	0.354**	0.041	0.255
	(0.056)	(0.141)	(0.045)	(0.261)
S Only * High Share	0.099	-0.053	-0.012	0.161
	(0.073)	(0.178)	(0.059)	(0.236)
S + AS * High Share	0.098	0.046	0.015	0.204
	(0.071)	(0.187)	(0.061)	(0.248)
S + MH * High Share	0.109	-0.177	0.023	0.214
	(0.072)	(0.182)	(0.056)	(0.178)
S + AS + MH * High Share	0.045	-0.427**	0.037	0.420
	(0.073)	(0.176)	(0.062)	(0.359)
All Inter Zero p-val	0.471	0.073	0.940	0.692
Adjusted R^2	0.01	0.01	0.01	0.01
N	1814	353	1636	1261

Note: This table shows the main results with treatment interacted with an indicator for whether the share of the firm's customers in the past 30 days that buy Turkish-made products is above the median. p-values are computed using randomisation inference. Specifically, we compute the randomised randomisation-t p-value from Young (2019) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. The outcome in Column 1 is the same as Column 1 of Table 1. The outcome in Column 2 is the same as Column 2 of Table 1. The outcome in Column 3 is the same as Column 1 of Table 2. The outcome in Column 4 is the same as Column 6 of Table 4.

Table C5: Heterogeneity by Sex of Owner

	Horizontal Dummy	High Quality Dummy	Any Supp in Türkiye	Profit Sales Index (1%)
	(1)	(2)	(3)	(4)
Search Only	0.020	0.111	-0.028	0.439**
	(0.050)	(0.128)	(0.037)	(0.216)
Search + AS	-0.045	0.148	-0.010	0.240
	(0.048)	(0.127)	(0.038)	(0.206)
Search + MH	0.026	-0.033	-0.056	0.005
	(0.050)	(0.120)	(0.036)	(0.152)
Search + AS + MH	-0.035	0.040	0.039	0.904***
	(0.048)	(0.122)	(0.041)	(0.321)
S Only * Female	0.189***	0.052	0.090	-0.574**
	(0.070)	(0.173)	(0.058)	(0.232)
S + AS * Female	0.191***	0.017	0.134**	-0.313
	(0.069)	(0.177)	(0.060)	(0.242)
S + MH * Female	0.143**	0.200	0.097*	-0.047
	(0.070)	(0.170)	(0.056)	(0.182)
S + AS + MH * Female	0.185***	0.046	0.060	-0.822**
	(0.070)	(0.173)	(0.062)	(0.343)
All Inter Zero p-val	0.021	0.768	0.222	0.029
Adjusted R^2	0.02	-0.01	0.02	0.05
N	1859	359	1680	1298

Note: This table shows the main results with treatment interacted with an indicator for whether the firm owner is female. p-values are computed using randomisation inference. Specifically, we compute the randomised randomisation-t p-value from Young (2019) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. The outcome in Column 1 is the same as Column 1 of Table 1. The outcome in Column 2 is the same as Column 2 of Table 1. The outcome in Column 3 is the same as Column 1 of Table 2. The outcome in Column 4 is the same as Column 6 of Table 4.

Table C6: Heterogeneity by Baseline Turkish Supplier Group

	Horizontal Dummy	High Quality Dummy	Any Supp in Türkiye	Profit Sales Index (1%)
	(1)	(2)	(3)	(4)
Search Only	0.107***	0.184*	0.053**	0.177
	(0.040)	(0.097)	(0.027)	(0.127)
Search + AS	0.033	0.245**	0.054**	0.196
	(0.039)	(0.099)	(0.028)	(0.148)
Search + MH	0.092**	0.133	0.028	-0.044
	(0.040)	(0.096)	(0.026)	(0.093)
Search + AS + MH	0.043	0.125	0.095***	0.438**
	(0.039)	(0.100)	(0.030)	(0.182)
S Only * In Turkish Group	0.029	-0.173	-0.144	-0.221
	(0.088)	(0.191)	(0.089)	(0.314)
S + AS * In Turkish Group	0.024	-0.295	-0.033	-0.547**
	(0.086)	(0.210)	(0.088)	(0.239)
S + MH * In Turkish Group	0.039	-0.221	-0.116	0.042
	(0.088)	(0.200)	(0.088)	(0.267)
S + AS + MH * In Turkish Group	0.055	-0.237	-0.126	0.421
	(0.087)	(0.192)	(0.090)	(0.564)
All Inter Zero p-val	0.983	0.656	0.415	0.111
Adjusted R^2	0.01	-0.00	0.10	0.01
N	1840	357	1662	1286

Note: This table shows the main results with treatment interacted with an indicator for whether the firm was in a supplier WhatsApp group with a supplier based in Türkiye at baseline. p-values are computed using randomisation inference. Specifically, we compute the randomised randomisation-t p-value from Young (2019) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. The outcome in Column 1 is the same as Column 1 of Table 1. The outcome in Column 2 is the same as Column 2 of Table 1. The outcome in Column 3 is the same as Column 1 of Table 2. The outcome in Column 4 is the same as Column 6 of Table 4.

Appendix D - Robustness Checks

We pre-specified that we would use the regression specification specified in Equation (2). Nonetheless, in this section, we run analysis using additional specifications to verify that results are not driven by this (pre-specified) choice. We first replicate all of the main tables in the analysis using the following simpler regression specification that does not include any covariates:

$$y_i = \alpha + \sum_{j=1}^{4} \beta_j T_{ji} + \varepsilon_i,$$

where, as before, Pure Control is the omitted category.

Then, we run a specification that is the same as Equation (2) but also includes supplier fixed effects. Specifically, suppliers were divided into 10 buckets of 3 suppliers each, and then firms were randomly allocated among the buckets that matched their sub-sector. We therefore include fixed effects for the bucket that a firm was assigned to. This is not defined for the Pure Control group as they were not matched with any suppliers, so we must exclude them from the regression (indeed, the indicator for Pure Control is perfectly collinear with bucket fixed effects). We thus present regressions for the main outcomes where Search Only is the omitted category (and Pure Control is dropped altogether), both with and without bucket fixed effects, in Table D5.

Table D1: Access to Foreign Goods (No Covariates)

	Horizontal	Vertical				Price
	Has Product $\geq$ 3 Criteria	High Qual Dummy	Qual Score (/50)	Made in Turkey	Index	Price (USD)
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Pooled						
Treatment	0.080***	0.107	-0.636	0.145**	0.379***	1.656**
	(0.027)	(0.068)	(0.591)	(0.068)	(0.138)	(0.840)
Panel B: Individual Treatments						
Search Only	0.115***	0.139*	-0.177	0.112	0.389**	2.444**
	(0.035)	(0.084)	(0.683)	(0.081)	(0.169)	(0.991)
	[0.005]	[0.231]	[0.951]	[0.315]	[0.084]	[0.051]
Search + Adverse Selection	0.049	0.154*	-0.026	0.108	0.358**	1.409
	(0.035)	(0.087)	(0.845)	(0.085)	(0.173)	(1.049)
	[0.199]	[0.229]	[0.966]	[0.315]	[0.084]	[0.287]
Search + Moral Hazard	0.100***	0.076	-1.284	0.154*	0.360**	1.746*
	(0.035)	(0.084)	(0.843)	(0.080)	(0.165)	(0.976)
	[0.010]	[0.577]	[0.357]	[0.139]	[0.084]	[0.171]
Search + AS + MH	0.055	0.062	-0.934	0.203**	0.405**	0.932
	(0.035)	(0.085)	(0.846)	(0.081)	(0.176)	(0.994)
	[0.199]	[0.577]	[0.549]	[0.053]	[0.084]	[0.347]
Control Mean	0.307	0.431	43.064	0.477	0.000	19.990
% Increase (Pooled)	26.1%	24.8%	-1.5%	30.4%	N/A	8.3%
All Coefs Equal p-val	0.158	0.609	0.445	0.549	0.992	0.358
Adjusted R^2	0.01	0.00	0.32	0.09	0.00	0.23
N	1859	359	359	361	359	642

Note: p-values are computed using randomisation inference. Specifically, we compute the randomised randomisation-t p-value from Young (2019) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. We also report Romano-Wolf multiple-testing adjusted p-values in square brackets (Romano and Wolf, 2005). To test the null that the trust treatments have no effect, we report at the bottom of the table the p-value for an F-test that all coefficients are equal, computed by permuting the F-statistic. Panel A shows the coefficient from a regression on an indicator that pools all treated groups. Panel B shows the coefficients corresponding to treatment indicators for each of the four treatment arms.

Column 1 is an indicator that is one if the firm has a good that matches at least 3 horizontal criteria. Column 2 is an indicator for whether the good's quality score is above the median product-group quality score. Column 3 is the raw quality score. Column 4 is an indicator for whether the good is made in Turkey, primarily inferred based on the label. See the text for full details of how this outcome is constructed. Column 5 is the Anderson (2008) index combining the vertical outcomes to account for multiple hypothesis testing across outcomes. Column 6 is the price in USD, which is only measured conditional on the firm finding a good matching at least three horizontal criteria.

Table D2: Supplier Relationships (Followup Survey) (No Covariates)

	Regular Suppliers in Türkiye			Previous Suppliers		
	Any Sup in Türkiye	Num Sup in Türkiye	Index	Num Sup Total	Num Sup in Senegal	Ended with Sup
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Pooled						
Treatment	0.035	0.079**	0.121*	-0.032	-0.092	0.070***
	(0.023)	(0.038)	(0.062)	(0.193)	(0.191)	(0.022)
Panel B: Individual Treatments						
Search Only	0.018	0.069	0.085	0.092	0.088	0.062**
	(0.029)	(0.055)	(0.080)	(0.269)	(0.266)	(0.029)
	[0.761]	[0.336]	[0.477]	[0.867]	[0.918]	[0.052]
Search + Adverse Selection	0.056*	0.077*	0.146*	-0.154	-0.236	0.067**
	(0.030)	(0.048)	(0.084)	(0.253)	(0.249)	(0.029)
	[0.150]	[0.245]	[0.214]	[0.867]	[0.722]	[0.050]
Search + Moral Hazard	-0.001	-0.009	0.009	-0.197	-0.168	0.091***
	(0.029)	(0.043)	(0.078)	(0.248)	(0.244)	(0.029)
	[0.971]	[0.829]	[0.910]	[0.832]	[0.825]	[0.006]
Search + AS + MH	0.069**	0.183***	0.249***	0.139	-0.049	0.057**
	(0.031)	(0.062)	(0.086)	(0.258)	(0.251)	(0.029)
	[0.085]	[0.007]	[0.016]	[0.867]	[0.918]	[0.052]
Control Mean	0.167	0.222	0.000	3.700	3.213	0.135
% Increase (Pooled)	21.0%	35.6%	N/A	-0.9%	-2.9%	51.9%
All Coefs Equal p-val	0.072	0.012	0.050	0.487	0.642	0.707
Adjusted R^2	0.00	0.01	0.01	-0.00	-0.00	0.00
N	1680	1680	1680	1681	1681	1671

Note: p-values are computed using randomisation inference. Specifically, we compute the randomised randomisation-t p-value from Young (2019) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. We also report Romano-Wolf multiple-testing adjusted p-values in square brackets (Romano and Wolf, 2005). To test the null that the trust treatments have no effect, we report at the bottom of the table the p-value for an F-test that all coefficients are equal, computed by permuting the F-statistic. Panel A shows the coefficient from a regression on an indicator that pools all treated groups. Panel B shows the coefficients corresponding to treatment indicators for each of the four treatment arms.

Column 1 is 1 if the firm says that they have a regular supplier in Türkiye. Column 2 is the number of regular suppliers in Türkiye. Column 3 is the Anderson (2008) index combining the previous two columns to account for multiple hypothesis testing across outcomes. Column 4 is the total number of regular suppliers. Column 5 is the number of regular suppliers in Senegal. Column 6 is 1 if the firm has ended a relationship with a regular supplier in the past 3 months. A regular supplier is defined as a supplier from whom the firm has made two or more orders with the intention to continue the relationship.

Table D3: Order Value (Mobile Money Data) (No Covariates)

	Entire Period			Post Mystery Shopping		
	Total Order Value (USD)	Num Weeks Ordered	Avg Order Value (USD)	Total Order Value (USD)	Num Weeks Ordered	Avg Order Value (USD)
	(1)	(2)	(3)	(4)	(5)	(6)
Search + Adverse Selection	3.58	-0.03	15.05	2.56	-0.06	26.22
	(5.20)	(0.11)	(10.00)	(4.13)	(0.10)	(17.91)
	[0.930]	[0.990]	[0.454]	[0.806]	[0.915]	[0.450]
Search + Moral Hazard	4.67	0.02	9.95	5.45	0.04	15.04
	(6.05)	(0.12)	(7.82)	(5.24)	(0.11)	(9.74)
	[0.930]	[0.990]	[0.549]	[0.680]	[0.929]	[0.450]
Search + AS + MH	12.63	-0.00	35.66	15.79	0.02	58.61
	(19.04)	(0.15)	(28.71)	(18.92)	(0.14)	(32.68)
	[0.930]	[0.990]	[0.549]	[0.806]	[0.929]	[0.378]
Search Only Mean	14.31	0.36	39.24	7.92	0.23	33.95
% Increase (AS)	25.0%	-8.1%	38.4%	32.3%	-25.4%	77.2%
% Increase (MH)	32.6%	5.8%	25.3%	68.9%	17.0%	44.3%
% Increase (AS+MH)	88.3%	-1.4%	90.9%	199.3%	9.8%	172.6%
All Coefs Zero p-val	0.894	0.989	0.371	0.730	0.863	0.238
Adjusted R^2	-0.00	-0.00	0.02	-0.00	-0.00	0.06
N	1500	1500	538	1500	1500	351

Note: p-values are computed using randomisation inference. Specifically, we compute the randomised randomisation-t p-value from Young (2019) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. We also report Romano-Wolf multiple-testing adjusted p-values in square brackets (Romano and Wolf, 2005). To test the null that the trust treatments have no effect, we report at the bottom of the table the p-value for an F-test that all coefficients are equal, computed by permuting the F-statistic. This table shows the coefficients corresponding to treatment indicators for each of the three treatment groups with trust treatments, where Search Only is the omitted category.

Column 1 is the total value of transfers to study suppliers. Column 2 is the total number of weeks in which a firm made a transfer to a study supplier. Column 3 is the mean value transferred to study suppliers in weeks with a positive transaction value. Column 4 is the total value of transfers to study suppliers, excluding the mystery shopping period. Column 5 is the total number of weeks in which a firm made a transfer to a study supplier, excluding the mystery shopping period. Column 6 is the mean value transferred to study suppliers in weeks with a positive transaction value, excluding the mystery shopping period. Mystery shopping generally took place during within 2-3 weeks of the firm entering the study. The data covers up to 22 months after being added to the study. Columns 3and 6are at the firm-week level, with standard errors clustered by firm. All values are in USD.

Table D4: Profit and Sales (No Covariates)

	Raw			Winsorized (1%)		
	Profit (USD)	Sales (USD)	Index	Profit (USD)	Sales (USD)	Index
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Pooled						
Treatment	57.3*	204.0*	0.311**	24.1	83.0	0.172**
	(28.8)	(108.1)	(0.101)	(20.9)	(83.6)	(0.075)
Panel B: Individual Treatments						
Search Only	.566	254	.194*	-5.26	107	.208*
	(29.1)	(187)	(.114)	(26.2)	(125)	(.108)
	[0.984]	[0.377]	[0.203]	[0.847]	[0.582]	[0.132]
Search + Adverse Selection	30.6	56	.194	18.4	38.1	.124
	(36.4)	(119)	(.122)	(31.5)	(111)	(.099)
	[0.644]	[0.660]	[0.204]	[0.775]	[0.725]	[0.352]
Search + Moral Hazard	-27.4	-103	.037	-27.4	-121	6.7e-03
	(24.8)	(103)	(.082)	(24.8)	(93.6)	(.078)
	[0.565]	[0.516]	[0.702]	[0.543]	[0.394]	[0.935]
Search + AS + MH	244***	640**	.873***	120***	325**	.367***
	(93.2)	(282)	(.31)	(45.2)	(152)	(.135)
	[0.011]	[0.050]	[0.004]	[0.019]	[0.089]	[0.027]
Control Mean	188.3	609.5	0.000	188.3	609.5	0.000
% Increase (Pooled)	30.4%	33.5%	N/A	12.8%	13.6%	N/A
All Coefs Equal p-val	0.032	0.027	0.036	0.022	0.018	0.024
Adjusted R^2	0.01	0.01	0.01	0.01	0.01	0.01
N	1351	1378	1298	1351	1378	1298

Note: p-values are computed using randomisation inference. Specifically, we compute the randomised randomisation-t p-value from Young (2019) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. We also report Romano-Wolf multiple-testing adjusted p-values in square brackets (Romano and Wolf, 2005). To test the null that the trust treatments have no effect, we report at the bottom of the table the p-value for an F-test that all coefficients are equal, computed by permuting the F-statistic. Panel A shows the coefficient from a regression on an indicator that pools all treated groups. Panel B shows the coefficients corresponding to treatment indicators for each of the four treatment arms.

Column 1 is total profit from the past 30 days in USD. Column 2 is total sales from the past 30 days in USD. Column 3 is the Anderson (2008) index combining the previous two columns to account for multiple hypothesis testing across outcomes. Column 4 is total profit from the past 30 days in USD, winsorizing the top 1%. Column 5 is total sales from the past 30 days in USD, winsorizing the top 1%. Column 6 is the Anderson (2008) index combining the previous two columns to account for multiple hypothesis testing across outcomes. Profit is measured using the survey question from De Mel, McKenzie, and Woodruff (2009). Sales is measured using a similar survey question.

Table D5: Robustness to Supplier Fixed Effects

	Horizontal Dummy		High Quality Dummy		Any Supp in Türkiye		Ord Value (MoMo)		Profit Sales Index	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Treatments vs Search Only										
Search + Adverse Selection	-.068*	-.076**	.049	.052	.03	.029	4.18	2.43	-.078	-.07
	(.035)	(.035)	(.085)	(.089)	(.029)	(.03)	(5.28)	(5.52)	(.118)	(.118)
	[0.132]	[0.083]	[0.562]	[0.605]	[0.479]	[0.522]	[0.910]	[0.973]	[0.508]	[0.552]
Search + Moral Hazard	-.025	-.027	-.087	-.071	-.017	-.017	4.79	3.89	-.202**	-.196*
	(.035)	(.035)	(.086)	(.088)	(.027)	(.028)	(6.31)	(6.55)	(.102)	(.103)
	[0.464]	[0.435]	[0.485]	[0.605]	[0.528]	[0.552]	[0.910]	[0.973]	[0.124]	[0.147]
Search + AS + MH	-.061*	-.062*	-.107	-.107	.054*	.053*	12.9	12.1	.165	.154
	(.035)	(.035)	(.089)	(.091)	(.029)	(.03)	(19.1)	(18.5)	(.153)	(.152)
	[0.143]	[0.138]	[0.485]	[0.496]	[0.156]	[0.172]	[0.910]	[0.973]	[0.466]	[0.514]
Search Only Mean	0.422	0.422	0.564	0.564	0.185	0.185	14.31	14.31	0.208	0.208
Supplier Bucket FEs	N	Y	N	Y	N	Y	N	Y	N	Y
All Coefs Zero p-val	0.195	0.131	0.234	0.309	0.066	0.086	0.879	0.956	0.025	0.026
Adjusted R^2	0.05	0.05	0.01	0.03	0.14	0.14	0.01	0.01	-0.03	-0.02
N	1498	1498	292	291	1338	1338	1500	1500	1032	1032

Note: p-values are computed using randomisation inference. Specifically, we compute the randomised randomisation-t p-value from Young (2019) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. We also report Romano-Wolf multiple-testing adjusted p-values in square brackets (Romano and Wolf, 2005). To test the null that the trust treatments have no effect, we report at the bottom of the table the p-value for an F-test that all coefficients are equal, computed by permuting the F-statistic. Panel A shows the coefficient from a regression on an indicator that pools all trust treated groups, where Search Only is the omitted category. Panel B shows the coefficients corresponding to treatment indicators for each of the three treatment groups with trust treatments. All regressions include any covariates selected by Double Lasso (Belloni, Chernozhukov, and Hansen, 2014), as well as stratum fixed effects and the outcome measured at baseline (if available).

Column 1 is the same as Column 1 of Table 1. Column 2 is the same as the previous column but includes supplier bucket fixed effects. Column 3 is the same as Column 2 of Table 1. Column 4 is the same as the previous column but includes supplier bucket fixed effects. Column 5 is the same as Column 1 of Table 2. Column 6 is the same as the previous column but includes supplier bucket fixed effects. Column 7 is the same as Column 4 of Table 3. Column 8 is the same as the previous column but with supplier bucket fixed effects. Column 9 is the same as Column 3 of Table 4. Column 10 is the same as the previous column but with supplier bucket fixed effects. The suppliers were divided into 10 buckets of 3 suppliers each, and firms were randomly allocated across buckets that matched their sub-sector. The regressions in this table exclude Pure Control as they were not matched with any suppliers and so supplier bucket fixed effects are not well-defined. All comparisons are therefore with reference to Search Only.

Appendix E - Further Details on Solving and Estimating the Model

In this Appendix, we provide more detail on our algorithm to numerically solve and estimate the model.

Solving the Model

As is often the case in dynamic optimisation, the original infinite horizon program is very difficult to work with directly. It is much more tractable to find a way to work with a recursive formulation. The challenge is that, unlike standard dynamic problems encountered in macro, we have two constraints (the DEC and DICC) that are forward looking, and, in particular, forward-looking to an infinite horizon.

We follow the approach of Marcet and Marimon (2019). The idea is that the original Lagrangean can be rewritten recursively as a pseudo planner's problem, where the Pareto weights are state variables that evolve endogenously to completely summarise historical binding constraints. Intuitively, if the DICC is binding in period 0, which implies that the agent must be delivered a certain amount of utils at some point in the future, the Pareto weight on the agent increases over time to ensure that the planner delivers precisely the required amount of utils. This allows the problem to be written recursively because the principal can trade off the benefit of making a constraint "more binding" today against the cost of increasing next period's Pareto weight on the agent. The recursive formulation delivers a Saddle Point Functional Equation, which is analogous to the familiar Bellman Equation but for saddle point problems, which satisfies a number of familiar properties that permit the use of dynamic programming techniques.

We first rewrite our problem as a recursive Lagrangean and thus derive the Saddle Point Functional Equation. To avoid cumbersome algebra and focus on the core intuition of the method, the derivations that follow assume that the firm can only purchase from one supplier at a time. We write the full version, allowing for purchases from both a new and existing supplier, at the end. Define $y(q_t, \tau_t, \mu_t) \equiv (1 - \mu_t(1-\lambda))r(q_t) + \mu_t(1-\lambda)r(0) - \tau_t$. Then, the original dynamic program is as follows

$$\begin{aligned}
\max_{\{q_t\},\{\tau_t\}} & y(q_0, \tau_0, \mu_0) + \delta\mu_0(1-\lambda)\bar{U} \\
& + \delta(1-\mu_0(1-\lambda))[y(q_1, \tau_1, \mu_1) + \delta\mu_1(1-\lambda)\bar{U} \\
& + \delta(1-\mu_1(1-\lambda))[y(q_2, \tau_2, \mu_2) + \delta\mu_2(1-\lambda)\bar{U} \\
& + \delta(1-\mu_2(1-\lambda))[\ldots
\end{aligned}$$

subject to

$$\sum_{n=1}^{\infty} \delta^n(\tau_{t+n} - cq_{t+n}) \geq \xi c q_t \qquad \forall t$$
$$U_t \geq \bar{U} \qquad \forall t$$
$$\tau_t \geq c q_t \qquad \forall t,$$

and with μ_t evolving according to Bayes' Rule. Rewriting the objective function as an infinite sum and including the constraints with Lagrange multipliers, the program can be expressed as follows

$$\begin{aligned} L = \min_{\{\rho_t\},\{\eta_t\},\{\gamma_t\}} \max_{\{q_t\},\{\tau_t\}} & \sum_{t=0}^{\infty} \delta^t \left(1 - \mu_0(1-\lambda^t)\right) \left((1 - \mu_t(1-\lambda)r(q_t) + \mu_t(1-\lambda)r(0) - \tau_t + \delta\mu_t(1-\lambda)\bar{U}\right) \\ & + \sum_{t=0}^{\infty} \delta^t \rho_t \left[\sum_{s=t+1}^{\infty} \delta^{s-t}(\tau_s - cq_s s) - \xi c q_t \right] \\ & + \sum_{t=0}^{\infty} \delta^t (1 - \mu_0(1-\lambda^t))\eta_t \left[\sum_{s=t}^{\infty} \delta^{s-t} \left(1 - \mu_s(1-\lambda^{s-t})\right) \left((1 - \mu_s(1-\lambda)r(q_s) + \mu_s(1-\lambda)r(0) - \tau_s + \delta\mu_s(1 - \right.\right. \\ & + \sum_{t=0}^{\infty} \delta^t \gamma_t \left[\tau_t - cq_t.\right] \end{aligned}$$

Then, with some algebra, we can collect the Lagrange terms directly inside the first infinite sum to express the Lagrangean as a function of "Pareto weights", Lagrange multipliers, state variables, and time-invariant functions.

$$\begin{aligned} L = \min_{\{\rho_t\},\{\eta_t\},\{\gamma_t\}} \max_{\{q_t\},\{\tau_t\}} \sum_{t=0}^{\infty} \delta^t \big[& (\zeta_t^b + \eta_t) h_0^b(q_t, \tau_t, \mu_t) + \eta_t h_1^b(q_t, \tau_t, \mu_t) \\ & + (\zeta_t^s + \gamma_t) h_0^s(q_t, \tau_t) + \rho_t h_1^s(q_t, \tau_t) \big] \end{aligned}$$

where $\zeta_{t+1}^b \equiv (\zeta_t^b + \eta_t)(1 - \mu_t(1-\lambda))$, $\zeta_{t+1}^s \equiv \zeta_t^s + \rho_t$, and

$$\begin{aligned} h_0^b(q_t, \tau_t, \mu_t) &\equiv \left[(1 - \mu_t(1-\lambda))r(q_t) + \mu_t(1-\lambda)r(0) - \tau_t + \delta\mu_t(1-\lambda)\bar{U}\right] \\ h_1^b(q_t, \tau_t, \mu_t) &\equiv -\bar{U} \\ h_0^s(q_t, \tau_t) &\equiv \tau_t - cq_t \\ h_1^s(q_t, \tau_t) &\equiv -\xi c q_t \end{aligned}$$

ζ_t^b and ζ_t^s behave like Pareto weights and are equal to the sum of all prior Lagrange multipliers from the forward-looking constraints for the principal (the buyer) and the agent (the seller), respectively. This is the sense in which they fully summarise the shadow cost of constraints from earlier periods of the problem. For example, if DICC is "very binding" in early periods, meaning ρ_t is large for early t, then this is reflected in a large value of ζ_t^s for later t, which causes the "planner" to endogenously choose a high value of τ_t and thus give the agent utility. Note that there is a small difference relative to how the weights evolve in Marcet and Marimon (2019) as the buyer's weight here includes the term $(1-\mu_t(1-\lambda))$ in the evolution, which appears because the Adverse Selection problem means that we only reach the next period in the relationship with probability $(1-\mu_t(1-\lambda))$.

Distinguishing between the foreign good, q_t, and the existing supplier q_{et}, we can then write this recursively as a Saddle Point Functional Equation (SPFE) as follows,

$$\begin{aligned} W(\zeta_t^b, \zeta_t^s, \mu_t) = \min_{\eta_t, \rho_t, \gamma_t} \max_{q_t, q_{et}, \tau_t} \quad & (\zeta_t^b + \eta_t) h_0^b(q_t, q_{et}, \tau_t, \mu_t) + \eta_t h_1^b(q_t, \tau_t, \mu_t) \\ & + (\zeta_t^s + \gamma_t) h_0^s(q_t, \tau_t) + \rho_t h_1^s(q_t, \tau_t) + \delta W(\zeta_{t+1}^b, \zeta_{t+1}^s, \mu_{t+1}), \end{aligned}$$

with

$$\begin{aligned} h_0^b(q_t, \tau_t, \mu_t) &\equiv \left[(1-\mu_t(1-\lambda)) r(q_t, q_{et}) + \mu_t(1-\lambda) r(0, q_{et}) - \tau_t - p_{et} q_{et} + \delta \mu_t (1-\lambda) \bar{U}\right] \\ h_1^b(q_t, \tau_t, \mu_t) &\equiv -\bar{U} \\ h_0^s(q_t, \tau_t) &\equiv \tau_t - c q_t \\ h_1^s(q_t, \tau_t) &\equiv -\xi c q_t \end{aligned}$$

subject to

$$\begin{aligned} \zeta_{t+1}^b &= (\zeta_t^b + \eta_t)(1-\mu_t(1-\lambda)) \\ \zeta_{t+1}^s &= \zeta_t^s + \rho_t, \end{aligned}$$

and the evolution of μ_t. For the evolution of μ_t, to prevent a discontinuity in learning when going from $q_t = 0$ to any positive value, while also ensuring the principal's choice set remains convex, we impose a technical assumption that for q_t below some small $\underline{q}$, posteriors after observing a high quality realisation are given by $\mu_{t+1} = \mu_{t+1}^{\text{Bayes}} + (\mu_t - \mu_{t+1}^{\text{Bayes}}) \cdot (1 - q_t/\underline{q})^p$, where μ_{t+1}^{Bayes} is the posterior implied by Bayes' Rule and $p \in (1, 2)$. We set $p = 1.4$ as numbers around this value tended to perform well in numerical experiments. Making learning a continuous function of q_t ensures the problem satisfies appropriate regularity conditions as it makes the buyer's choice set convex. Intuitively, it captures the idea of a minimum order size (which is the case in our empirical setting), and we set $\underline{q}$ to 6 as packets of 6 are a very common minimum order size in our empirical setting. This learning function makes little difference in practice as it only applies in rare cases where the optimal q is below $\underline{q}$, but it is important to ensure the problem satisfies the appropriate regularity conditions.

The Saddle Point Functional Equation is analogous to the Bellman Equation for saddle point problems, and Marcet and Marimon (2019) prove that–under some regularity conditions–

it has the usual desirable properties associated with dynamic programming problems. In particular, this means that we can obtain solutions to the original problem by using dynamic programming techniques to solve for the value function and policy functions associated with the SPFE.

We solve the model by iterating on the value function, W. Since W is homogeneous of degree one in the Pareto weights, we can write $W(\zeta_t^b, \zeta_t^s, \mu_t) = \zeta_t^b W(1, \frac{\zeta_t^s}{\zeta_t^b}, \mu_t)$. This means that we can eliminate one state variable. We define a discrete grid over the two state variables and linearly interpolate over the grid.[41] Within each iteration, we solve the $\min \max$ as follows. First, the FOC with respect to τ_t gives implies $\zeta_t^b + \eta_t = \zeta_t^s + \gamma_t$. Since we know from Appendix B that we will never have $\eta_t > 0$ and $\gamma_t > 0$ simultaneously, it must then be that $\eta_t = \max(\zeta_t^s - \zeta_t^b, 0)$ and $\gamma_t = \max(\zeta_t^b - \zeta_t^s, 0)$. We can then obtain τ_t as either cq_t if $\gamma_t > 0$ or $(1-\mu_t(1-\lambda))r(q_t, q_e) + \mu_t(1-\lambda)r(0, q_e) + \delta\mu_t(1-\lambda)\bar{U} - p_e q_e + \delta(1-\mu_t(1-\lambda))\frac{\partial W}{\partial \zeta_{t+1}^b} - \bar{U}$ if $\eta_t > 0$. (We discuss the edge-case where $\eta_t = \gamma_t = 0$ below). To obtain q_t, q_{et}, for a given ρ_t we can solve for q_t and q_{et} analytically using the FOCs if $q_t > \underline{q}$; otherwise we have to solve for them numerically as a term $\frac{\partial W}{\partial \mu_{t+1}}$ appears in this region as the choice of q affects μ_{t+1} through the smoothed learning function. The FOC with respect to ρ_t includes $\frac{\partial W}{\partial \zeta_{t+1}^s}$, which must be evaluated numerically and thus we cannot solve for ρ_t analytically. We thus solve for it numerically in an outer loop, with the solver for q_t, q_{et} in the inner loop.

Once we have obtained the value function, we iterate forward from the initial conditions at $t = 0$ to get the solution path. Since the agent's utility is linear in τ_t, the value function has a kink in the neighbourhood of $\zeta_t^b = \zeta_t^s$. This kink is also the point at which the surplus shifts from the buyer to the seller, and the period where it occurs is exactly $T^* - 1$ (see Appendix B), where τ_t is interior. We resolve this by doing the following. First, since $\pi_t = 0$ for all $t < T^* - 1$, we know that $V_0 = \delta^t V_t$ for any $t < T^* - 1$. Starting with $t = 1$, we can calculate $V_0 - \delta^t V_t$ for each t, and find $T^* - 1$ by checking when this object switches sign. Second, once we know $T^* - 1$, we iterate forward normally until $T^* - 2$, at which point we set ρ_{T^*-2} to be such that $\zeta_{T^*-1}^b = \zeta_{T^*-1}^s$ and we calculate τ_{T^*-1} as the value that ensures that promises are consistent.

Estimation

The above sub-section describes how we solve the model for a given guess of the parameters. In order to estimate the parameters, we need to find the parameters that best match the empirical moments, and thus solve the model for many combinations of parameters. We do this in two steps. First, we solve the model for a grid of $(\mu_0, \xi, c, \delta, \lambda)$, where c is the cost of supplier abroad (varying c and ψ are isomorphic and one can simply map between them by setting $\psi = 1/c$), and linearly interpolate over the grid. This gives *functions* for all of the relevant theoretical objects, such as $y_t(\mu_0, \xi, c, \delta, \lambda, \bar{U})$, which means that we do not need to further solve the model as we can simply evaluate these functions at a given $(\mu_0, \xi, c, \delta, \lambda)$. The model converges for 99.6% of the parameter combinations, and

[41] We experimented with both linear interpolation and cubic splines, and generally found that linear interpolation performed better.

we impute the values for the combinations that do not converge using a simple nearest-neighbour means algorithm.

Second, we use the above functions for estimation. We define 11 theoretical moments. The first four are the treatment effects on winsorized profit, i.e., the coefficients in Column 4 of Table 4. The next there are the post-mystery shopping total value ordered treatment effects in Column 1 of Table 3. For these three moments, because the mobile money data does not capture all transactions (and is likely top-censored), we do not match the coefficients directly but rather the relative treatment effects—i.e., the coefficients divided by the control mean reported in the table. The next four moments are the four treatment effects on the survey outcome of whether a firm has a regular supplier in Turkey, i.e., the coefficients in Column 1 in Table 2. This gives 11 treatment effect moments which provide exogenous variation to identify the parameters governing the frictions. We calculate the variance-covariance matrix of these 11 empirical moments by drawing 1,000 bootstrap sub-samples of the experimental data and use the inverse of this as the weighting matrix.

As the moments only identify differences, we ensure that the model matches the control means through the choice of the z distribution and the ψ_μ parameter. For z, since our assumption on the baseline equilibrium is that each firm can be represented as having one supplier, the model implies that profit for a firm i with one input is given by $\pi_i = z_i^\sigma \frac{(\sigma-1)^{\sigma-1}}{\sigma^\sigma} p_i^{1-\sigma}$, where p_i is their input price. The input price is a function of z: firms with $z < \bar{z}$ have p_l; other firms have an endogenous price determined by their reservation utility from the search game (which is a function of z). We thus have a many-to-one function that maps z onto p, which implies that there is a one-to-one mapping from z to π given by $\pi(z) = z^\sigma \frac{(\sigma-1)^{\sigma-1}}{\sigma^\sigma} p(z)^{1-\sigma}$. We thus draw at random from the empirical π_i distribution of the control group and invert this mapping to obtain an empirical distribution of z that rationalizes the baseline control distribution (and thus control mean). As firms often report profit in round numbers, there are several mass points in this distribution, so to improve numerical stability in the optimization routine, whenever we draw z we jitter it by adding $u \sim U[-1, 1]$. For ψ_μ, note that the share of firms importing in the control group is determined by the share of firms with $z > \bar{z}$. Thus, for a given guess of the parameters, we can solve for ψ_μ that sets this share to match the control mean. Intuitively, for any guess of the parameters, we can always shift the location of the distribution of match-specific productivities of the suppliers until exactly a share d falls above the cutoff.

The two dimensions of heterogeneity are the firm-specific productivities, z, and the match-specific productivities, ψ. For distribution of match-specific productivity, we directly estimate the parameters of the LogNormal, (ψ_μ, ψ_σ). We winsorize the upper tail such that $c = 1/\psi$ never falls below 2. Note that the absolute level is not meaningful—the model is homogenous of degree $1 - \sigma$ in c and p_l, so all that matters is the ratio c/p_l, and since we calibrate $p_l = 12.29$, a relative price of less than one sixth is already very low. We do this both because heavy tails make estimation sensitive to implementation details of the SMM and because the value function becomes very large as c becomes small (causing numerical issues with convergence). We actually compute the moments by Monte Carlo integration using $\Omega = 10,000$ draws of ψ.[42]

As this is a non-stationary dynamic model, the questions of what is baseline and how changes in the environment affect existing relationships are non-trivial. We assume that the baseline equilibrium is the long-run steady state of the model. That means that all firms with $z \geq \bar{z}$, where $\bar{z}$ is the endogenous cutoff above which a firm will engage in search for a supplier, have already searched and are in the "terminal" phase of the relationship. Recall from Section 3 that the optimal contract features a unique T^* such that the firm earns exactly their outside option (which is the maximum of re-searching or buying locally forever) for all $t > T^*$, which we refer to as the terminal phase. Thus, all firms with $z \leq \bar{z}$ earn $U_l = \frac{(\sigma-1)^{\sigma-1}}{\sigma^\sigma} z^\sigma p_l^{1-\sigma}/(1-\delta)$, while all firms with $z > \bar{z}$ earn $\bar{U}(z) > U_l$, where $\bar{U}(z)$ is the expected value of participating in the search game, defined implicitly by $\int_\psi \max\{U(\psi, z) - \bar{U}(z), 0\} dF(\psi) = s$. For simplicity, we simply assume that these firms earning $\bar{U}(z)$ also have access to a technology that allows them to frictionlessly purchase at a constant price $p_e(z)$ defined such that $\bar{U}(z) = \frac{(\sigma-1)^{\sigma-1}}{\sigma^\sigma} z^\sigma p_e(z)^{1-\sigma}/(1-\delta)$, i.e., the constant price that gives the same utility as they are currently earning. This simplifies the model as they can then combine orders from this technology with orders from the new supplier without us having to take a stand on how or when they might renegotiate their existing dynamic contract. Thus, the baseline is defined by firms with $z < \bar{z}$ having existing price $p_e = p_l$, while firms with $z > \bar{z}$ have existing price $p_e(z) < p_l$. Using the fact that the model is homogeneous of degree $1-\sigma$ in p_e and c, we can then evaluate the interpolated objects from the grid of solved model runs.

For the theoretical moments, we calculate the theoretical profit in each treatment group, with the treatments implemented as described in the main text. For profit, we calculate the mean profit in the first three periods, $\frac{1}{3}\sum_{t=0}^{2} y_t$, as the survey asked about profits over the past 30 days and typically took place within 2-3 months after recruitment. For the binary outcome relationships, we set this to 1 if they already have an existing supplier abroad (i.e., firms with $z > \bar{z}$) and 0 if they have $z < \bar{z}$ and they choose $q_{f0}(c_{\text{min}}) = 0$, where c_{min} is the minimum of the three free cost draws from the search treatment (or, equivalently, the maximum of the three productivity draws, $\psi = 1/c$). For firms with $q_{f0} > 0$ and $z < \bar{z}$, we set the moment equal to the probability that the relationship reaches the third period (as a function of μ_0). Finally, for mobile money, we calculate $\sum_{t=3}^{18} \tau_t$, reflecting the total value ordered after mystery shopping finished (and, as discussed above, we divide the treatment means by the control–i.e., search only–mean). The final moments are shown in Figure A10.

[42] We experimented with different numbers of draws, and found that the value of the objective function becomes fairly stable across seeds starting at around Ω of 5,000.

Appendix F - Survey and Protocol Details

F.1 Firm Eligibility Criteria

To ensure the relevance of the study, we administered a short screening survey with the following eligibility criteria. Firm owners were included if they: (i) managed their own orders with suppliers; (ii) averaged at least five ready-to-wear sales per week over the past 30 days; (iii) had access to WhatsApp; and (iv) had used WhatsApp to sell to clients in the past 30 days, including through posts or status updates. Respondents who did not meet these criteria were thanked for their interest but deemed ineligible to participate.

Among those who agreed to participate, 85% of online sellers and 89% of physical-store sellers met the eligibility criteria. This indicates that our final sample represents a substantial share of these two key populations of interest, while being appropriately filtered to focus on active garment retailers engaged in a minimum level of digital commerce.

F.2 Further Detail on Supplier Recruitment, Eligibility Criteria, and Matching

Suppliers were recruited in three stages. First, we hired two surveyors in Istanbul to conduct a listing of Senegalese suppliers in Aksaray and Laleli, two important quarters with a large presence of exporters and export infrastructure to Francophone West Africa. The listing was a short survey that asked questions necessary to assess inclusion criteria for the study. The inclusion criteria were the following: a supplier must (1) sell ready-to-wear garments manufactured in Turkey, (2) live in Istanbul, (3) visit Senegal in person no more than once per year on average, (4) report making at least 5 sales per week via WhatsApp on average, (5) be willing to make a WhatsApp group to advertise their products for the purpose of the study. The listing was a short survey that asked the questions necessary to assess the inclusion criteria. Suppliers with physical stores were listed by a census, and suppliers without physical stores were listed by being approached in cargo stores (businesses that manage transportation logistics for exporting).

Second, among listed suppliers who met the inclusion criteria, we conducted a mystery shopping exercise 3-4 months later. We hired and trained a team of 6 firms in Senegal to make a total of 2 separate orders at each supplier (one order of 2 dresses, and another order of 2 polo shirts). The firms did not mention the study, and, if asked, said that they received the supplier's number from a friend. As our target sample size was 30 suppliers, we rolled out this mystery shopping among the 89 suppliers that met the inclusion criteria in waves (with order randomized), inviting suppliers that were sufficiently active in responding to the mystery shoppers to join the study, until closing recruitment once the target sample size was reached. The purpose of this exercise was to screen out suppliers that would not engage with new customers or were otherwise unresponsive, as such suppliers would not be relevant for the study. If any supplier had scammed us in this exercise then we would have excluded them from the study, too, but this did not happen in practice (perhaps because the reputational cost of doing so would greatly exceed the gain for orders of this size).

Third, we required participating suppliers to agree to respect a few benign conditions governing appropriate use of the supplier WhatsApp group (posting regularly in the group, posting items of varying quality levels, posting only items within their chosen sector, including prices with their posts), which almost all suppliers agreed to.

We divided the 30 suppliers into 10 buckets of 3 suppliers each. Three buckets corresponded to men's clothing, five buckets corresponded to women's clothing, and two buckets corresponded to shoes and accessories. In the Search treatment, which matched treated firms to suppliers, we randomly allocated treated firms to a bucket that matched their sub-sector. That is, for example, a firm whose main sub-sector is women's clothing was randomly allocated among the five women's clothing buckets. This means that suppliers should be balanced across the four treatment arms, and, in line with this, we show in Table D5 that controlling for supplier fixed effects makes no difference to the results.

F.3 Moral Hazard Treatment Details

The Moral Hazard treatment, as the model highlights, requires either shifting the suppliers' incentives or shifting firms' perceptions of suppliers' incentives. To do this, we read the information to treated firms at the end of the baseline survey to emphasize that study suppliers have strong incentives:

> I have one last piece of information to give you. As you know, you have been added to WhatsApp groups of Senegalese suppliers in Turkey.
>
> We work with many suppliers in our study. We want to assure you that they are motivated.
>
> We would like to collect feedback on these suppliers so that we can recommend the best ones in the future. To do this, we will ask the merchants in the study [such as yourself] to rate your experience with the suppliers we have presented to you on a scale of 1 to 5 on product arrival and quality. These reviews help identify the best suppliers, which is beneficial to them and allows us to continue recommending them to others. They are therefore motivated.
>
> If a supplier gets bad ratings, we will investigate and remove them from the study if they did not do a good job. They will therefore lose access to around 150 merchants if they do not do a good job.
>
> I will give you a phone number that you can use to give your rating or report a problem.
>
> Lastly, I want to emphasise that the suppliers are aware that they are being rated and that, if they receive bad ratings, they will be removed from the study. We can thus assure you that they are motivated.

F.4 Mystery Shopping Protocol

To evaluate firms' access to high-quality and differentiated foreign goods, we implemented a structured mystery shopping protocol. Surveyors contacted firms over WhatsApp posing as genuine customers interested in purchasing garments for an upcoming event. The goal was to simulate realistic sourcing and negotiation behavior while standardizing the interaction across sellers.

Initial contact. Two weeks after the baseline survey, trained surveyors messaged each firm over WhatsApp, referencing a product image and asking whether the firm had something similar. Firms had previously been told that they might be contacted by potential customers who were seeking high-quality goods.

Horizontal match criteria. The referenced product images were designed to include five horizontal characteristics that are orthogonal to quality (e.g., color, presence of graphic, sleeve style, neckline). If the firm proposed a good that met at least three of the five criteria, this was coded as a horizontal match. This is the primary outcome for the horizontal dimension.

Negotiation protocol. If a horizontal match was found, then the mystery shopper proceeded to negotiating the price and delivery details. All price interactions followed a standardized script:

- Surveyors first asked the seller for a quote (including delivery).
- They then proposed a counteroffer equal to 75% of the quoted price (rounded down to the nearest 500 FCFA), capped at 20,000 FCFA.
- The seller's final offer was recorded, along with whether they accepted payment on delivery or required an advance (and how much).

Purchase and fallback procedure. If a match was found, the seller agreed to a price at or below the 20,000 FCFA cap, and the seller accepted either 50% or full payment on delivery, the mystery shopper proceeded to buy the good in a randomly selected 80% of cases (due to our budget constraints). In the 20% of cases where the random assignment indicated to not buy the good, they explained that their plans had changed, and offered a nominal payment of about 2 USD (1,000 FCFA) as a gesture of appreciation. This fallback protocol was piloted to ensure natural exit and avoid upsetting sellers.

Rapid-order test. In a second stage, surveyors sent an image from a randomized set of "priority products" and asked if the seller could find the item within 24 hours. This assessed the firm's ability to respond quickly to buyer demand or to test new suppliers.

Secondary Outcomes. Surveyors recorded whether the merchant added them to WhatsApp status, if delivery date was confirmed, and if the merchant mentioned quality or reputation. They also tracked total time elapsed between contact and order.

Exit. Depending on the scenario (purchase/no purchase), shoppers used an exit script, including a fallback offer of 1,000 FCFA to compensate for the merchant's time in cases where no purchase occurred because of the 20% random draw.

Vertical quality scoring. Upon receipt, each good was evaluated by two professional tailors (and a shoemaker for footwear) using a structured 50-point scorecard developed for the study.

Each criterion was scored on a 0–2 or 0–3 scale (with a few exceptions), and we computed both the raw score and a binary high-quality indicator (equal to 1 if the item's score was above the median for its product type). As each product was independently assessed by both experts, we average the two assessments.

Origin recording. In addition to the quality scores, surveyors recorded whether the item was labeled as "Made in Türkiye," a strong signal of quality in this setting. When labels were absent, tailors independently evaluated origin based on design and construction cues. The resulting variable is a binary indicator equal to 1 if the label—or both experts—indicated Turkish origin.

F.5 Vertical Quality Scoring — Scorecard

Upon receipt, each good was evaluated by two professional tailors (and a shoemaker for footwear) using a structured 50-point scorecard developed for the study. We designed this scorecard with our hired experts. We also drew inspiration from Vitali (2024), who uses a similar approach to assess garment quality in Kampala.

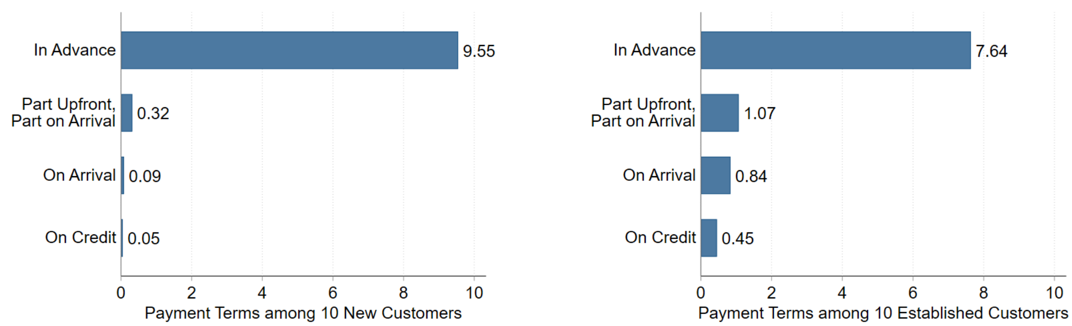
Table F1: Garment Quality Scorecard (50-point scale; higher = better)

Category	**Question**	**Polo**	**Dress**
Fabric	Is the fabric soft?	3	3
	Are there pills?	2	2
	Is it transparent under light?	4	4
	Are the threads consistent?	3	3
Sewing	Are the seams straight and neat?	3	3
	Do the seams come apart upon gentle pulling?	3	3
	Are there stray threads or loose stitches?	2	2
	Do the seams lie flat without puckering?	2	2
	Is the hem seam reinforced?	2	2
Shoulders	Do the shoulders fall correctly?	3	3
Collar and Buttons	Is the collar well-attached and flat?	3	–
	Are the buttons securely attached?	3	–
Darts	Are there darts on the bust and waist?	–	3
	Are the darts neatly sewn?	–	3
Colourfastness	Does it lose colour upon soaking?	7	7
Pattern	Does the pattern line up with the seams?	2	2
Hem	Is the hem well done?	4	4
Sleeves	Are the sleeves the same length and width?	4	4
Absence of Other Defects	Are there any other defects not covered above?	-10	-10
Total Score		**50**	**50**

Appendix G - Contract Terms and Relationship Dynamics

In this Appendix, we discuss descriptive evidence pertaining to the contract structure and relationship dynamics in the model and relate it to empirical findings in other settings. We discuss three features. First, for the contract structure itself, the model assumes that the buyer is required to pay for orders upfront. We make this assumption as this how the vast majority of contracts operate in our setting. As evidence for this, in a baseline survey that we conducted with 141 suppliers in Türkiye, we asked them, out of 10 customers, how many they would require to pay in full upfront, allow to pay on arrival, allow to pay on credit (i.e., at a later date than arrival), or pay some portion upfront and the remainder on arrival. We asked this separately for new and established customers. We plot the results in Figure G1. New customers are almost always required to pay in full upfront. This is relaxed slightly among established customers, although it is still the case that the majority of transactions involve upfront payment. Startz (2025) documents a similar payment structure among Nigerian importers, who prepay 97% of costs.

Figure G1: Contract Payment Terms out of 10 Customers



(a) New Customers

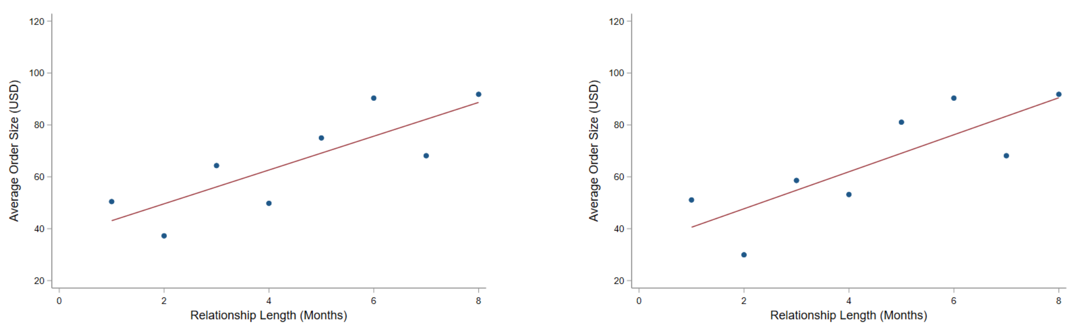
(b) Established Customers

Note: This figure shows the mean response to two survey questions about payment terms in a baseline survey with 141 suppliers in Turkey. The first question, shown in Panel (a), asks, out of 10 new customers, how many the supplier would require to pay upfront, allow them to pay on arrival, allow them to pay on credit (i.e., at a later date than arrival), or pay some portion upfront and the remainder on arrival. The second question, shown in Panel (b), is the same except for established customers.

Second, on relationship dynamics, the optimal contract in the model is non-stationary and, in particular, is backloaded in value, with both quantity and price rising over time. Quantity and/or value rising over time occurs under models of both adverse selection and moral hazard and is well-established in several empirical settings. For example, Monarch and Schmidt-Eisenlohr (2023) find that value traded increases with relationship length using US import data matched with foreign supplier identifiers, and Fitzgerald et al. (2024) shows that Irish exporters grow quantities in new markets over time. We find the same pattern in our relationship data. In Figure G2, we plot the mean value traded between study firms and study suppliers—i.e., new matches created in the study—as a function of relationship length. Panel (a) shows this unconditionally, and Panel (b) restricts to only relationships that lasted at least six months, to account for potential selection into relationship length. In both cases, average value traded approximately doubles over the course of the relationship.

The dynamic price path is significantly more complicated to evaluate empirically as the theoretical prediction is about *surplus* increasing—i.e., markups rather than prices *per se*. This means that one needs to control for changes in cost, which may change over the course of a relationship due to, for example, economies of scale (since quantities grow over time), learning by doing, or initial fixed costs. Monarch and Schmidt-Eisenlohr (2023) study prices in relationships in US imports, and find that prices decrease with relationship length, while Fitzgerald et al. (2024), who study markups directly, find that markups are flat over the course of an export spell for Irish exporters (though the data is at the market level rather than relationship level). Another empirical challenge is that, in general, both the buyer and seller have contracting problems (whose relative magnitude varies with the context) and there are two instruments available to provide incentives: the timing of the payment, and the allocation of future surplus. Whether markups grow or shrink thus depends on which party is primarily being incentivized by future surplus. This means that there is likely to be important heterogeneity across settings. In our setting, since we see in Figure G1 that the buyer's contracting problem is largely resolved in this setting through upfront payment, this leaves dynamic incentives for the seller side, which suggests that markups should at least be unlikely to decline. Unfortunately, our mobile payments data only contains value so ultimately we are not able to test this directly in our setting. That said, anecdotally, we heard often from suppliers that they provide new customer discounts and/or sell at cost in early phases of a relationship in order to allow new customers to observe quality and develop trust, i.e., investing in a reputation that they will later extract surplus from.

Figure G2: Average Value by Relationship Length



(a) Unconditional

(b) Conditional on Relationship Length

Note: This figure shows the mean value transferred from study firms to study suppliers—aggregated to the month level—in the mobile money data as a function of the number of months in which a given buyer-supplier pair has transacted. In Panel (b), to account for potential selection issues driven by the possibility that more optimistic firms are more likely to make both larger and repeat orders, we restrict the analysis to buyer-seller pairs with relationships lasting at least six months.